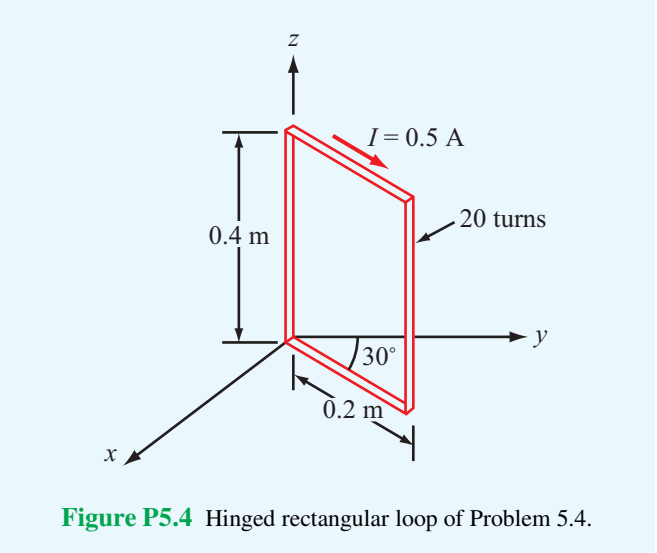


***5.4** The rectangular loop shown in **Fig. P5.4** consists of 20 closely wrapped turns and is hinged along the z axis. The plane of the loop makes an angle of 30° with the y axis, and the current in the windings is 0.5 A. What is the magnitude of the torque exerted on the loop in the presence of a uniform field $\mathbf{B} = \hat{y} 2.4 \text{ T}$? When viewed from above, is the expected direction of rotation clockwise or counterclockwise?



5.5 In a cylindrical coordinate system, a 2-m long straight wire carrying a current of 5 A in the positive z direction is located at $r = 4 \text{ cm}$, $\phi = \pi/2$, and $-1 \text{ m} \leq z \leq 1 \text{ m}$.

- (a)** If $\mathbf{B} = \hat{r} 0.2 \cos \phi \text{ (T)}$, what is the magnetic force acting on the wire?
- (b)** How much work is required to rotate the wire once about the z axis in the negative ϕ direction (while maintaining $r = 4 \text{ cm}$)?
- (c)** At what angle ϕ is the force a maximum?

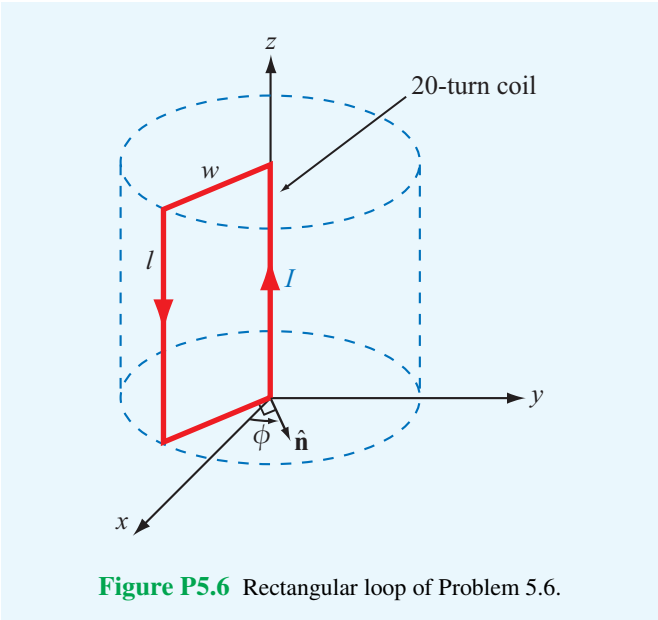
5.6 A 20-turn rectangular coil with sides $l = 30 \text{ cm}$ and $w = 10 \text{ cm}$ is placed in the y - z plane as shown in **Fig. P5.6**.

- (a)** If the coil, which carries a current $I = 10 \text{ A}$, is in the presence of a magnetic flux density

$$\mathbf{B} = 2 \times 10^{-2}(\hat{x} + \hat{y}2) \quad (\text{T}).$$

Determine the torque acting on the coil.

- (b)** At what angle ϕ is the torque zero?
- (c)** At what angle ϕ is the torque maximum? Determine its value.



Section 5-2: The Biot–Savart Law

***5.7** An $8 \text{ cm} \times 12 \text{ cm}$ rectangular loop of wire is situated in the x - y plane with the center of the loop at the origin and its long sides parallel to the x axis. The loop has a current of 50 A flowing clockwise (when viewed from above). Determine the magnetic flux density at the center of the loop.

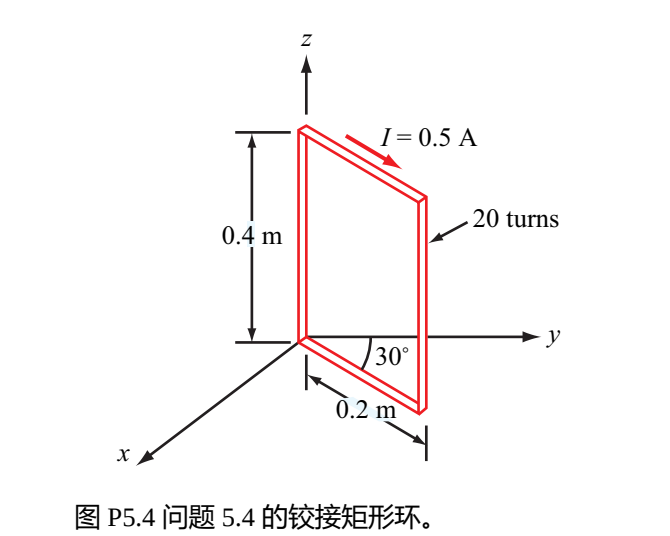
5.8 Use the approach outlined in Example 5-2 to develop an expression for the magnetic field $\mathbf{H}$ at an arbitrary point P due to the linear conductor defined by the geometry shown in **Fig. P5.8**. If the conductor extends between $z_1 = 3 \text{ m}$ and $z_2 = 7 \text{ m}$ and carries a current $I = 15 \text{ A}$, find $\mathbf{H}$ at $P = (2, \phi, 0)$.

***5.9** The loop shown in **Fig. P5.9** consists of radial lines and segments of circles whose centers are at point P . Determine the magnetic field $\mathbf{H}$ at P in terms of a , b , θ , and I .

5.10 An infinitely long, thin conducting sheet defined over the space $0 \leq x \leq w$ and $-\infty \leq y \leq \infty$ is carrying a current with a uniform surface current density $\mathbf{J}_s = \hat{y}5 \text{ (A/m)}$. Obtain an expression for the magnetic field at point $P = (0, 0, z)$ in Cartesian coordinates.

***5.11** An infinitely long wire carrying a 25 A current in the positive x direction is placed along the x axis in the vicinity of a 20-turn circular loop located in the x - y plane (**Fig. P5.11**). If the magnetic field at the center of the loop is zero, what is the direction and magnitude of the current flowing in the loop?

***5.4** 图 P5.4 所示的矩形环由 20 个紧密缠绕的匝组成，并沿 z 轴铰接。环路平面与 y 轴成 30° 角，绕组中的电流为 0.5 A。在存在均匀磁场 $\mathbf{B} = \hat{y} 2.4 \text{ T}$ 的情况下，施加在环路上的扭矩大小是多少？从上方观察时，预期的旋转方向是顺时针还是逆时针？



5.5 在圆柱坐标系中，在 z 正方向上承载 5 A 电流的 2 米长直导线位于 $r = 4 \text{ cm}$ 、 $\phi = \pi/2$ 和 $-1 \text{ m} \leq z \leq 1 \text{ m}$ 处。

(a) 如果 $\mathbf{B} = \hat{r} 0.2 \cos \phi \text{ (T)}$ ，则作用在导线上的磁力是多少？**(b)** 将导线绕 z 轴沿负 ϕ 方向旋转一次需要多少功（同时保持 $r = 4 \text{ cm}$ ）？

(c) 在什么角度 ϕ 时力最大？

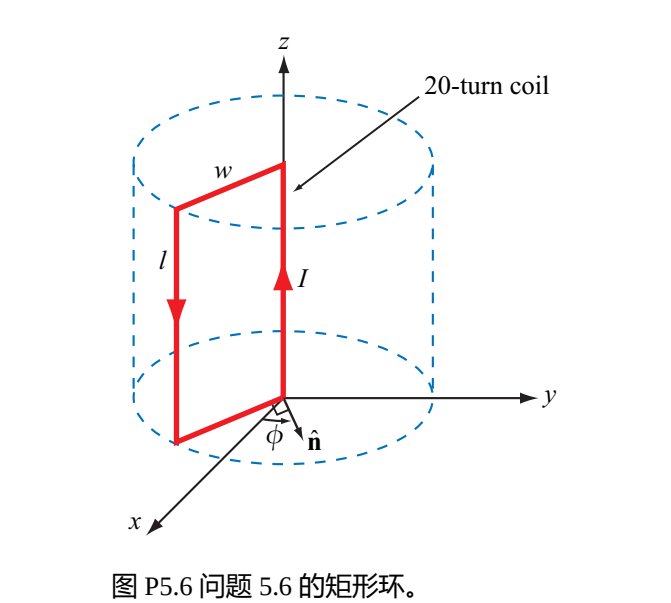
5.6 将边长 $l = 30 \text{ cm}$ 、 $w = 10 \text{ cm}$ 的 20 匝矩形线圈放置在 y - z 平面上，如图 P5.6 所示。

(a) 如果承载电流 $I = 10 \text{ A}$ 的线圈存在磁通密度

$$\mathbf{B} = 2 \times 10^{-2}(\hat{x} + \hat{y}2) \quad (\text{T}).$$

确定作用在线圈上的扭矩。

- (b)** 在什么角度 ϕ 处扭矩为零？
- (c)** 在什么角度 ϕ 时扭矩最大？确定其价值。



第 5-2 节：毕奥-萨伐尔定律

***5.7** 将一个 8 厘米 $\times$ 12 厘米的矩形线圈放置在 x - y 平面上，线圈的中心位于原点，其长边平行于 x 轴。该环路有 50 A 的电流顺时针流动（从上方观察）。确定环路中心的磁通密度。

5.8 使用例 5-2 中概述的方法，建立由图 P5.8 所示的几何形状定义的线性导体在任意点 P 处的磁场 $\mathbf{H}$ 的表达式。如果导体在 $z_1 = 3 \text{ m}$ 和 $z_2 = 7 \text{ m}$ 之间延伸并承载电流 $I = 15 \text{ A}$ ，则在 $P = (2, \phi, 0)$ 处找到 $\mathbf{H}$ 。

***5.9** 图 P5.9 所示的环路由径向线和圆心位于 P 点的圆段组成。用 a 、 b 、 θ 和 I 确定 P 处的磁场 $\mathbf{H}$ 。

5.10 定义在空间 $0 \leq x \leq w$ 和 $-\infty \leq y \leq \infty$ 上的无限长、薄的导电片承载着均匀表面电流密度 $\mathbf{J}_s = \hat{y}5 \text{ (A/m)}$ 的电流。获得笛卡尔坐标中 $P = (0, 0, z)$ 点的磁场表达式。

***5.11** 将一根在 x 正方向上承载 25 A 电流的无限长导线沿 x 轴放置在位于 x - y 平面的 20 匝圆环附近（图 P5.11）。如果环路中心的磁场为零，则环路中流动的电流的方向和大小是多少？

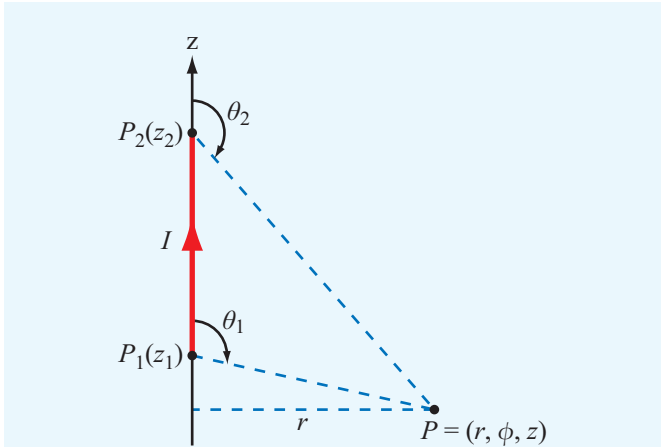


Figure P5.8 Current-carrying linear conductor of Problem 5.8.

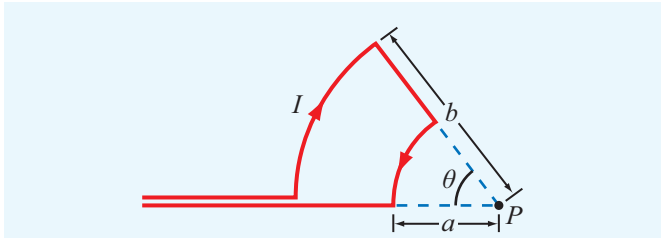


Figure P5.9 Configuration of Problem 5.9.

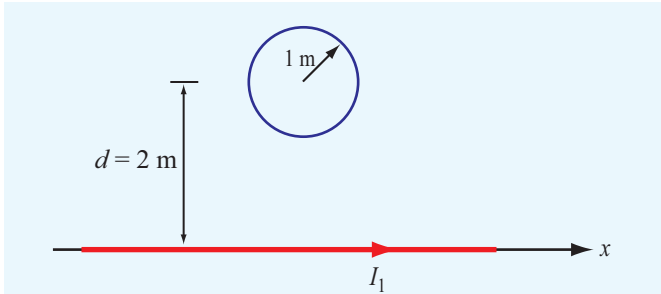


Figure P5.11 Circular loop next to a linear current (Problem 5.11).

5.12 Two infinitely long, parallel wires are carrying 6 A currents in opposite directions. Determine the magnetic flux density at point P in Fig. P5.12.

*5.13 A long, East–West-oriented power cable carrying an unknown current I is at a height of 8 m above the Earth’s

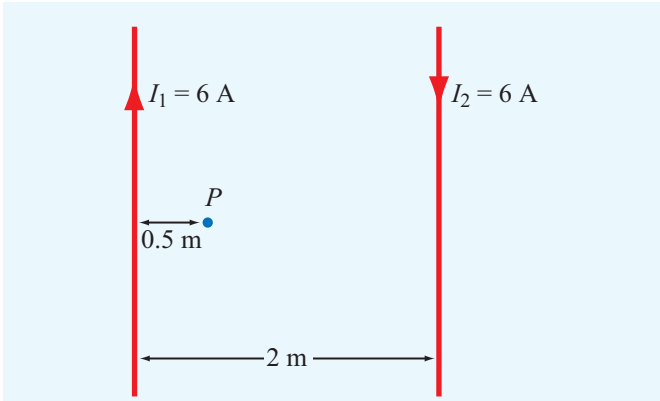


Figure P5.12 Arrangement for Problem 5.12.

surface. If the magnetic flux density recorded by a magnetic-field meter placed at the surface is $15 \mu\text{T}$ when the current is flowing through the cable and $20 \mu\text{T}$ when the current is zero, what is the magnitude of I ?

5.14 Two parallel, circular loops carrying a current of 40 A each are arranged as shown in Fig. P5.14. The first loop is situated in the x – y plane with its center at the origin, and the second loop’s center is at $z = 2 \text{ m}$. If the two loops have the same radius $a = 3 \text{ m}$, determine the magnetic field at

- (a) $z = 0$
- (b) $z = 1 \text{ m}$
- (c) $z = 2 \text{ m}$

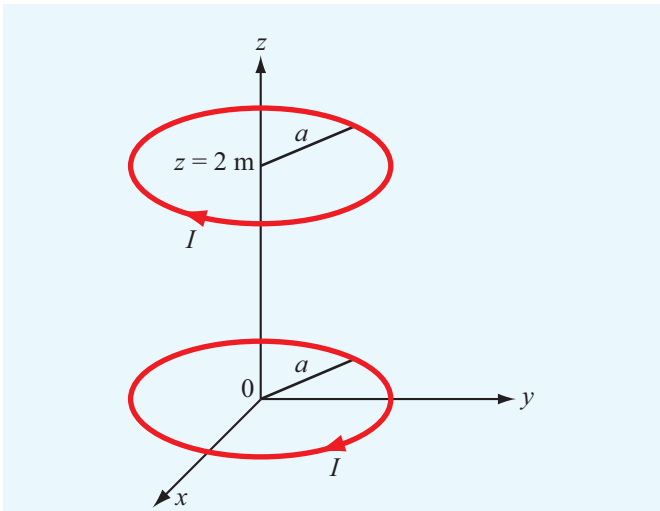


Figure P5.14 Parallel circular loops of Problem 5.14.

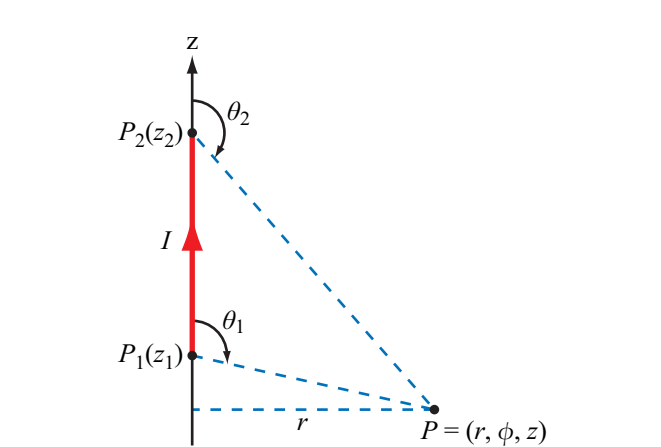


图 P5.8 问题 5.8 的载流线性导体。

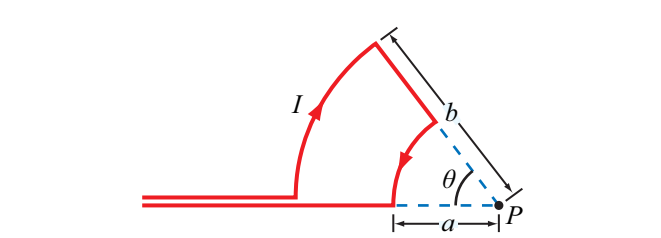


图 P5.9 问题 5.9 的配置。

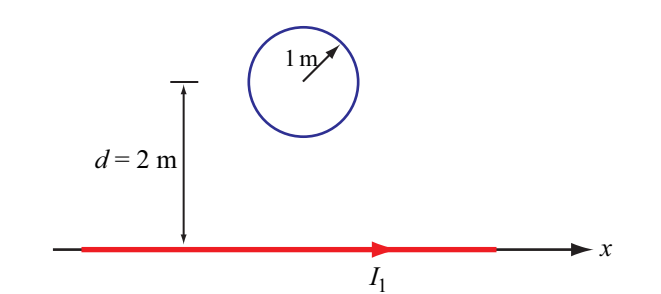


图 P5.11 线性电流旁边的圆形环路 (问题 5.11)。

5.12 两根无限长的平行电线在相反方向上承载 6 A 电流。确定图 P5.12 中 P 点的磁通密度。

*5.13 一条东西向的长电力电缆，载有未知电流 I ，位于距地球 8 m 的高度

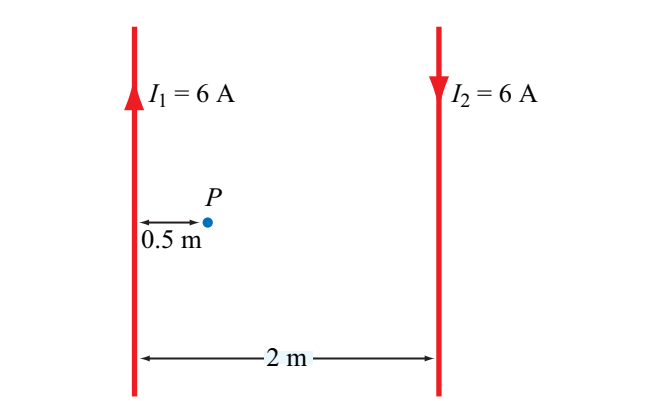


图 P5.12 问题 5.12 的排列。

表面。如果当电流流过电缆时，放置在表面的磁场计记录的磁通密度为 $15 \mu\text{T}$ ，当电流为零时为 $20 \mu\text{T}$ ，那么 I 的大小是多少？

5.14 两个并联的圆形回路，每个回路承载 40 A 的电流，如图 P5.14 所示布置。第一个回路位于 x – y 平面内，其中心位于原点，第二个回路的中心位于 $z = 2 \text{ m}$ 。如果两个回路具有相同的半径 $a = 3 \text{ m}$ ，则确定磁场

- (a) $z = 0$ (b)
-) $z = 1 \text{ m}$ (c)
-) $z = 2 \text{ m}$

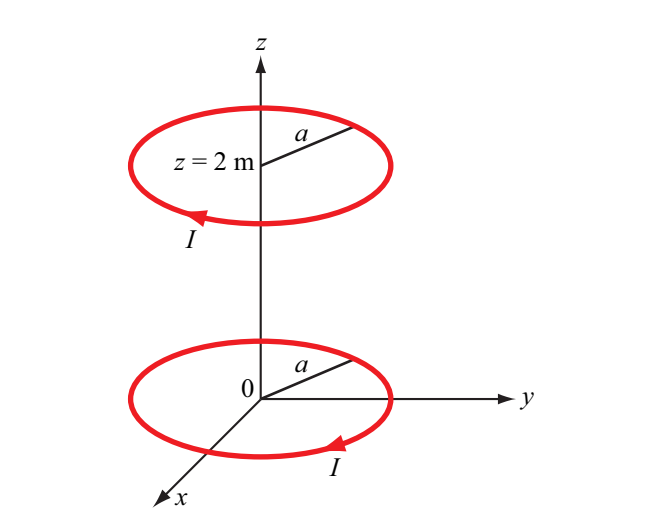
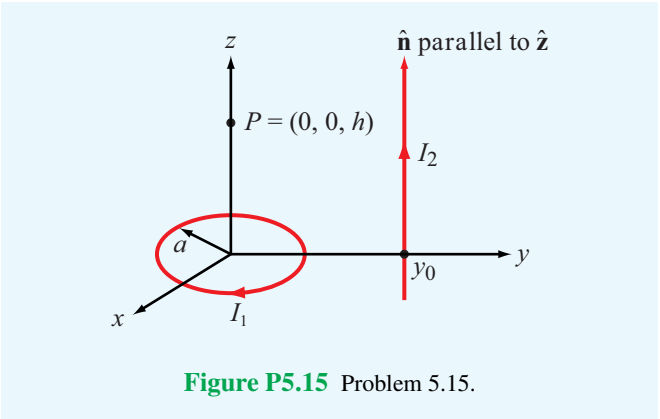


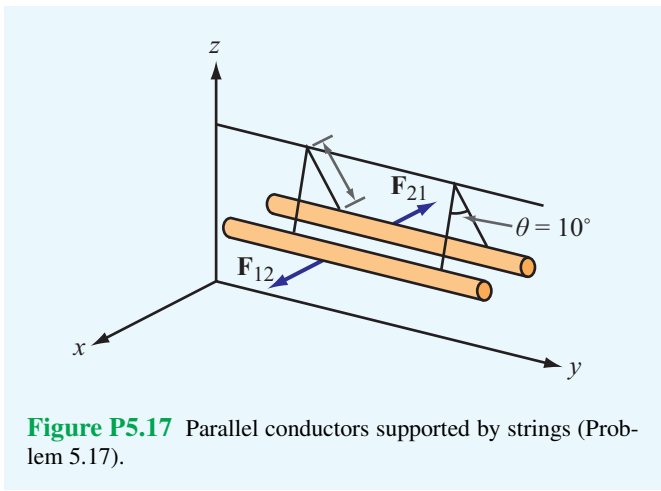
图 P5.14 问题 5.14 的平行循环。

5.15 A circular loop of radius a carrying current I_1 is located in the x - y plane as shown in **Fig. P5.15**. In addition, an infinitely long wire carrying current I_2 in a direction parallel with the z axis is located at $y = y_0$.

- (a) Determine $\mathbf{H}$ at $P = (0, 0, h)$.
(b) Evaluate $\mathbf{H}$ for $a = 3$ cm, $y_0 = 10$ cm, $h = 4$ cm, $I_1 = 10$ A, and $I_2 = 20$ A.



5.17 In the arrangement shown in **Fig. P5.17**, each of the two long, parallel conductors carries a current I , is supported by 8-cm long strings, and has a mass per unit length of 1.2 g/cm. Due to the repulsive force acting on the conductors, the angle θ between the supporting strings is 10° . Determine the magnitude of I and the relative directions of the currents in the two conductors.



***5.16** The long, straight conductor shown in **Fig. P5.16** lies in the plane of the rectangular loop at a distance $d = 0.1$ m. The loop has dimensions $a = 0.2$ m and $b = 0.5$ m, and the currents are $I_1 = 20$ A and $I_2 = 30$ A. Determine the net magnetic force acting on the loop.

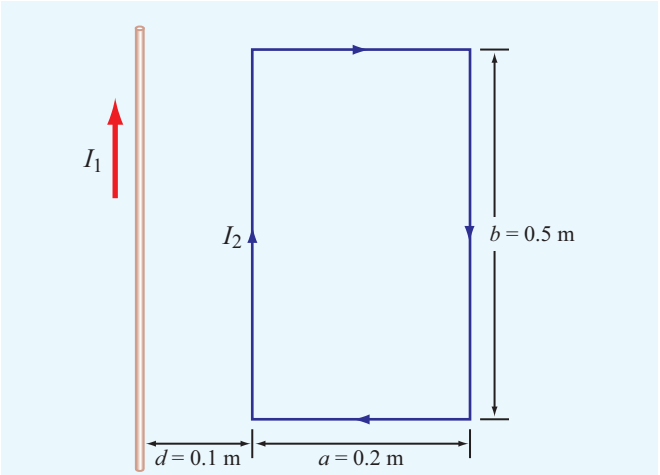


Figure P5.16 Current loop next to a conducting wire (Problem 5.16).

5.18 An infinitely long, thin conducting sheet of width w along the x direction lies in the x - y plane and carries a current I in the $-y$ direction. Determine the following:

- (a) The magnetic field at a point P midway between the edges of the sheet and at a height h above it (**Fig. P5.18**).
(b) The force per unit length exerted on an infinitely long wire passing through point P and parallel to the sheet if the current through the wire is equal in magnitude but opposite in direction to that carried by the sheet.

5.19 Three long, parallel wires are arranged as shown in **Fig. P5.19**. Determine the force per unit length acting on the wire carrying I_3 .

***5.20** A square loop placed as shown in **Fig. P5.20** has 2 m sides and carries a current $I_1 = 5$ A. If a straight, long conductor carrying a current $I_2 = 10$ A is introduced and placed just above the midpoints of two of the loop's sides, determine the net force acting on the loop.

5.15 载有电流 I_1 的半径为 a 的圆环位于 x - y 平面内，如图 P5.15 所示。此外，在与 z 轴平行的方向上承载电流 I_2 的无限长导线位于 $y = y_0$ 处。

- (a) 确定 $P = (0, 0, h)$ 处的 $\mathbf{H}$ 。(b) 评估 $a = 3$ cm、 $y_0 = 10$ cm、 $h = 4$ cm、 $I_1 = 10$ A 和 $I_2 = 20$ A 的 $\mathbf{H}$ 。

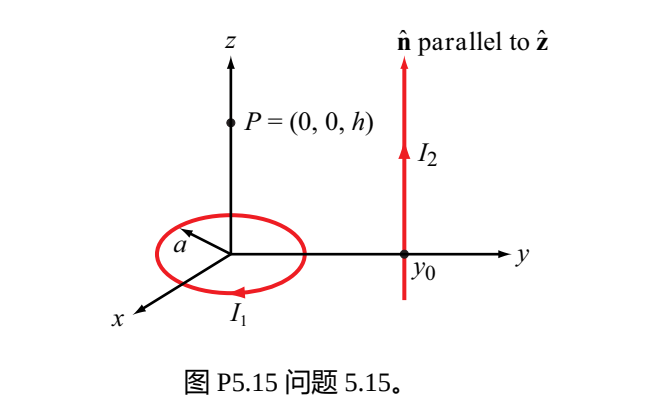


图 P5.15 问题 5.15。

***5.16** 图 P5.16 所示的长直导体位于矩形环平面内，距离 $d = 0.1$ m。该环路的尺寸为 $a = 0.2$ m 和 $b = 0.5$ m，电流为 $I_1 = 20$ A 和 $I_2 = 30$ A。确定作用在环路上的净磁力。

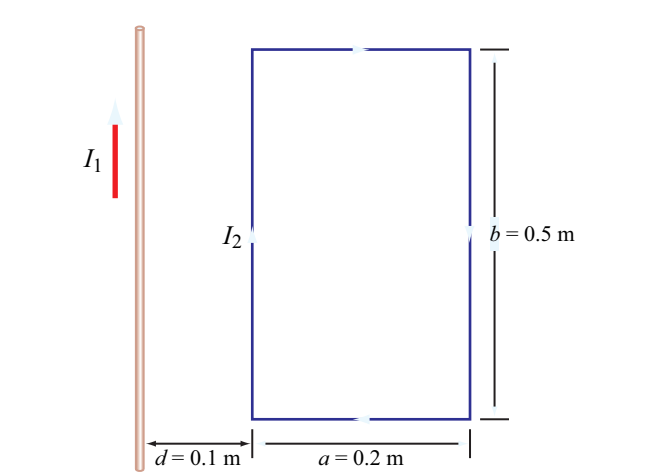


图 P5.16 导线旁边的电流环路 (问题 5.16)。

5.17 在图 P5.17 所示的布置中，两个长的平行导体中的每一个都承载电流 I ，由 8 厘米长的细绳支撑，单位长度的质量为 1.2 克/厘米。由于排斥力作用在导体上，支撑弦之间的角度 θ 为 10° 。确定 I 的大小以及两个导体中电流的相对方向。

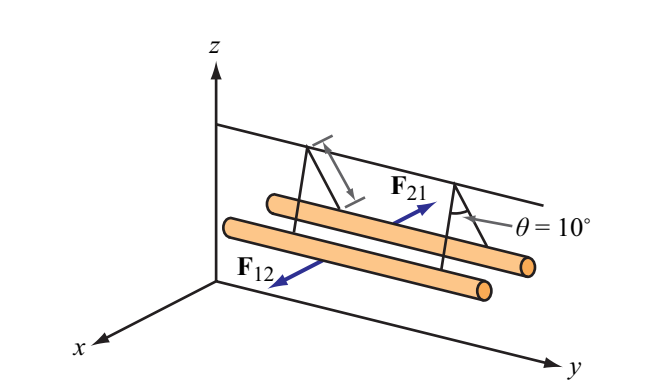


图 P5.17 由串支撑的平行导体 (问题 5.17)。

5.18 沿 x 方向宽度为 w 的无限长、薄导电片位于 x - y 平面内，并在 $-y$ 方向上承载电流 I 。确定以下内容：

- (a) 片材边缘中间 P 点及其上方高度 h 处的磁场 (图 P5.18)。
(b) 如果通过导线的电流大小相等但方向与片材所承载的电流方向相反，则施加在穿过点 P 并平行于片材的无限长导线上的每单位长度的力。

5.19 三根长的平行导线的排列如图 P5.19 所示。确定作用在承载 I_3 的导线上的每单位长度的力。

***5.20** 如图 P5.20 所示放置的方形环路有 2 m 边，承载电流 $I_1 = 5$ A。如果引入承载电流 $I_2 = 10$ A 的直长导体并将其放置在环路两侧中点的正上方，请确定作用在环路上的合力。

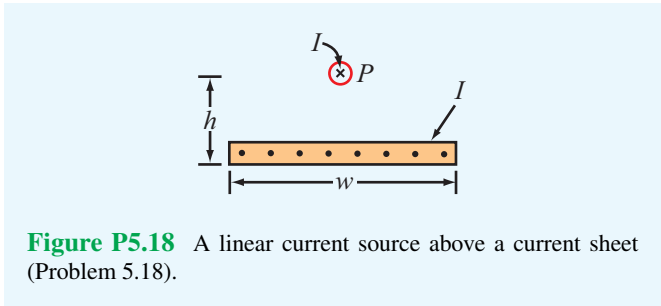


Figure P5.18 A linear current source above a current sheet (Problem 5.18).

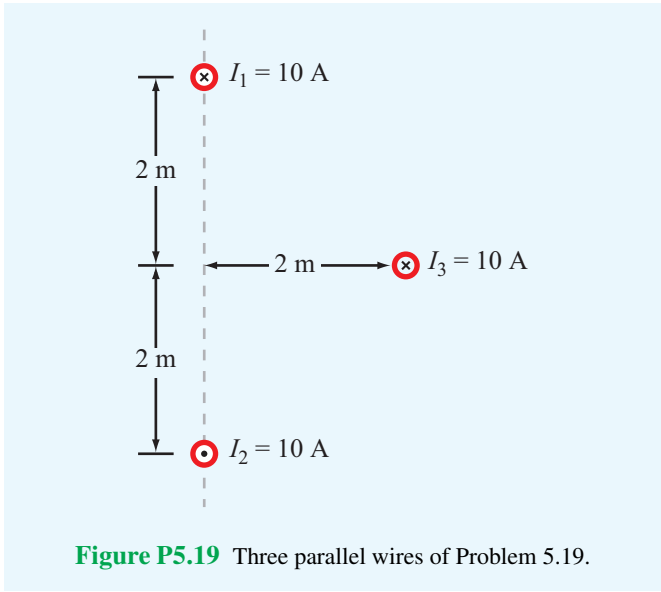


Figure P5.19 Three parallel wires of Problem 5.19.

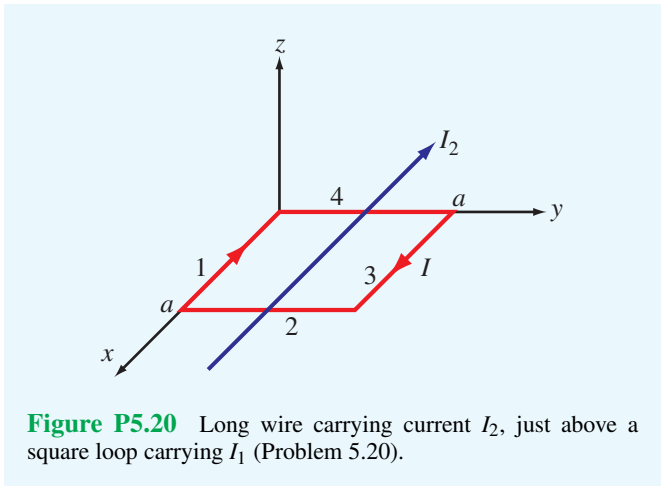


Figure P5.20 Long wire carrying current I_2 , just above a square loop carrying I_1 (Problem 5.20).

Section 5-3: Maxwell's Magnetostatic Equations

5.21 Current I flows along the positive z direction in the inner conductor of a long coaxial cable and returns through the outer conductor. The inner conductor has radius a , and the inner and outer radii of the outer conductor are b and c , respectively.

- (a) Determine the magnetic field in each of the following regions: $0 \leq r \leq a$, $a \leq r \leq b$, $b \leq r \leq c$, and $r \geq c$.
- (b) Plot the magnitude of $\mathbf{H}$ as a function of r over the range from $r = 0$ to $r = 10$ cm given that $I = 10$ A, $a = 2$ cm, $b = 4$ cm, and $c = 5$ cm.

5.22 A long cylindrical conductor whose axis is coincident with the z axis has a radius a and carries a current characterized by a current density $\mathbf{J} = \hat{\mathbf{z}}J_0/r$, where J_0 is a constant and r is the radial distance from the cylinder's axis. Obtain an expression for the magnetic field $\mathbf{H}$ for

- (a) $0 \leq r \leq a$
- (b) $r > a$

5.23 Repeat Problem 5.22 for a current density $\mathbf{J} = \hat{\mathbf{z}}J_0e^{-r}$.

*5.24 In a certain conducting region, the magnetic field is given in cylindrical coordinates by

$$\mathbf{H} = \hat{\boldsymbol{\phi}} \frac{4}{r} [1 - (1 + 3r)e^{-3r}].$$

Find the current density $\mathbf{J}$.

5.25 A cylindrical conductor whose axis is coincident with the z axis has an internal magnetic field given by

$$\mathbf{H} = \hat{\boldsymbol{\phi}} \frac{2}{r} [1 - (4r + 1)e^{-4r}] \quad (\text{A/m}) \quad \text{for } r \leq a,$$

where a is the conductor's radius. If $a = 5$ cm, what is the total current flowing in the conductor?

Section 5-4: Vector Magnetic Potential

5.26 With reference to Fig. 5-10:

- * (a) Derive an expression for the vector magnetic potential $\mathbf{A}$ at a point P located at a distance r from the wire in the x - y plane.
- (b) Derive $\mathbf{B}$ from $\mathbf{A}$. Show that your result is identical with the expression given by Eq. (5.29), which was derived by applying the Biot-Savart law.

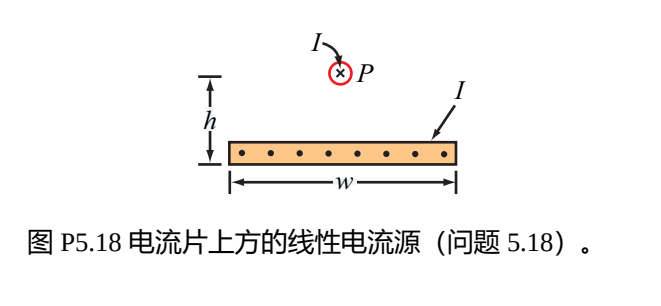


图 P5.18 电流片上方的线性电流源 (问题 5.18)。

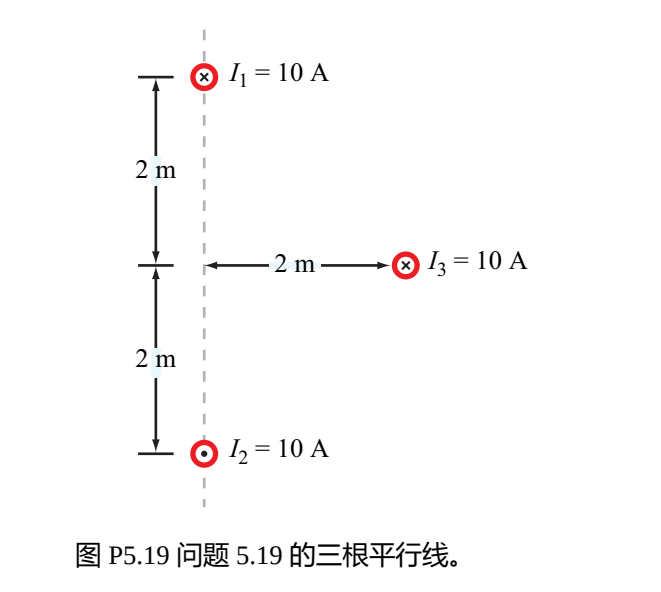


图 P5.19 问题 5.19 的三根平行线。

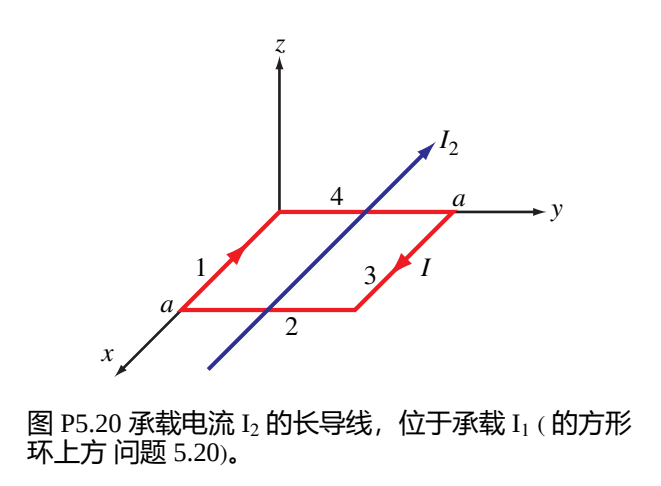


图 P5.20 承载电流 I_2 的长导线，位于承载 I_1 的方形环上方 (问题 5.20)。

第 5-3 节：麦克斯韦静磁方程

5.21 电流 I 在长同轴电缆的内导体中沿 z 轴正方向流动，并通过外导体返回。内导体的半径为 a ，外导体的内半径和外半径分别为 b 和 c 。

- (a) 确定以下每个区域的磁场： $0 \leq r \leq a$ 、 $a \leq r \leq b$ 、 $b \leq r \leq c$ 和 $r \geq c$ 。
- (b) 假定 $I = 10$ A、 $a = 2$ cm、 $b = 4$ cm 和 $c = 5$ cm，在 $r = 0$ 到 $r = 10$ cm 范围内绘制 H 的幅度作为 r 的函数。

5.22 轴线与 z 轴重合的长圆柱形导体，其半径为 a ，并承载由电流密度 $\mathbf{J} = \hat{\mathbf{z}}J_0/r$ 表征的电流，其中 J_0 是常数， r 是距圆柱轴线的径向距离。获得磁场 $\mathbf{H}$ 的表达式

- (a) $0 \leq r \leq a$
- (b) $r > a$

5.23 对于电流密度 $\mathbf{J} = \hat{\mathbf{z}}J_0e^{-r}$ 重复问题 5.22

*5.24 在某个导电区域，磁场在柱坐标中由下式给出：

$$\mathbf{H} = \hat{\boldsymbol{\phi}} \frac{4}{r} [1 - (1 + 3r)e^{-3r}].$$

求电流密度 $\mathbf{J}$ 。

5.25 轴线与 z 轴重合的圆柱形导体，其内部磁场为：

$$\mathbf{H} = \hat{\boldsymbol{\phi}} \frac{2}{r} [1 - (4r + 1)e^{-4r}] \quad (\text{A/m}) \quad \text{for } r \leq a,$$

其中 a 是导体的半径。如果 $a = 5$ cm，导体中流动的总电流是多少？

第 5-4 节：矢量磁势

5.26 参考图 5-10：

- * (a) 导出 x - y 平面中距导线距离 r 的点 P 处的矢量磁势 $\mathbf{A}$ 的表达式。
- (b) 从 $\mathbf{A}$ 导出 $\mathbf{B}$ 。证明你的结果与式 (1) 给出的表达式相同。(5.29)，这是通过应用毕奥-萨伐尔定律推导出来的。

5.27 In a given region of space, the vector magnetic potential is given by $\mathbf{A} = \hat{\mathbf{x}}5 \cos \pi y + \hat{\mathbf{z}}(2 + \sin \pi x)$ (Wb/m).

- * (a)** Determine $\mathbf{B}$.
- (b)** Use Eq. (5.66) to calculate the magnetic flux passing through a square loop with 0.25-m long edges if the loop is in the x - y plane, its center is at the origin, and its edges are parallel to the x and y axes.
- (c)** Calculate Φ again using Eq. (5.67).

5.28 A uniform current density given by

$$\mathbf{J} = \hat{\mathbf{z}}J_0 \quad (\text{A/m}^2)$$

gives rise to a vector magnetic potential

$$\mathbf{A} = -\hat{\mathbf{z}} \frac{\mu_0 J_0}{4} (x^2 + y^2) \quad (\text{Wb/m}).$$

- (a)** Apply the vector Poisson’s equation to confirm the above statement.
- (b)** Use the expression for $\mathbf{A}$ to find $\mathbf{H}$.
- (c)** Use the expression for $\mathbf{J}$ in conjunction with Ampère’s law to find $\mathbf{H}$. Compare your result with that obtained in part (b).

*** 5.29** A thin current element extending between $z = -L/2$ and $z = L/2$ carries a current I along $+\hat{\mathbf{z}}$ through a circular cross-section of radius a .

- (a)** Find $\mathbf{A}$ at a point P located very far from the origin (assume R is so much larger than L that point P may be considered to be at approximately the same distance from every point along the current element).
- (b)** Determine the corresponding $\mathbf{H}$.

Section 5-5: Magnetic Properties of Materials

5.30 In the model of the hydrogen atom proposed by Bohr in 1913, the electron moves around the nucleus at a speed of 2×10^6 m/s in a circular orbit of radius 5×10^{-11} m. What is the magnitude of the magnetic moment generated by the electron’s motion?

*** 5.31** Iron contains 8.5×10^{28} atoms/m³. At saturation, the alignment of the electrons’ spin magnetic moments in iron can contribute 1.5 T to the total magnetic flux density $\mathbf{B}$. If the spin magnetic moment of a single electron is 9.27×10^{-24} (A·m²), how many electrons per atom contribute to the saturated field?

Section 5-6: Magnetic Boundary Conditions

5.32 The x - y plane separates two magnetic media with magnetic permeabilities μ_1 and μ_2 (**Fig. P5.32**). If there is no surface current at the interface and the magnetic field in medium 1 is

$$\mathbf{H}_1 = \hat{\mathbf{x}}H_{1x} + \hat{\mathbf{y}}H_{1y} + \hat{\mathbf{z}}H_{1z},$$

find:

- (a)** $\mathbf{H}_2$
- (b)** θ_1 and θ_2
- (c)** Evaluate $\mathbf{H}_2$, θ_1 , and θ_2 for $H_{1x} = 2$ (A/m), $H_{1y} = 0$, $H_{1z} = 4u$ (A/m), $\mu_1 = \mu_0$, and $\mu_2 = 4\mu_0$

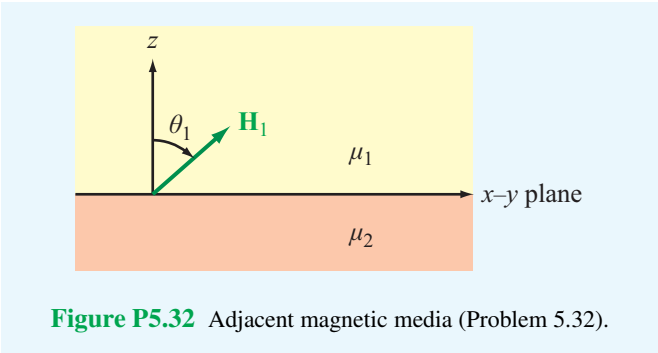


Figure P5.32 Adjacent magnetic media (Problem 5.32).

*** 5.33** Given that a current sheet with surface current density $\mathbf{J}_s = \hat{\mathbf{x}}8$ (A/m) exists at $y = 0$, the interface between two magnetic media, and $\mathbf{H}_1 = \hat{\mathbf{z}}11$ (A/m) in medium 1 ($y > 0$), determine $\mathbf{H}_2$ in medium 2 ($y < 0$).

5.34 In **Fig. P5.34**, the plane defined by $x - y = 1$ separates medium 1 of permeability μ_1 from medium 2 of permeability μ_2 . If no surface current exists on the boundary and

$$\mathbf{B}_1 = \hat{\mathbf{x}}2 + \hat{\mathbf{y}}3 \quad (\text{T}),$$

find $\mathbf{B}_2$ and then evaluate your result for $\mu_1 = 5\mu_2$. *Hint:* Start by deriving the equation for the unit vector normal to the given plane.

*** 5.35** The plane boundary defined by $z = 0$ separates air from a block of iron. If $\mathbf{B}_1 = \hat{\mathbf{x}}4 - \hat{\mathbf{y}}6 + \hat{\mathbf{z}}8$ in air ($z \geq 0$), find $\mathbf{B}_2$ in iron ($z \leq 0$) given that $\mu = 5000\mu_0$ for iron.

5.27 在给定的空间区域中， 矢量磁势由 $\mathbf{A} = \hat{\mathbf{x}}5 \cos \pi y + \hat{\mathbf{z}}(2 + \sin \pi x)$ (Wb/m) 给出。

- * (a)** 确定 $\mathbf{B}$.
- (b)** 使用等式。 (5.66) 计算通过具有 0.25 米长边的方形环的磁通量， 如果该环位于 x - y 平面内， 其中心位于原点， 且其边缘平行于 x 轴和 y 轴。
- (c)** 使用等式再次计算 Φ 。 (5.67) 。

5.28 均匀电流密度由下式给出

$$\mathbf{J} = \hat{\mathbf{z}}J_0 \quad (\text{A/m}^2)$$

产生矢量磁势

$$\mathbf{A} = -\hat{\mathbf{z}} \frac{\mu_0 J_0}{4} (x^2 + y^2) \quad (\text{Wb/m}).$$

- (a)** 应用矢量泊松方程来证实上述说法。
- (b)** 使用 $\mathbf{A}$ 的表达式求 $\mathbf{H}$ 。
- (c)** 使用 $\mathbf{J}$ 的表达式结合 Ampère 定律求 $\mathbf{H}$ 。 将你的结果与 (b) 部分得到的结果进行比较。

*** 5.29** 在 $z = -L/2$ 和 $z = L/2$ 之间延伸的细电流元件沿着 $+\hat{\mathbf{z}}$ 通过半径为 a 的圆形横截面承载电流 I 。

- (a)** 在距离原点很远的点 P 处找到 $\mathbf{A}$ （假设 R 远大于 L ， 以至于点 P 可以被认为与沿电流微元的每个点的距离大致相同）。
- (b)** 确定相应的 $\mathbf{H}$ 。

第 5-5 节： 材料的磁性

5.30 在玻尔 1913 年提出的氢原子模型中， 电子在半径为 5×10^{-11} m 的圆形轨道上以 2×10^6 m/s 的速度绕原子核运动。 电子运动产生的磁矩的大小是多少？

*** 5.31** 铁含有 8.5×10^{28} 原子/m³。 饱和时， 铁中电子自旋磁矩的排列可以为总磁通密度 $\mathbf{B}$ 贡献 1.5 T。 如果单个电子的自旋磁矩为 9.27×10^{-24} (A·m²)， 则每个原子有多少电子对饱和场有贡献？

第 5-6 节： 磁边界条件

5.32 x - y 平面将两种磁导率分别为 μ_1 和 μ_2 (的磁介质分开 (图 5.32) 。 **P5.32**)。 如果界面处没有表面电流且介质 1 中的磁场为

$$\mathbf{H}_1 = \hat{\mathbf{x}}H_{1x} + \hat{\mathbf{y}}H_{1y} + \hat{\mathbf{z}}H_{1z},$$

发现：

- (a)** $\mathbf{H}_2$
- (b)** θ_1 和 θ_2
- (c)** 计算 $\mathbf{H}_2$ 、 θ_1 和 θ_2 ， 以获得 $H_{1x} = 2$ (A/m)、 $H_{1y} = 0$ 、 $H_{1z} = 4u$ (A/m)、 $\mu_1 = \mu_0$ 和 $\mu_2 = 4\mu_0$

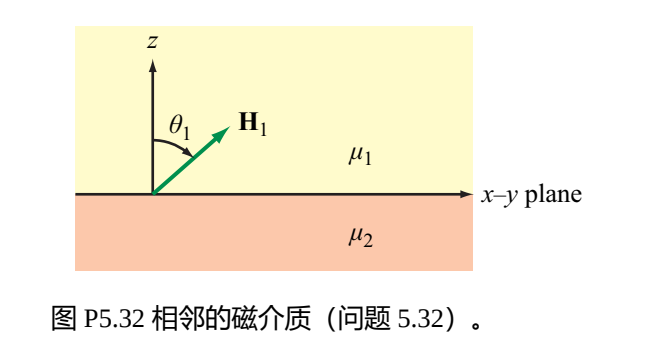


图 P5.32 相邻的磁介质 (问题 5.32) 。

*** 5.33** 假设在两种磁性介质之间的界面 $y = 0$ 处存在表面电流密度为 $\mathbf{J}_s = \hat{\mathbf{x}}8$ (A/m) 的电流片， 以及介质 1 ($y > 0$) 中的 $\mathbf{H}_1 = \hat{\mathbf{z}}11$ (A/m)， 则确定介质 2 ($y < 0$) 中的 $\mathbf{H}_2$ 。

5.34 在图 P5.34 中， 由 $x - y = 1$ 定义的平面将渗透率 μ_1 的介质 1 与渗透率 μ_2 的介质 2 分开。 如果边界上不存在表面电流且

$$\mathbf{B}_1 = \hat{\mathbf{x}}2 + \hat{\mathbf{y}}3 \quad (\text{T}),$$

找到 $\mathbf{B}_2$ ， 然后评估 $\mu_1 = 5\mu_2$ 的结果。 提示： 首先推导垂直于给定平面的单位向量的方程。

*** 5.35** 由 $z = 0$ 定义的平面边界将空气与铁块分开。 如果 $\mathbf{B}_1 = \hat{\mathbf{x}}4 - \hat{\mathbf{y}}6 + \hat{\mathbf{z}}8$ 在空气中 ($z \geq 0$)， 则在铁 ($z \leq 0$) 中找到 $\mathbf{B}_2$ ， 前提是铁的 $\mu = 5000\mu_0$ 。

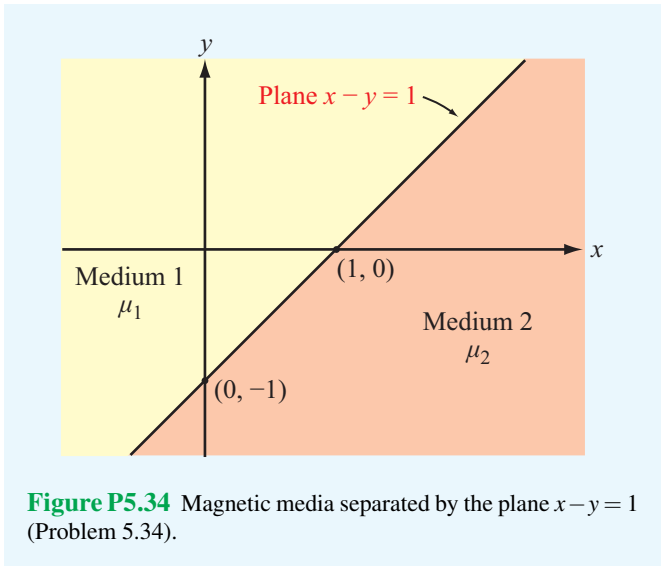


Figure P5.34 Magnetic media separated by the plane $x - y = 1$ (Problem 5.34).

5.36 Show that, if no surface current densities exist at the parallel interfaces shown in Fig. P5.36, the relationship between θ_4 and θ_1 is independent of μ_2 .

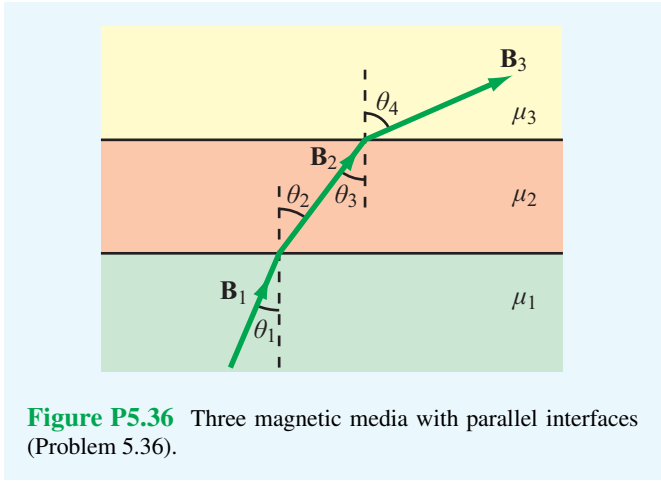


Figure P5.36 Three magnetic media with parallel interfaces (Problem 5.36).

Sections 5-7 and 5-8: Inductance and Magnetic Energy

- *5.37 Obtain an expression for the self-inductance per unit length for the parallel wire transmission line of Fig. 5-28(a) in terms of a , d , and μ , where a is the radius of the wires, d is the axis-to-axis distance between the wires, and μ is the permeability of the medium in which they reside.
- 5.38 A solenoid with a length of 20 cm and a radius of 5 cm consists of 400 turns and carries a current of 12 A. If $z = 0$ represents the midpoint of the solenoid, generate a plot for $|\mathbf{H}(z)|$ as a function of z along the axis of the solenoid for the range $-20 \text{ cm} \leq z \leq 20 \text{ cm}$ in 1 cm steps.
- 5.39 In terms of the dc current I , how much magnetic energy is stored in the insulating medium of a 3-m long, air-filled section of a coaxial transmission line given that the radius of the inner conductor is 5 cm and the inner radius of the outer conductor is 10 cm?
- *5.40 Determine the mutual inductance between the circular loop and the linear current shown in Fig. P5.40.

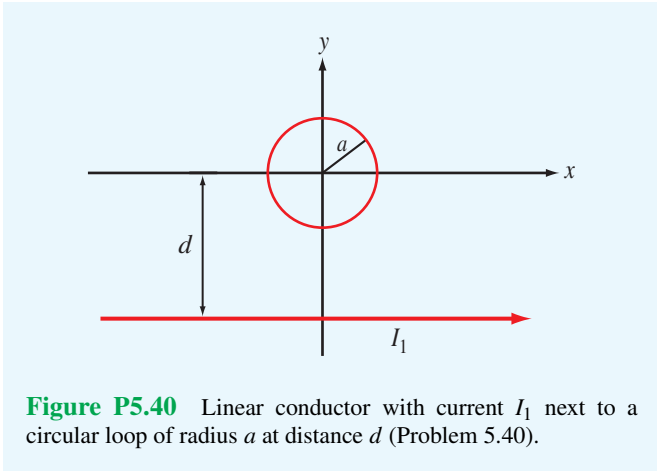


Figure P5.40 Linear conductor with current I_1 next to a circular loop of radius a at distance d (Problem 5.40).

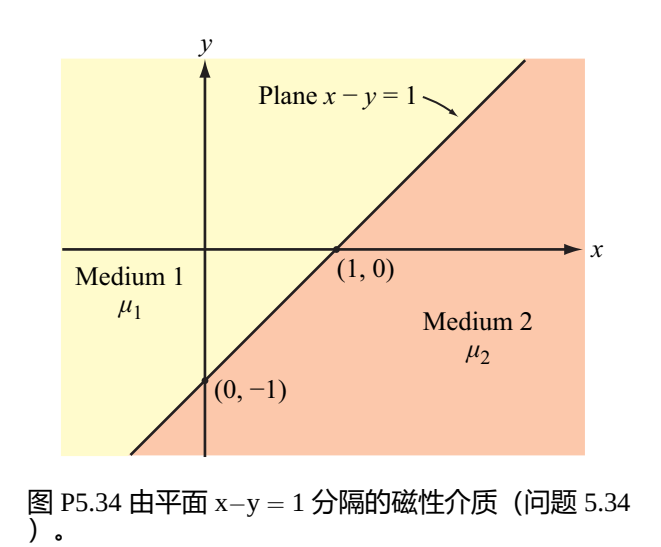


图 P5.34 由平面 $x - y = 1$ 分隔的磁性介质 (问题 5.34)。

5.36 证明, 如果图 P5.36 所示的平行界面处不存在表面电流密度, 则 θ_4 和 θ_1 之间的关系与 μ_2 无关。

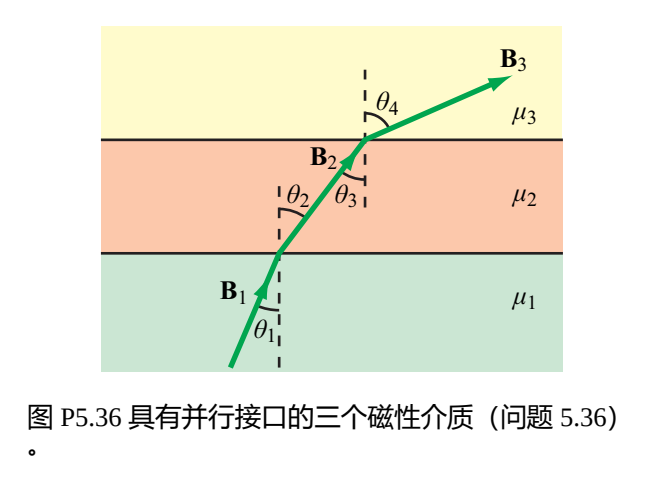


图 P5.36 具有并行接口的三个磁性介质 (问题 5.36)。

第 5-7 节和 5-8 节: 电感和磁能

- *5.37 求出图 5-28 (a) 中平行导线传输线单位长度自感的表达式, 用 a 、 d 和 μ 表示, 其中 a 是导线的半径, d 是导线之间的轴距, μ 是导线所在介质的磁导率。
- 5.38 长度为 20 厘米、半径为 5 厘米的螺线管由 400 匝组成, 电流为 12 A。如果 $z = 0$ 代表螺线管的中点, 则在 $-20 \text{ cm} \leq z \leq 20 \text{ cm}$ 范围内, 生成 $|\mathbf{H}(z)|$ 与沿螺线管轴 z 的函数关系图以 1 厘米为步长。
- 5.39 就直流电流 I 而言, 假设内导体的半径为 5 cm, 外导体的内半径为 10 cm, 同轴传输线的 3 米长的充气部分的绝缘介质中存储了多少磁能?
- *5.40 确定图 P5.40 所示的环形回路和线性电流之间的互感。

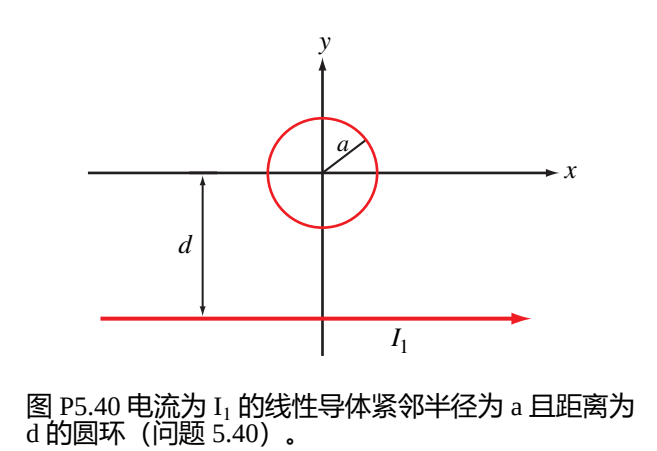
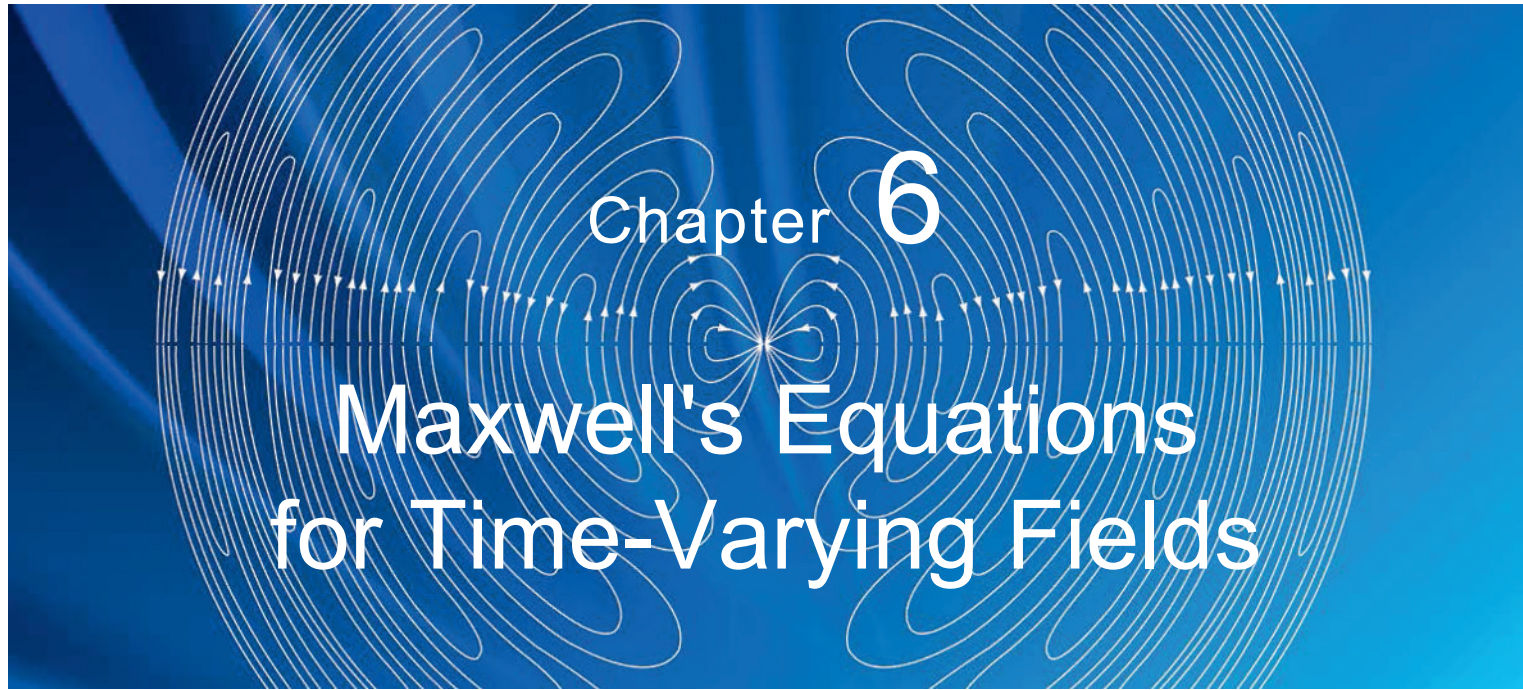


图 P5.40 电流为 I_1 的线性导体紧邻半径为 a 且距离为 d 的圆环 (问题 5.40)。



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Objectives

Upon learning the material presented in this chapter, you should be able to:

1. Apply Faraday’s law to compute the voltage induced by a stationary coil placed in a time-varying magnetic field or moving in a medium containing a magnetic field.
2. Describe the operation of the electromagnetic generator.
3. Calculate the displacement current associated with a time-varying electric field.
4. Calculate the rate at which charge dissipates in a material with known ϵ and σ .



章节内容

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目标

学习本章介绍的材料后, 您应该能够:

1. 应用法拉第定律计算放置在时变磁场中或在包含磁场的介质中移动的固定线圈所感应的电压。
2. 描述电磁发电机的工作原理。
3. 计算与时变电场相关的位移电流。
4. 计算已知 ϵ 和 σ 的材料中电荷耗散的速率。

Dynamic Fields

Electric charges induce electric fields and electric currents induce magnetic fields. As long as the charge and current distributions remain constant in time, so will the fields they induce. If the charges and currents vary in time, the electric and magnetic fields vary accordingly. Additionally, however, the electric and magnetic fields become coupled and travel through space in the form of electromagnetic waves. Examples of such waves include light, x-rays, infrared, gamma rays, and radio waves (see Fig. 1-16).

To study time-varying electromagnetic phenomena, we need to consider the entire set of Maxwell’s equations simultaneously. These equations, first introduced in the opening section of Chapter 4, are given in both differential and integral form in Table 6-1. In the static case ($\partial/\partial t = 0$), we use the first pair of Maxwell’s equations to study electric phenomena (Chapter 4) and the second pair to study magnetic phenomena (Chapter 5). In the dynamic case ($\partial/\partial t \neq 0$), the coupling that exists between the electric and magnetic fields, as expressed by the second and fourth equations in Table 6-1, prevents such decomposition. The first equation represents Gauss’s law for electricity, and it is equally valid for static and dynamic fields. The same is true for the third equation, which is Gauss’s law for magnetism. By contrast, the second and fourth equations—Faraday’s and Ampère’s laws—are of a totally different nature. Faraday’s law expresses the fact that a time-varying magnetic field gives rise to an electric field. Conversely, Ampère’s law states that a time-varying electric field must be accompanied by a magnetic field.

Some statements in this and succeeding chapters contradict conclusions reached in Chapter 4 and 5 as those pertained to the special case of static charges and dc currents. The behavior of dynamic fields reduces to that of static ones when $\partial/\partial t$ is set to zero.

We begin this chapter by examining Faraday’s and Ampère’s laws and some of their practical applications. We then combine Maxwell’s equations to obtain relations among the charge and current sources, ρ_v and $\mathbf{J}$; the scalar and vector potentials, V and $\mathbf{A}$; and the electromagnetic fields, $\mathbf{E}$, $\mathbf{D}$, $\mathbf{H}$, and $\mathbf{B}$. We do so for the most general time-varying case and for the specific case of sinusoidal-time variations.

6-1 Faraday’s Law

The close connection between electricity and magnetism was established by Oersted, who demonstrated that a wire carrying an electric current exerts a force on a compass needle and that the needle always turns to point in the $\hat{\phi}$ direction when the current is along the $\hat{\mathbf{z}}$ direction. The force acting on the compass needle is due to the magnetic field produced by the current in the wire. Following this discovery, *Faraday hypothesized that if a current produces a magnetic field, then the converse should also be true: A magnetic field should produce a current in a wire*. To test his hypothesis, he conducted numerous experiments in his laboratory in London over a period of about 10 years—all aimed at making magnetic fields induce currents in wires. Similar work was being carried out by Henry in Albany, New York. Wires were placed next to permanent magnets or current-carrying loops of all different sizes, but no

时变场

电荷产生电场，电流产生磁场。只要电荷和电流的分布不随时间变化，它们产生的场也不随时间变化。若电荷和电流随时间变化，电场和磁场也相应变化；同时，电场与磁场相互耦合，并可作为电磁波在空间中传播。光、X 射线、红外线、 γ 射线及无线电波均属电磁波（见图 1-16）。

研究时变电磁现象时，必须同时考虑完整的麦克斯韦方程组。这些方程已在第4章开头引入，其微分形式和积分形式列于表 6-1。静态情况下 ($\partial/\partial t = 0$)，可用前两个方程研究电现象（第4章），用后两个方程研究磁现象（第5章）。在时变情况下 ($\partial/\partial t \neq 0$)，表中第二、第四个方程所描述的电磁耦合使这种分开处理不再适用。第一个方程为电场的高斯定律，对静态场和时变场均成立；第三个方程为磁场的高斯定律，同样如此。第二、第四个方程分别为法拉第定律和安培–麦克斯韦定律，其性质有所不同：法拉第定律表明时变磁场产生电场；安培–麦克斯韦定律则表明时变电场必然伴随磁场。

本章及后续章节中的某些论述看似与第4、5章的结论不同，是因为那些结论仅适用于静止电荷和直流电流。当令 $\partial/\partial t = 0$ 时，时变场的关系便退化为静态场的关系。

本章首先讨论法拉第定律、安培–麦克斯韦定律及其应用；随后联立麦克斯韦方程，建立电荷、电流源 ρ_v 、 $\mathbf{J}$ ，标量势 V 、矢量势 $\mathbf{A}$ ，以及电磁场 $\mathbf{E}$ 、 $\mathbf{D}$ 、 $\mathbf{H}$ 、 $\mathbf{B}$ 之间的关系。讨论既包括一般的时变情况，也包括随时间作正弦变化的特殊情况。

6-1 法拉第定律

奥斯特揭示了电与磁的密切联系：载流导线对磁针施加作用力；当电流沿 +z 方向时，磁针转向方位角 ϕ 增大的方向。磁针受到的作用来自导线中电流产生的磁场。随后，法拉第提出假设：既然电流能够产生磁场，反过来磁场是否也能在导线中产生电流？为检验这一假设，他在伦敦实验室进行了约十年的大量实验，试图利用磁场在导线中产生感应电流。同一时期，亨利也在纽约州奥尔巴尼开展类似研究。他们将导线放在永久磁铁或各种尺寸的载流回路旁边，但并未……（下页续）

Table 6-1 Maxwell’s equations.

Reference	Differential Form	Integral Form	
Gauss’s law	$\nabla \cdot \mathbf{D} = \rho_v$	$\oint_S \mathbf{D} \cdot d\mathbf{s} = Q$	(6.1)
Faraday’s law	$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$	$\oint_C \mathbf{E} \cdot d\mathbf{l} = -\int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s}$	(6.2)*
No magnetic charges (Gauss’s law for magnetism)	$\nabla \cdot \mathbf{B} = 0$	$\oint_S \mathbf{B} \cdot d\mathbf{s} = 0$	(6.3)
Ampère’s law	$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$	$\oint_C \mathbf{H} \cdot d\mathbf{l} = \int_S \left(\mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} \right) \cdot d\mathbf{s}$	(6.4)
*For a stationary surface S .			

表 6-1 麦克斯韦方程组

定律	微分形式	积分形式	
电场的高斯定律	$\nabla \cdot \mathbf{D} = \rho_v$	$\oint_S \mathbf{D} \cdot d\mathbf{s} = Q$	(6.1)
法拉第定律	$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$	$\oint_C \mathbf{E} \cdot d\mathbf{l} = -\int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s}$	(6.2)*
不存在磁荷 (磁场的高斯定律)	$\nabla \cdot \mathbf{B} = 0$	$\oint_S \mathbf{B} \cdot d\mathbf{s} = 0$	(6.3)
安培–麦克斯韦定律	$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$	$\oint_C \mathbf{H} \cdot d\mathbf{l} = \int_S \left(\mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} \right) \cdot d\mathbf{s}$	(6.4)
* 适用于静止曲面 S 。			

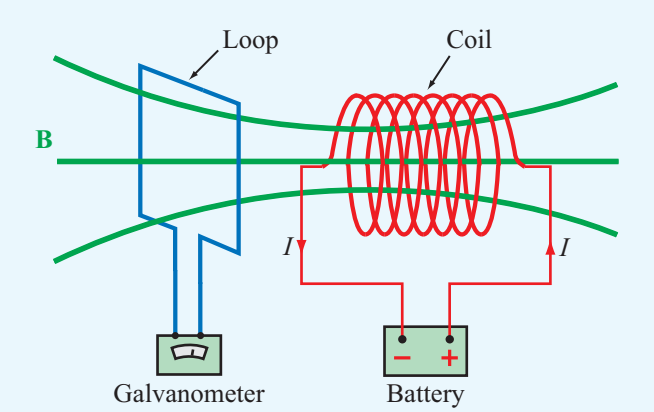


Figure 6-1 The galvanometer (predecessor of the ammeter) shows a deflection whenever the magnetic flux passing through the square loop changes with time.

currents were ever detected. Eventually, these experiments led to the following discovery by both Faraday and Henry.

► Magnetic fields can produce an electric current in a closed loop, but *only* if the magnetic flux linking the surface area of the loop changes with time. The key to the induction process is *change*. ◀

To elucidate the induction process, consider the arrangement shown in **Fig. 6-1**. A conducting loop connected to a galvanometer (a sensitive instrument used in the 1800s to detect current flow) is placed next to a conducting coil connected to a battery. The current in the coil produces a magnetic field **B** whose lines pass through the loop. In Section 5-4, we defined the magnetic flux Φ passing through a loop as the integral of the normal component of the magnetic flux density over the surface area of the loop, S , or

$$\Phi = \int_S \mathbf{B} \cdot d\mathbf{s} \quad (\text{Wb}). \quad (6.5)$$

Under stationary conditions, the dc current in the coil produces a constant magnetic field **B**, which in turn produces a constant flux through the loop. When the flux is constant, no current is detected by the galvanometer. However, when the battery is disconnected, thereby interrupting the flow of current in the coil, the magnetic field drops to zero, and the consequent change in magnetic flux causes a momentary deflection of the galvanometer needle. When the battery is reconnected, the galvanometer again exhibits a momentary deflection, but in the

opposite direction. Thus, current is induced in the loop when the magnetic flux changes, and the direction of the current depends on whether the flux increases (when the battery is being connected) or decreases (when the battery is being disconnected). It was further discovered that current can also flow in the loop while the battery is connected to the coil if the loop turns or moves either closer to or away from the coil. The physical movement of the loop changes the amount of flux linking its surface S , even though the field **B** due to the coil has not changed.

A galvanometer is a predecessor of the voltmeter and ammeter. When a galvanometer detects the flow of current through the coil, it means that a voltage has been induced across the galvanometer terminals. This voltage is called the **electromotive force (emf)**, V_{emf} , and the process is called **electromagnetic induction**. The emf induced in a closed conducting loop of N turns is given by

$$V_{\text{emf}} = -N \frac{d\Phi}{dt} = -N \frac{d}{dt} \int_S \mathbf{B} \cdot d\mathbf{s} \quad (\text{V}). \quad (6.6)$$

Even though the results leading to Eq. (6.6) were also discovered independently by Henry, Eq. (6.6) is attributed to Faraday and known as **Faraday's law**. The significance of the negative sign in Eq. (6.6) is explained in the next section.

We note that the derivative in Eq. (6.6) is a total time derivative that operates on the magnetic field **B**, as well as the differential surface area ds . Accordingly, an emf can be generated in a closed conducting loop under any of the following three conditions:

1. A time-varying magnetic field linking a stationary loop; the induced emf is then called the **transformer emf**, $V_{\text{emf}}^{\text{tr}}$.
2. A moving loop with a time-varying area (relative to the normal component of **B**) in a static field **B**; the induced emf is then called the **motional emf**, $V_{\text{emf}}^{\text{m}}$.
3. A moving loop in a time-varying field **B**.

The total emf is given by

$$V_{\text{emf}} = V_{\text{emf}}^{\text{tr}} + V_{\text{emf}}^{\text{m}} \quad (6.7)$$

with $V_{\text{emf}}^{\text{m}} = 0$ if the loop is stationary [case (1)] and $V_{\text{emf}}^{\text{tr}} = 0$ if **B** is static [case (2)]. For case (3), both terms are important. Each of the three cases is examined separately in the following sections.

6-2 Stationary Loop in a Time-Varying Magnetic Field

The stationary, single-turn, conducting, circular loop with contour C and surface area S shown in **Fig. 6-2(a)** is exposed

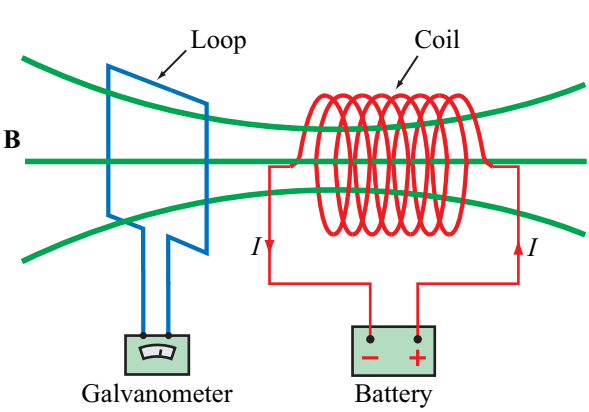


图 6-1 每当穿过方形环路的磁通量随时间变化时，检流计（电流表的前身）就会显示出偏转。

曾经检测到电流。最终，这些实验导致了法拉第和亨利的以下发现。

► 磁场可以在闭环中产生电流，但前提是连接环路表面区域的磁通量随时间变化。归纳过程的关键是改变。 ◀

为了阐明归纳过程，请考虑图 6-1 中所示的布置。连接到检流计（一种 1800 年代用于检测电流的灵敏仪器）的导电环被放置在连接到电池的导电线圈旁边。线圈中的电流产生磁场 **B**，其磁力线穿过环路。在第 5-4 节中，我们将穿过环路的磁通量 Φ 定义为环路表面上磁通密度法向分量 S 的积分，或

$$\Phi = \int_S \mathbf{B} \cdot d\mathbf{s} \quad (\text{Wb}). \quad (6.5)$$

在静止条件下，线圈中的直流电流产生恒定磁场 **B**，进而产生通过环路的恒定磁通。当通量恒定时，检流计检测不到电流。然而，当电池断开时，线圈中的电流中断，磁场降至零，磁通量随之变化，导致检流计针瞬时偏转。当电池重新连接时，检流计再次表现出瞬时偏转，但在

相反的方向。因此，当磁通量变化时，回路中会感应出电流，电流的方向取决于磁通量是增加（连接电池时）还是减少（当电池断开时）。进一步发现，当电池连接到线圈时，如果环路转动或靠近或远离线圈，电流也会在环路中流动。环路的物理运动改变了连接其表面 S 的磁通量，尽管线圈产生的磁场 **B** 并未改变。

检流计是电压表和电流表的前身。当检流计检测到流过线圈的电流时，这意味着检流计端子上已感应出电压。该电压称为电动势 (emf) V_{emf} ，该过程称为电磁感应。 N 匝闭合导电回路中感应的电动势由下式给出

$$V_{\text{emf}} = -N \frac{d\Phi}{dt} = -N \frac{d}{dt} \int_S \mathbf{B} \cdot d\mathbf{s} \quad (\text{V}). \quad (6.6)$$

即使结果导致方程。(6.6) 也是由亨利独立发现的，方程。(6.6) 归因于法拉第，被称为法拉第定律。方程中负号的意义。(6.6) 将在下一节中解释。

我们注意到方程中的导数。(6.6) 是作用于磁场 **B** 以及微分表面积 ds 的总时间导数。因此，在以下三个条件中的任何一个条件下，闭合导电回路中都会产生电动势：

1. 连接静止环路的时变磁场；感应电动势称为变压器电动势， $V_{\text{emf}}^{\text{tr}}$ 。
2. 静态场 **B** 中具有时变面积（相对于 **B** 的法向分量）的移动环路；感应电动势称为运动电动势， $V_{\text{emf}}^{\text{m}}$ 。
3. 时变场中的移动环 **B**。

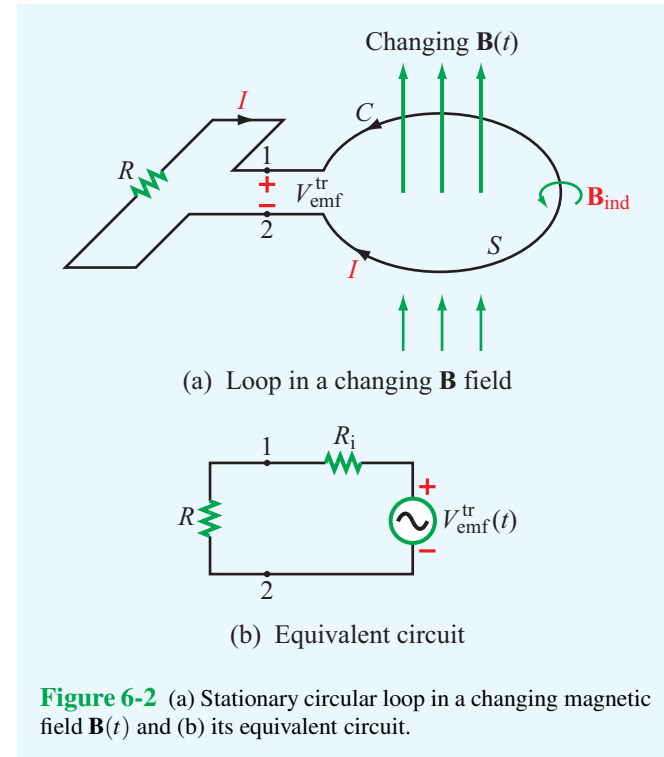
总电动势由下式给出

$$V_{\text{emf}} = V_{\text{emf}}^{\text{tr}} + V_{\text{emf}}^{\text{m}} \quad (6.7)$$

如果循环是静态的，则 $V_{\text{emf}}^{\text{m}} = 0$ [情况 (1)]；如果 **B** 是静态的，则 $V_{\text{emf}}^{\text{tr}} = 0$ [情况 (2)]。对于情况 (3)，这两个术语都很重要。以下各节将分别研究这三种情况。

6-2 时变磁场中的静止环路

暴露出如图 6-2(a) 所示的具有轮廓 C 和表面积 S 的固定单匝导电圆形回路



to a time-varying magnetic field $\mathbf{B}(t)$. As stated earlier, the emf induced when S is stationary and the field is time varying is called the **transformer emf** and is denoted $V_{\text{emf}}^{\text{tr}}$. Since the loop is stationary, d/dt in Eq. (6.6) now operates on $\mathbf{B}(t)$ only. Hence,

$$V_{\text{emf}}^{\text{tr}} = -N \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s}, \quad (\text{transformer emf}) \quad (6.8)$$

where the full derivative d/dt has been moved inside the integral and changed into a partial derivative $\partial/\partial t$ to signify that it operates on $\mathbf{B}$ only. The transformer emf is the voltage difference that would appear across the small opening between terminals 1 and 2, even in the absence of the resistor R . That is, $V_{\text{emf}}^{\text{tr}} = V_{12}$, where V_{12} is the open-circuit voltage across the open ends of the loop. Under dc conditions, $V_{\text{emf}}^{\text{tr}} = 0$. For the loop shown in Fig. 6-2(a) and the associated definition for $V_{\text{emf}}^{\text{tr}}$ given by Eq. (6.8), the direction of $d\mathbf{s}$, which is the loop's differential surface normal, can be chosen to be either upward or downward. The two choices are associated with opposite designations of the polarities of terminals 1 and 2 in Fig. 6-2(a).

Right-Hand Rule for $V_{\text{emf}}^{\text{tr}}$

The connection between the direction of $d\mathbf{s}$ and the polarity of $V_{\text{emf}}^{\text{tr}}$ is governed by the following right-hand rule: If $d\mathbf{s}$ points along the thumb of the right hand, then the direction of the contour C indicated by the four fingers is such that it always passes across the opening from the positive terminal of $V_{\text{emf}}^{\text{tr}}$ to the negative terminal.

If the loop has an internal resistance R_i , the circuit in Fig. 6-2(a) can be represented by the equivalent circuit shown in Fig. 6-2(b), in which case the current I flowing through the circuit is given by

$$I = \frac{V_{\text{emf}}^{\text{tr}}}{R + R_i}. \quad (6.9)$$

For good conductors, R_i usually is very small, and it may be ignored in comparison with practical values of R .

► The polarity of $V_{\text{emf}}^{\text{tr}}$ and hence the direction of I is governed by **Lenz's law**, which states that the current in the loop is always in a direction that opposes the change of magnetic flux $\Phi(t)$ that produced I . ◀

The current I induces a magnetic field of its own, $\mathbf{B}_{\text{ind}}$, with a corresponding flux Φ_{ind} . The direction of $\mathbf{B}_{\text{ind}}$ is governed by the right-hand rule: If I is in a clockwise direction, then $\mathbf{B}_{\text{ind}}$ points downward through S ; conversely, if I is in a counter-clockwise direction, then $\mathbf{B}_{\text{ind}}$ points upward through S . If the original field $\mathbf{B}(t)$ is increasing, which means that $d\Phi/dt > 0$, then according to Lenz's law I has to be in the direction shown in Fig. 6-2(a) in order for $\mathbf{B}_{\text{ind}}$ to be in opposition to $\mathbf{B}(t)$. Consequently, terminal 2 would be at a higher potential than terminal 1, and $V_{\text{emf}}^{\text{tr}}$ would have a negative value. However, if $\mathbf{B}(t)$ were to remain in the same direction but to decrease in magnitude, then $d\Phi/dt$ would become negative, the current would have to reverse direction, and its induced field $\mathbf{B}_{\text{ind}}$ would be in the same direction as $\mathbf{B}(t)$ in order to oppose the change (decrease) of $\mathbf{B}(t)$. In that case, $V_{\text{emf}}^{\text{tr}}$ would be positive.

► It is important to remember that $\mathbf{B}_{\text{ind}}$ serves to oppose the change in $\mathbf{B}(t)$ and not necessarily $\mathbf{B}(t)$ itself. ◀

Despite the presence of the small opening between terminals 1 and 2 of the loop in Fig. 6-2(a), we shall treat the loop as a closed path with contour C . We do this in order to establish the link between $\mathbf{B}$ and the electric field $\mathbf{E}$ associated with the induced emf, $V_{\text{emf}}^{\text{tr}}$. Also, at any point along the loop, the

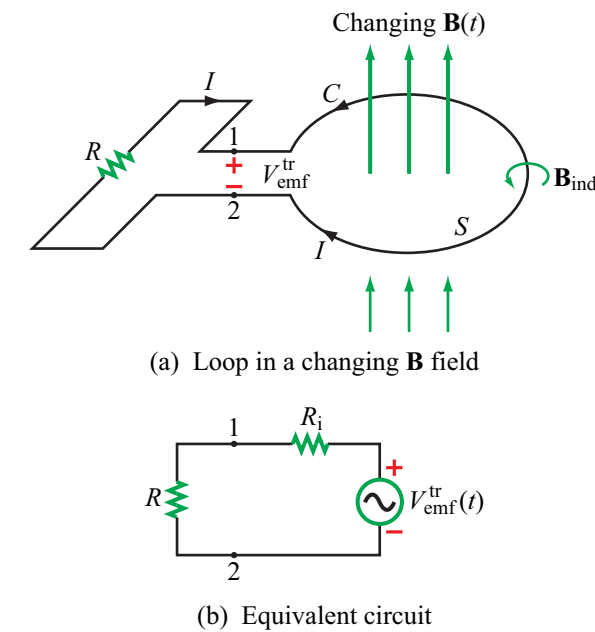


图 6-2 (a) 变化磁场 $\mathbf{B}(t)$ 中的静止圆环和 (b) 其等效电路。

为时变磁场 $\mathbf{B}(t)$ 。如前所述，当 S 静止并且磁场随时间变化时感应的电动势称为变压器电动势，并表示为 $V_{\text{emf}}^{\text{tr}}$ 。由于循环是平稳的，式中的 d/dt (6.6) 现在仅适用于 $\mathbf{B}(t)$ 。因此，

$$V_{\text{emf}}^{\text{tr}} = -N \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s}, \quad (\text{transformer emf}) \quad (6.8)$$

其中全导数 d/dt 已移至积分内部并更改为偏导数 $\partial/\partial t$ 以表示它仅对 $\mathbf{B}$ 进行运算。变压器电动势是端子 1 和 2 之间的小开口上出现的电压差，即使没有电阻器 R 。即 $V_{\text{emf}}^{\text{tr}} = V_{12}$ ，其中 V_{12} 是环路开路端的开路电压。在直流条件下， $V_{\text{emf}}^{\text{tr}} = 0$ 。对于图 6-2(a) 中所示的循环以及由方程给出的 $V_{\text{emf}}^{\text{tr}}$ 的相关定义：(6.8) 中， $d\mathbf{s}$ 的方向是环的微分面法线，可以选择向上或向下。这两种选择与图 6-2(a) 中端子 1 和 2 极性的相反指定相关。

V 的右手定则_{emf}

$d\mathbf{s}$ 的方向和 $V_{\text{emf}}^{\text{tr}}$ 的极性之间的关系受以下右手定则约束：如果 $d\mathbf{s}$ 沿着右手的拇指指向，则四个手指所指示的轮廓 C 的方向总是穿过从 $V_{\text{emf}}^{\text{tr}}$ 的正极端子到负极端子的开口。

如果回路有一个内阻 R_i ，图6-2(a)中的电路可以用图6-2(b)所示的等效电路表示，此时流过电路的电流 I 为

$$I = \frac{V_{\text{emf}}^{\text{tr}}}{R + R_i}. \quad (6.9)$$

对于良导体， R_i 通常很小，与 R 的实际值相比可以忽略不计。

► $V_{\text{emf}}^{\text{tr}}$ 的极性以及 I 的方向受楞次定律控制，该定律规定环路中的电流始终处于与产生 I 的磁通量 $\Phi(t)$ 变化相反的方向。◀

电流 I 感应出自己的磁场 $\mathbf{B}_{\text{ind}}$ ，并具有相应的磁通量 Φ_{ind} 。 $\mathbf{B}_{\text{ind}}$ 的方向受右手定则约束：如果 I 沿顺时针方向，则 $\mathbf{B}_{\text{ind}}$ 通过 S 指向向下；反之，如果 I 沿逆时针方向，则 $\mathbf{B}_{\text{ind}}$ 通过 S 指向上方。如果原始场 $\mathbf{B}(t)$ 递增，即 $d\Phi/dt > 0$ ，则根据楞次定律， I 必须沿图 6-2(a) 所示方向， $\mathbf{B}_{\text{ind}}$ 才能与 $\mathbf{B}(t)$ 相反。因此，端子 2 的电位将高于端子 1，并且 $V_{\text{emf}}^{\text{tr}}$ 将具有负值。然而，如果 $\mathbf{B}(t)$ 保持相同方向但幅度减小，则 $d\Phi/dt$ 将变为负值，电流必须反转方向，并且其感应场 $\mathbf{B}_{\text{ind}}$ 将与 $\mathbf{B}(t)$ 方向相同，以抵抗 $\mathbf{B}(t)$ 的变化（减小）。在这种情况下， $V_{\text{emf}}^{\text{tr}}$ 将为正。

► 重要的是要记住， $\mathbf{B}_{\text{ind}}$ 用于反对 $\mathbf{B}(t)$ 的变化，而不一定是 $\mathbf{B}(t)$ 本身。◀

尽管图 6-2(a) 中环路的端子 1 和 2 之间存在小开口，但我们仍应将环路视为具有轮廓 C 的闭合路径。这样做是为了在 $\mathbf{B}$ 和与感应电动势 $V_{\text{emf}}^{\text{tr}}$ 相关的电场 $\mathbf{E}$ 之间建立联系。此外，在循环中的任何一点，

field E is related to the current I flowing through the loop. For contour C , $V_{\text{emf}}^{\text{tr}}$ is related to $\mathbf{E}$ by

$$V_{\text{emf}}^{\text{tr}} = \oint_C \mathbf{E} \cdot d\mathbf{l}. \quad (6.10)$$

For $N = 1$ (a loop with one turn), equating Eqs. (6.8) and (6.10) gives

$$\oint_C \mathbf{E} \cdot d\mathbf{l} = - \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s}, \quad (6.11)$$

which is the integral form of Faraday’s law given in **Table 6-1**. We should keep in mind that the direction of the contour C and the direction of $d\mathbf{s}$ are related by the right-hand rule.

By applying Stokes’s theorem to the left-hand side of Eq. (6.11), we have

$$\int_S (\nabla \times \mathbf{E}) \cdot d\mathbf{s} = - \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s}, \quad (6.12)$$

and in order for the two integrals to be equal for all possible choices of S , their integrands must be equal, which gives

$$\nabla \times \mathbf{E} = - \frac{\partial \mathbf{B}}{\partial t}. \quad \textbf{(Faraday’s law)} \quad (6.13)$$

This differential form of Faraday’s law states that a time-varying magnetic field induces an electric field $\mathbf{E}$ whose curl is equal to the negative of the time derivative of $\mathbf{B}$. Even though the derivation leading to Faraday’s law started out by considering the field associated with a physical circuit, Eq. (6.13) applies at any point in space, whether or not a physical circuit exists at that point.

Example 6-1: Inductor in a Changing Magnetic Field

An inductor is formed by winding N turns of a thin conducting wire into a circular loop of radius a . The inductor loop is in the x - y plane with its center at the origin and connected to a resistor R , as shown in **Fig. 6-3**. In the presence of a magnetic field $\mathbf{B} = B_0(\hat{\mathbf{y}}2 + \hat{\mathbf{z}}3) \sin \omega t$, where ω is the angular frequency, find

- (a) the magnetic flux linking a single turn of the inductor,
- (b) the transformer emf given that $N = 10$, $B_0 = 0.2$ T, $a = 10$ cm, and $\omega = 10^3$ rad/s,
- (c) the polarity of $V_{\text{emf}}^{\text{tr}}$ at $t = 0$, and
- (d) the induced current in the circuit for $R = 1$ k Ω (assume the wire resistance to be much smaller than R).

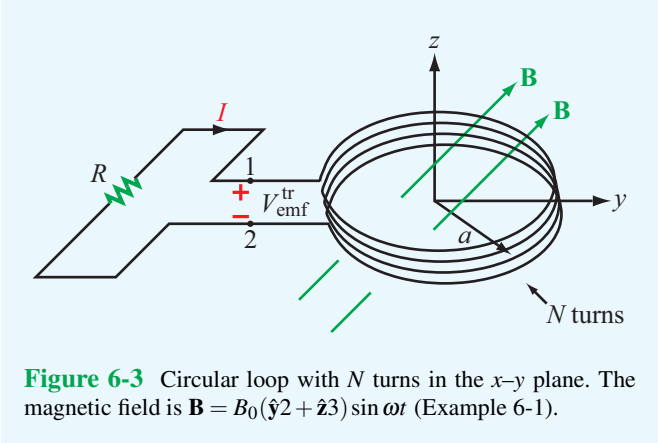


Figure 6-3 Circular loop with N turns in the x - y plane. The magnetic field is $\mathbf{B} = B_0(\hat{\mathbf{y}}2 + \hat{\mathbf{z}}3) \sin \omega t$ (Example 6-1).

Solution: (a) With $d\mathbf{s}$ defined upwards (along $+\hat{\mathbf{z}}$ direction), the magnetic flux linking each turn of the inductor is

$$\Phi = \int_S \mathbf{B} \cdot d\mathbf{s} = \int_S [B_0(\hat{\mathbf{y}}2 + \hat{\mathbf{z}}3) \sin \omega t] \cdot \hat{\mathbf{z}} d\mathbf{s} = 3\pi a^2 B_0 \sin \omega t.$$

(b) To find $V_{\text{emf}}^{\text{tr}}$, we can apply Eq. (6.8) or apply the general expression given by Eq. (6.6) directly. The latter approach gives

$$\begin{aligned} V_{\text{emf}}^{\text{tr}} &= -N \frac{d\Phi}{dt} = -\frac{d}{dt} (3\pi N a^2 B_0 \sin \omega t) \\ &= -3\pi N \omega a^2 B_0 \cos \omega t. \end{aligned}$$

Having chosen the direction of $d\mathbf{s}$ to be upwards, the polarity of $V_{\text{emf}}^{\text{tr}}$ is governed by the right-hand rule requiring the direction of the contour C in **Fig. 6-3** to pass across the gap from the positive terminal of $V_{\text{emf}}^{\text{tr}}$ to the negative terminal. That is,

$$V_{\text{emf}}^{\text{tr}} = V_1 - V_2 = -3\pi N \omega a^2 B_0 \cos \omega t.$$

For $N = 10$, $a = 0.1$ m, $\omega = 10^3$ rad/s, and $B_0 = 0.2$ T,

$$V_{\text{emf}}^{\text{tr}} = -188.5 \cos 10^3 t \quad (\text{V}).$$

(c) The flux passing through each loop in the upward direction is $\Phi(t) = 3\pi a^2 B_0 \sin \omega t$. At $t = 0$, $\Phi(0) = 0$, but

$$\left. \frac{d\Phi}{dt} \right|_{t=0} = 3\pi a^2 \omega B_0 \cos \omega t_0 \Big|_{t=0} = 3\pi a^2 \omega B_0 > 0.$$

Hence, even though the flux itself is zero at $t = 0$, its time derivative is positive. Therefore, the flux is increasing at $t = 0$. Lenz’s law requires the direction of the current I to be such that the flux it induces, Φ_{ind} , opposes the change in Φ . Since Φ is defined to be upwards, and it is increasing at $t = 0$,

磁场 $\mathbf{B}$ 与流过回路的电流 I 相关。对于轮廓 C , $V_{\text{emf}}^{\text{tr}}$ 与 $\mathbf{E}$ 的关系为

$$V_{\text{emf}}^{\text{tr}} = \oint_C \mathbf{E} \cdot d\mathbf{l}. \quad (6.10)$$

对于 $N = 1$ (一圈循环), 等式: (6.8) 和 (6.10) 给出

$$\oint_C \mathbf{E} \cdot d\mathbf{l} = - \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s}, \quad (6.11)$$

这是表6-1中给出的法拉第定律的积分形式。我们应该记住, 轮廓 C 的方向和 $d\mathbf{s}$ 的方向通过右手定则相关。

将斯托克斯定理应用于方程的左侧。 (6.11), 我们有

$$\int_S (\nabla \times \mathbf{E}) \cdot d\mathbf{s} = - \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s}, \quad (6.12)$$

为了使两个积分对于 S 的所有可能选择都相等, 它们的被积函数必须相等, 这给出

$$\nabla \times \mathbf{E} = - \frac{\partial \mathbf{B}}{\partial t}. \quad \textbf{(Faraday’s law)} \quad (6.13)$$

法拉第定律的这种微分形式指出, 随时间变化的磁场会感应出一个电场 $\mathbf{E}$, 其旋度等于 $\mathbf{B}$ 的时间导数的负值。尽管法拉第定律的推导是从考虑与物理电路相关的场开始的, 方程 (6.13) 适用于空间中的任何一点, 无论该点是否存在物理电路。

例 6-1: 变化磁场中的电感器

电感器是将一根细导线缠绕 N 匝, 形成半径为 a 的圆环。电感环路位于 x - y 平面上, 中心位于原点, 并连接电阻器 R , 如图 6-3 所示。在存在磁场 $\mathbf{B} = B_0(\hat{\mathbf{y}}2 + \hat{\mathbf{z}}3) \sin \omega t$ 的情况下, 其中 ω 是角频率, 找到

(a) 连接单匝电感器的磁通量, (b) 变压器电动势, 假设 $N = 10$ 、 $B_0 = 0.2$ T、 $a = 10$ cm 和 $\omega = 10^3$ rad/s, (c) $t = 0$ 时 $V_{\text{emf}}^{\text{tr}}$ 的极性, 以及 (d) R 电路中的感应电流 = 1 k Ω (假设导线电阻远小于 R)。

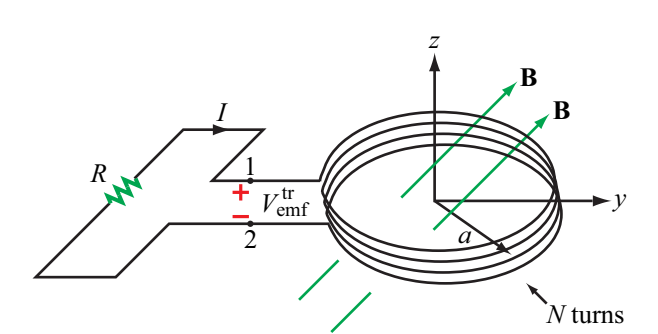


图 6-3 在 x - y 平面上有 N 匝的圆环。磁场为 $\mathbf{B} = B_0(\hat{\mathbf{y}}2 + \hat{\mathbf{z}}3) \sin \omega t$ (例 6-1)。

解: (a) $d\mathbf{s}$ 向上定义 (沿 $+\hat{\mathbf{z}}$ 方向), 连接电感器每匝的磁通量为

$$\Phi = \int_S \mathbf{B} \cdot d\mathbf{s} = \int_S [B_0(\hat{\mathbf{y}}2 + \hat{\mathbf{z}}3) \sin \omega t] \cdot \hat{\mathbf{z}} d\mathbf{s} = 3\pi a^2 B_0 \sin \omega t.$$

(b) 要找到 $V_{\text{emf}}^{\text{tr}}$, 我们可以应用方程: (6.8) 或应用式 (6.8) 给出的一般表达式。 (6.6) 直接。后一种方法给出

$$\begin{aligned} V_{\text{emf}}^{\text{tr}} &= -N \frac{d\Phi}{dt} = -\frac{d}{dt} (3\pi N a^2 B_0 \sin \omega t) \\ &= -3\pi N \omega a^2 B_0 \cos \omega t. \end{aligned}$$

选择 $d\mathbf{s}$ 的方向为向上后, $V_{\text{emf}}^{\text{tr}}$ 的极性受右手法则控制, 要求图 6-3 中轮廓 C 的方向穿过从 $V_{\text{emf}}^{\text{tr}}$ 正极端子到负极端子的间隙。那是,

$$V_{\text{emf}}^{\text{tr}} = V_1 - V_2 = -3\pi N \omega a^2 B_0 \cos \omega t.$$

对于 $N = 10$ 、 $a = 0.1$ m、 $\omega = 10^3$ rad/s 和 $B_0 = 0.2$ T,

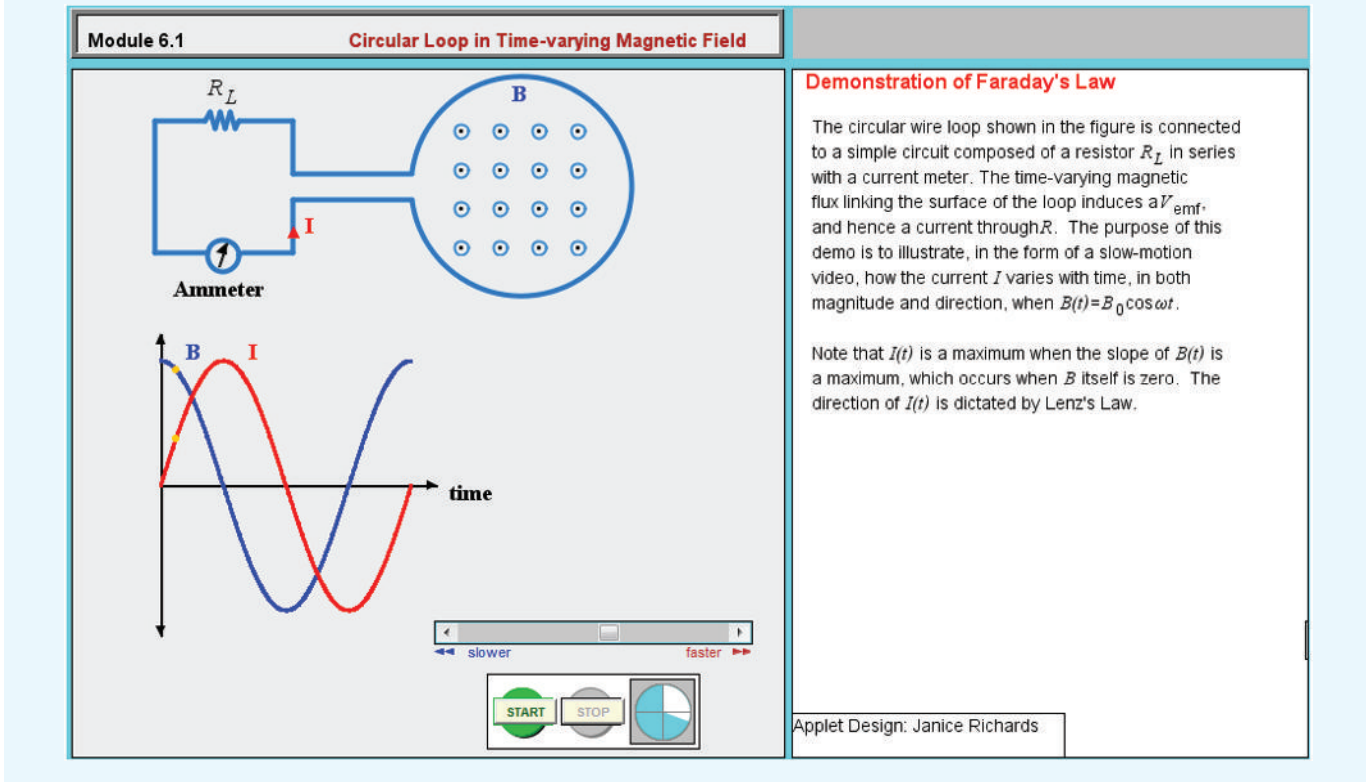
$$V_{\text{emf}}^{\text{tr}} = -188.5 \cos 10^3 t \quad (\text{V}).$$

(c) 向上通过每个环路的通量为 $\Phi(t) = 3\pi a^2 B_0 \sin \omega t$ 。在 $t = 0$ 、 $\Phi(0) = 0$ 时, 但是

$$\left. \frac{d\Phi}{dt} \right|_{t=0} = 3\pi a^2 \omega B_0 \cos \omega t_0 \Big|_{t=0} = 3\pi a^2 \omega B_0 > 0.$$

因此, 即使通量本身在 $t = 0$ 处为零, 其时间导数也是正的。Therefore, the flux is increasing at $t = 0$. Lenz’s law requires the direction of the current I to be such that the flux it induces, Φ_{ind} , opposes the change in Φ .由于 Φ 被定义为向上, 并且它在 $t = 0$ 处增加,

Module 6.1 Circular Loop in Time-Varying Magnetic Field Faraday's law of induction is demonstrated by simulating the current induced in a loop in response to the change in magnetic flux flowing through it.



Φ_{ind} should be downward through the loop in order to slow down the increase in Φ . Hence, I should be in the direction shown in **Fig. 6-3**.

Since I flows through a resistor from its positive voltage terminal to its negative voltage terminal, it follows that V_{12} should be negative at $t = 0$, which it is:

$$V_{\text{emf}}^{\text{tr}} = V_1 - V_2 = -188.5 \quad (\text{V}).$$

(d) The current I is given by

$$I = \frac{V_2 - V_1}{R} = \frac{188.5}{10^3} \cos 10^3 t = 0.19 \cos 10^3 t \quad (\text{A}).$$

Exercise 6-1: For the loop shown in **Fig. 6-3**, what is $V_{\text{emf}}^{\text{tr}}$ if $\mathbf{B} = \hat{\mathbf{y}}B_0 \cos \omega t$?

Answer: $V_{\text{emf}}^{\text{tr}} = 0$ because $\mathbf{B}$ is orthogonal to the loop's surface normal $d\mathbf{s}$. (See **EM**.)

Exercise 6-2: Suppose that the loop of Example 6-1 is replaced with a 10-turn square loop centered at the origin and having 20 cm sides oriented parallel to the x and y axes. If $\mathbf{B} = \hat{\mathbf{z}}B_0 x^2 \cos 10^3 t$ and $B_0 = 100 \text{ T}$, find the current in the circuit.

Answer: $I = -133 \sin 10^3 t \text{ (mA)}$. (See **EM**.)

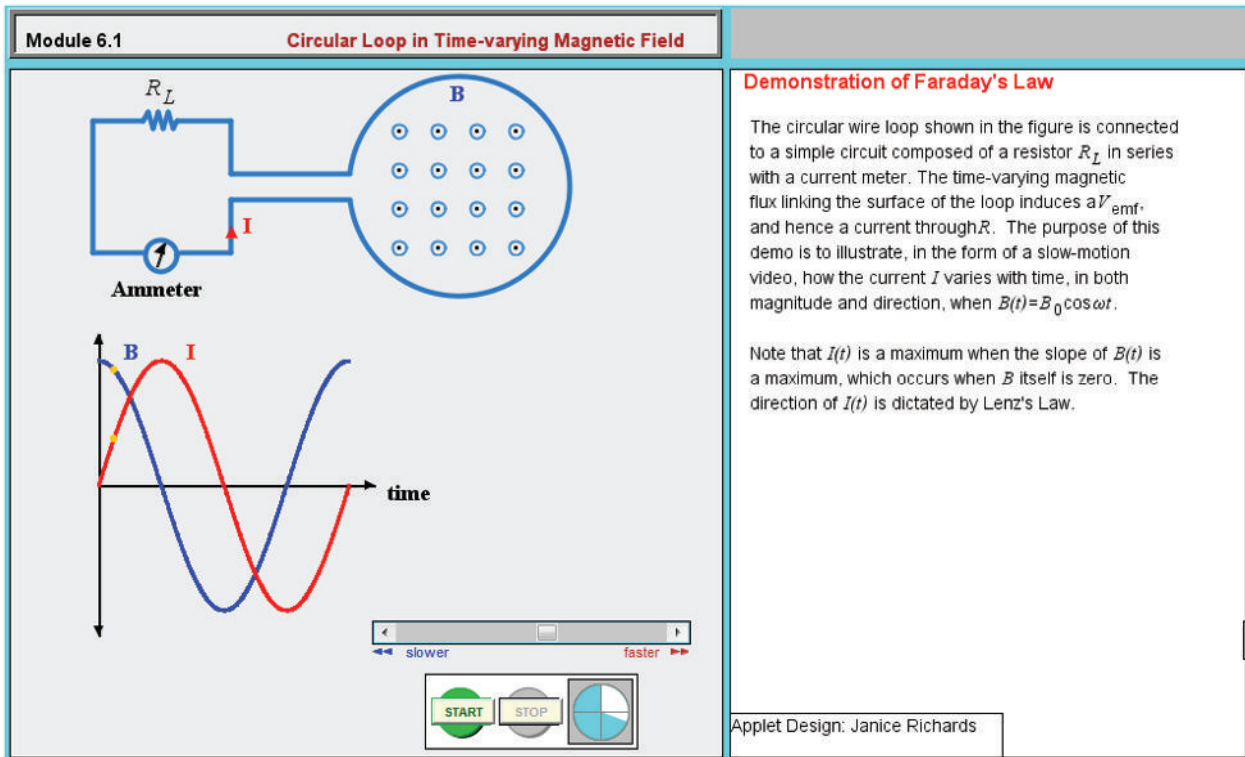
Example 6-2: Lenz's Law

Determine voltages V_1 and V_2 across the 2Ω and 4Ω resistors shown in **Fig. 6-4**. The loop is located in the x - y plane, its area is 4 m^2 , the magnetic flux density is $\mathbf{B} = -\hat{\mathbf{z}}0.3t \text{ (T)}$, and the internal resistance of the wire may be ignored.

Solution: The flux flowing through the loop is

$$\Phi = \int_S \mathbf{B} \cdot d\mathbf{s} = \int_S (-\hat{\mathbf{z}}0.3t) \cdot \hat{\mathbf{z}} ds = -0.3t \times 4 = -1.2t \quad (\text{Wb}),$$

模块 6.1 时变磁场中的圆环 法拉第感应定律通过模拟环路中响应流过磁通量变化而感应的电流来演示。



Φ_{ind} 应该通过循环向下，以减缓 Φ 的增加。因此， I 应该处于图 6-3 所示的方向。

由于 I 从电阻器的正电压端子流向负电压端子，因此 V_{12} 在 $t = 0$ 处应为负值，即：

$$V_{\text{emf}}^{\text{tr}} = V_1 - V_2 = -188.5 \quad (\text{V}).$$

(d) 电流 I 由下式给出

$$I = \frac{V_2 - V_1}{R} = \frac{188.5}{10^3} \cos 10^3 t = 0.19 \cos 10^3 t \quad (\text{A}).$$

练习 6-1: 对于图 6-3 所示的循环，如果 $\mathbf{B} = \hat{\mathbf{y}}B_0 \cos \omega t$ ，则 $V_{\text{emf}}^{\text{tr}}$ 是多少？

答案: $V_{\text{emf}}^{\text{tr}} = 0$ ，因为 $\mathbf{B}$ 与环的表面法线 $d\mathbf{s}$ 正交。(参见 **EM**。)

练习 6-2: 假设示例 6-1 中的环路被替换为以原点为中心、边长为 20 厘米、平行于 x 轴和 y 轴的 10 匝方形环路。如果 $\mathbf{B} = \hat{\mathbf{z}}B_0 x^2 \cos 10^3 t$ 且 $B_0 = 100 \text{ T}$ ，求电路中的电流。

答案: $I = -133 \sin 10^3 t \text{ (mA)}$ 。(参见 **EM**。)

例 6-2: 楞次定律

确定图 6-4 所示的 2Ω 和 4Ω 电阻两端的电压 V_1 和 V_2 。环路位于 x - y 平面上，其面积为 4 m^2 ，磁通密度为 $\mathbf{B} = -\hat{\mathbf{z}}0.3t \text{ (T)}$ ，导线的内阻可以忽略不计。

解: 流经环路的磁通为

$$\Phi = \int_S \mathbf{B} \cdot d\mathbf{s} = \int_S (-\hat{\mathbf{z}}0.3t) \cdot \hat{\mathbf{z}} ds = -0.3t \times 4 = -1.2t \quad (\text{Wb}),$$

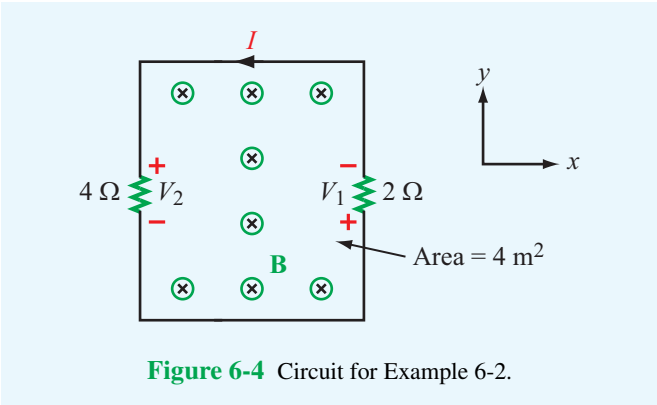


Figure 6-4 Circuit for Example 6-2.

and the corresponding transformer emf is

$$V_{\text{emf}}^{\text{tr}} = -\frac{d\Phi}{dt} = 1.2 \quad (\text{V}).$$

Since the magnetic flux through the loop is along the $-z$ direction (into the page) and increases in magnitude with time t , Lenz's law states that the induced current I should be in a direction such that the magnetic flux density $\mathbf{B}_{\text{ind}}$ it induces counteracts the direction of change of Φ . Hence, I has to be in the direction shown in the circuit because the corresponding $\mathbf{B}_{\text{ind}}$ is along the $+z$ direction in the region inside the loop area. This, in turn, means that V_1 and V_2 are positive voltages.

The total voltage of 1.2 V is distributed across two resistors in series. Consequently,

$$I = \frac{V_{\text{emf}}^{\text{tr}}}{R_1 + R_2} = \frac{1.2}{2 + 4} = 0.2 \text{ A},$$

and

$$\begin{aligned} V_1 &= IR_1 = 0.2 \times 2 = 0.4 \text{ V}, \\ V_2 &= IR_2 = 0.2 \times 4 = 0.8 \text{ V}. \end{aligned}$$

- Concept Question 6-1:** Explain Faraday's law and the function of Lenz's law.
- Concept Question 6-2:** Under what circumstances is the net voltage around a closed loop equal to zero?
- Concept Question 6-3:** Suppose the magnetic flux density linking the loop of Fig. 6-4 (Example 6-2) is given by $\mathbf{B} = -\hat{z}0.3e^{-t}$ (T). What would the direction of the current be, relative to that shown in Fig. 6-4, for $t \geq 0$?

6-3 The Ideal Transformer

The transformer shown in Fig. 6-5(a) consists of two coils wound around a common magnetic core. The primary coil has N_1 turns and is connected to an ac voltage source $V_1(t)$. The secondary coil has N_2 turns and is connected to a load resistor R_L . In an ideal transformer the core has infinite permeability ($\mu = \infty$), and the magnetic flux is confined within the core.

► The directions of the currents flowing in the two coils, I_1 and I_2 , are defined such that, when I_1 and I_2 are both positive, the flux generated by I_2 is opposite to that generated by I_1 . The transformer gets its name from the fact that it transforms currents, voltages, and impedances between its primary and secondary circuits, and vice versa. ◀

On the primary side of the transformer, the voltage source V_1 generates current I_1 in the primary coil, which establishes a flux Φ in the magnetic core. The flux Φ and voltage V_1 are

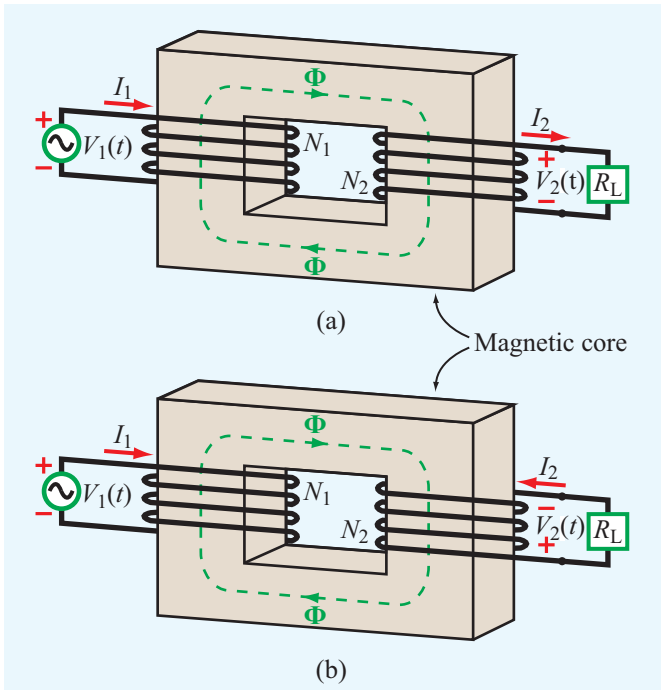


Figure 6-5 In a transformer, the directions of I_1 and I_2 are such that the flux Φ generated by one of them is opposite to that generated by the other. The direction of the secondary winding in (b) is opposite to that in (a), and so are the direction of I_2 and the polarity of V_2 .

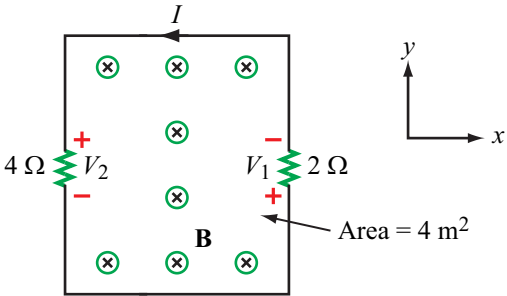


图 6-4 例 6-2 的电路。

相应的变压器电动势为

$$V_{\text{emf}}^{\text{tr}} = -\frac{d\Phi}{dt} = 1.2 \quad (\text{V}).$$

由于穿过环路的磁通量沿着 $-z$ 方向（进入页面），并且其大小随时间 t 增加，因此楞次定律指出，感应电流 I 的方向应使其感应的磁通量密度 $\mathbf{B}_{\text{ind}}$ 抵消 Φ 的变化方向。因此， I 必须沿电路中所示的方向，因为相应的 $\mathbf{B}_{\text{ind}}$ 在环路区域内沿 $+z$ 方向。这又意味着 V_1 和 V_2 是正电压。

1.2 V 的总电压分布在两个串联电阻上。最后，

$$I = \frac{V_{\text{emf}}^{\text{tr}}}{R_1 + R_2} = \frac{1.2}{2 + 4} = 0.2 \text{ A},$$

和

$$\begin{aligned} V_1 &= IR_1 = 0.2 \times 2 = 0.4 \text{ V}, \\ V_2 &= IR_2 = 0.2 \times 4 = 0.8 \text{ V}. \end{aligned}$$

概念问题 6-1：解释法拉第定律和楞次定律的作用。

概念题6-2：什么情况下闭环周围的净电压为零？

概念问题 6-3：假设连接图 6-4（例 6-2）的回路磁通密度由 $\mathbf{B} = -\hat{z}0.3e^{-t}$ (T) 给出。相对于图 6-4 所示，对于 $t \geq 0$ ，电流的方向是什么？

6-3 理想的变压器

图 6-5(a) 所示的变压器由缠绕在公共磁芯上的两个线圈组成。初级线圈有 N_1 匝，并连接到交流电压源 $V_1(t)$ 。次级线圈有 N_2 匝，并连接到负载电阻 R_L 。在理想的变压器中，磁芯具有无限的磁导率 ($\mu = \infty$)，并且磁通量被限制在磁芯内。

► 两个线圈 I_1 和 I_2 中流动的电流方向被定义为，当 I_1 和 I_2 均为正时， I_2 产生的磁通与 I_1 产生的磁通相反。变压器因其在初级和次级电路之间变换电流、电压和阻抗而得名，反之亦然。 ◀

在变压器的初级侧，电压源 V_1 在初级线圈中产生电流 I_1 ，从而在磁芯中建立磁通 Φ 。磁通 Φ 和电压 V_1 为

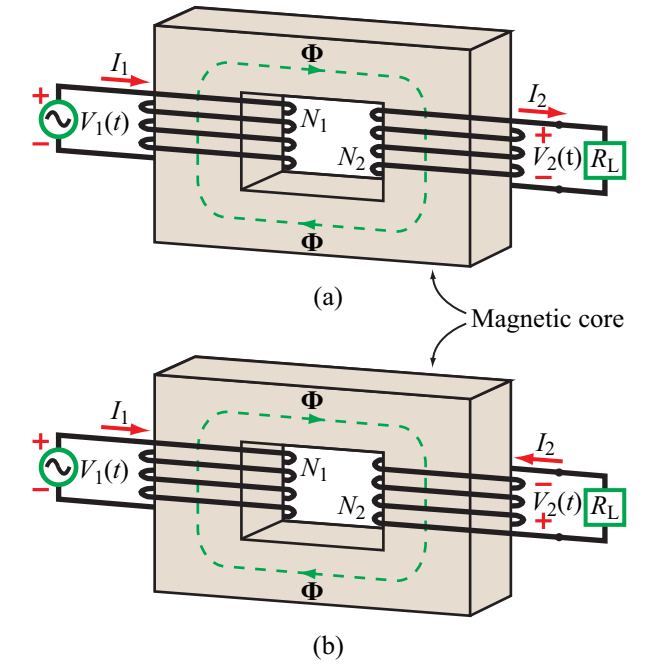


图 6-5 在变压器中， I_1 和 I_2 的方向使得其中一个产生的磁通 Φ 与另一个产生的磁通相反。(b)中次级绕组的方向与(a)中的相反， I_2 的方向和 V_2 的极性也相反。

6-4 MOVING CONDUCTOR IN A STATIC MAGNETIC FIELD

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related by Faraday's law:

$$V_1 = -N_1 \frac{d\Phi}{dt}. \quad (6.14)$$

A similar relation holds true on the secondary side:

$$V_2 = -N_2 \frac{d\Phi}{dt}. \quad (6.15)$$

The combination of Eqs. (6.14) and (6.15) gives

$$\frac{V_1}{V_2} = \frac{N_1}{N_2}. \quad (6.16)$$

In an ideal lossless transformer, all the instantaneous power supplied by the source connected to the primary coil is delivered to the load on the secondary side. Thus, no power is lost in the core, and

$$P_1 = P_2. \quad (6.17)$$

Since $P_1 = I_1 V_1$ and $P_2 = I_2 V_2$, and in view of Eq. (6.16), we have

$$\frac{I_1}{I_2} = \frac{N_2}{N_1}. \quad (6.18)$$

Thus, whereas the ratio of the voltages given by Eq. (6.16) is proportional to the corresponding turns ratio, the ratio of the currents is equal to the inverse of the turns ratio. If $N_1/N_2 = 0.1$, V_2 of the secondary circuit would be 10 times V_1 of the primary circuit, but I_2 would be only $I_1/10$.

The transformer shown in Fig. 6-5(b) is identical to that in Fig. 6-5(a) except for the direction of the windings of the secondary coil. Because of this change, the direction of I_2 and the polarity of V_2 in Fig. 6-5(b) are the reverse of those in Fig. 6-5(a).

The voltage and current in the secondary circuit in Fig. 6-5(a) are related by $V_2 = I_2 R_L$. To the input circuit, the transformer may be represented by an equivalent input resistance R_{in} , as shown in Fig. 6-6, defined as

$$R_{in} = \frac{V_1}{I_1}. \quad (6.19)$$

Use of Eqs. (6.16) and (6.18) gives

$$R_{in} = \frac{V_2}{I_2} \left(\frac{N_1}{N_2} \right)^2 = \left(\frac{N_1}{N_2} \right)^2 R_L. \quad (6.20)$$

When the load is an impedance Z_L and V_1 is a sinusoidal source, the phasor-domain equivalent of Eq. (6.20) is

$$Z_{in} = \left(\frac{N_1}{N_2} \right)^2 Z_L. \quad (6.21)$$

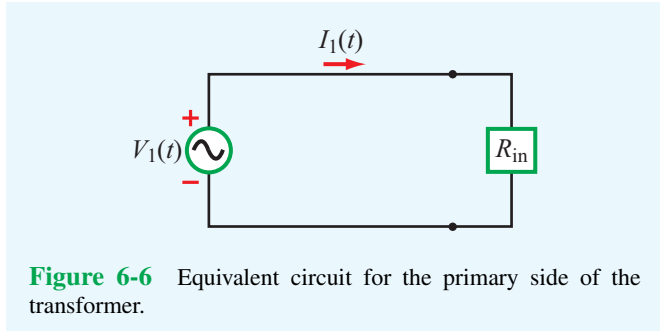


Figure 6-6 Equivalent circuit for the primary side of the transformer.

6-4 Moving Conductor in a Static Magnetic Field

Consider a wire of length l moving across a static magnetic field $\mathbf{B} = \hat{\mathbf{z}}B_0$ with constant velocity $\mathbf{u}$, as shown in Fig. 6-7. The conducting wire contains free electrons. From Eq. (5.3), the magnetic force $\mathbf{F}_m$ acting on a particle with charge q moving with velocity $\mathbf{u}$ in a magnetic field $\mathbf{B}$ is

$$\mathbf{F}_m = q(\mathbf{u} \times \mathbf{B}). \quad (6.22)$$

This magnetic force is equivalent to the electrical force that would be exerted on the particle by the electric field $\mathbf{E}_m$ given by

$$\mathbf{E}_m = \frac{\mathbf{F}_m}{q} = \mathbf{u} \times \mathbf{B}. \quad (6.23)$$

The field $\mathbf{E}_m$ generated by the motion of the charged particle is called a **motional electric field** and is in the direction perpendicular to the plane containing $\mathbf{u}$ and $\mathbf{B}$. For the wire shown in Fig. 6-7, $\mathbf{E}_m$ is along $-\hat{\mathbf{y}}$. The magnetic force acting

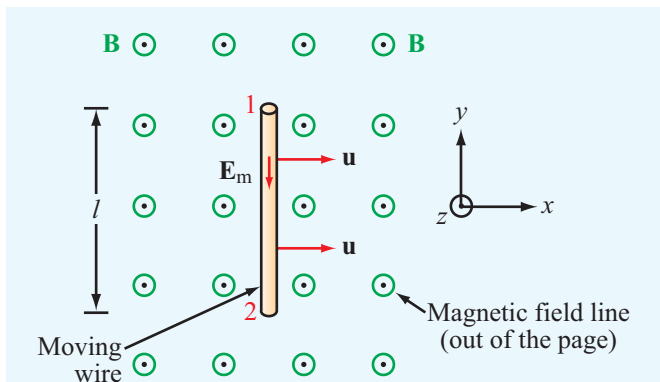


Figure 6-7 Conducting wire moving with velocity $\mathbf{u}$ in a static magnetic field.

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由法拉第定律关联:

$$V_1 = -N_1 \frac{d\Phi}{dt}. \quad (6.14)$$

类似的关系也适用于次级侧:

$$V_2 = -N_2 \frac{d\Phi}{dt}. \quad (6.15)$$

等式的组合。(6.14) 和 (6.15) 给出

$$\frac{V_1}{V_2} = \frac{N_1}{N_2}. \quad (6.16)$$

在理想的无损变压器中, 连接到初级线圈的电源提供的所有瞬时功率都被传送到次级侧的负载。因此, 核心不会损失任何能量, 并且

$$P_1 = P_2. \quad (6.17)$$

由于 $P_1 = I_1 V_1$ 和 $P_2 = I_2 V_2$, 并鉴于等式: (6.16), 我们有

$$\frac{I_1}{I_2} = \frac{N_2}{N_1}. \quad (6.18)$$

因此, 而方程给出的电压比。(6.16) 与相应的匝数比成正比, 电流之比等于匝数比的倒数。如果 $N_1/N_2 = 0.1$, 则次级电路的 V_2 将是初级电路 V_1 的 10 倍, 但 I_2 将仅为 $I_1/10$ 。

图6-5(b)所示的变压器与图6-5(a)所示的变压器相同, 只是次级线圈的绕组方向不同。由于这种变化, 图6-5(b)中 I_2 的方向和 V_2 的极性与图6-5(a)中的方向相反。

图6-5(a)中次级电路中的电压和电流之间的关系为 $V_2 = I_2 R_L$ 。对于输入电路来说, 变压器可以用等效输入电阻 R_{in} 来表示, 如图6-6所示, 定义为

$$R_{in} = \frac{V_1}{I_1}. \quad (6.19)$$

使用等式。(6.16) 和 (6.18) 给出

$$R_{in} = \frac{V_2}{I_2} \left(\frac{N_1}{N_2} \right)^2 = \left(\frac{N_1}{N_2} \right)^2 R_L. \quad (6.20)$$

当负载为阻抗 Z_L 且 V_1 为正弦源时, 等式的相量域等效为: (6.20) 是

$$Z_{in} = \left(\frac{N_1}{N_2} \right)^2 Z_L. \quad (6.21)$$

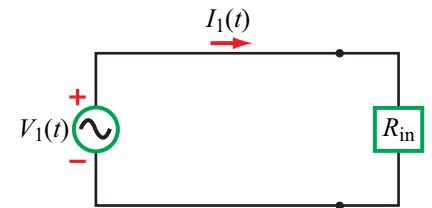


图 6-6 变压器初级侧的等效电路。

6-4 静磁场中移动的导体

考虑一根长度为 l 的导线以恒定速度 $\mathbf{u}$ 穿过静磁场 $\mathbf{B} = \hat{\mathbf{z}}B_0$, 如图 6-7 所示。导线含有自由电子。从方程。(5.3) 作用在磁场 $\mathbf{B}$ 中以速度 $\mathbf{u}$ 运动的带电荷 q 的粒子上的磁力 $\mathbf{F}_m$ 为

$$\mathbf{F}_m = q(\mathbf{u} \times \mathbf{B}). \quad (6.22)$$

该磁力相当于电场 $\mathbf{E}_m$ 施加在粒子上的电力, 如下所示:

$$\mathbf{E}_m = \frac{\mathbf{F}_m}{q} = \mathbf{u} \times \mathbf{B}. \quad (6.23)$$

带电粒子运动产生的场 $\mathbf{E}_m$ 称为运动电场, 其方向垂直于包含 $\mathbf{u}$ 和 $\mathbf{B}$ 的平面。对于图 6-7 所示的导线, $\mathbf{E}_m$ 沿 $-\hat{\mathbf{y}}$ 方向。磁力作用

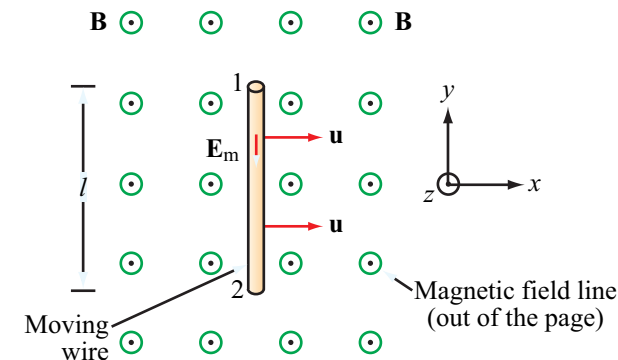


图 6-7 导线在静磁场中以速度 $\mathbf{u}$ 运动。

on the (negatively charged) electrons in the wire causes them to drift in the direction of $-\mathbf{E}_m$; that is, toward the wire end labeled 1 in Fig. 6-7. This, in turn, induces a voltage difference between ends 1 and 2, with end 2 being at the higher potential. The induced voltage is called a **motional emf**, V_{emf}^m , and is defined as the line integral of $\mathbf{E}_m$ between ends 2 and 1 of the wire,

$$V_{\text{emf}}^m = V_{12} = \int_2^1 \mathbf{E}_m \cdot d\mathbf{l} = \int_2^1 (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l}. \quad (6.24)$$

For the conducting wire, $\mathbf{u} \times \mathbf{B} = \hat{\mathbf{x}}u \times \hat{\mathbf{z}}B_0 = -\hat{\mathbf{y}}uB_0$ and $d\mathbf{l} = \hat{\mathbf{y}} dl$. Hence,

$$V_{\text{emf}}^m = V_{12} = -uB_0l. \quad (6.25)$$

In general, if any segment of a closed circuit with contour C moves with a velocity $\mathbf{u}$ across a static magnetic field $\mathbf{B}$, then the induced motional emf is given by

$$V_{\text{emf}}^m = \oint_C (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l}. \quad (\text{motional emf}) \quad (6.26)$$

► Only those segments of the circuit that cross magnetic field lines contribute to V_{emf}^m . ◀

Example 6-3: Sliding Bar

The rectangular loop shown in Fig. 6-8 has a constant width l , but its length x_0 increases with time as a conducting bar slides with uniform velocity $\mathbf{u}$ in a static magnetic field $\mathbf{B} = \hat{\mathbf{z}}B_0x$.

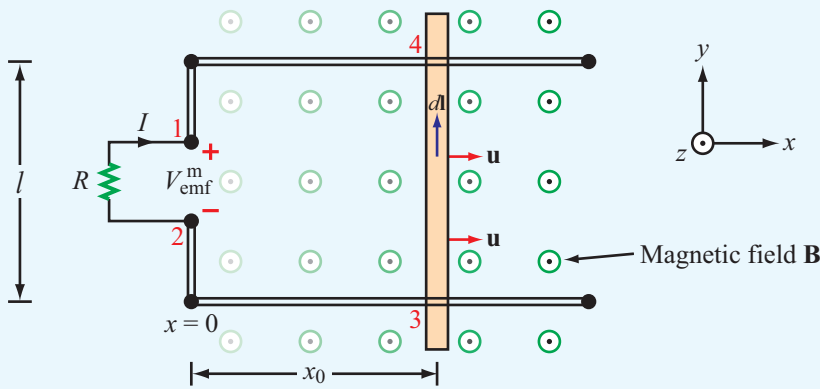


Figure 6-8 Sliding bar with velocity $\mathbf{u}$ in a magnetic field that increases linearly with x ; that is, $\mathbf{B} = \hat{\mathbf{z}}B_0x$ (Example 6-3).

Note that $\mathbf{B}$ increases linearly with x . The bar starts from $x = 0$ at $t = 0$. Find the motional emf between terminals 1 and 2 and the current I flowing through the resistor R . Assume that the loop resistance $R_l \ll R$.

Solution: This problem can be solved by using the motional emf expression given by Eq. (6.26) or by applying the general formula of Faraday's law. We now show that the two approaches yield the same result.

(a) Motional emf

The sliding bar, being the only part of the circuit that crosses the lines of the field $\mathbf{B}$, is the only part of contour 2341 that contributes to V_{emf}^m . Hence, at $x = x_0$, for example,

$$\begin{aligned} V_{\text{emf}}^m = V_{12} = V_{43} &= \int_3^4 (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l} \\ &= \int_3^4 (\hat{\mathbf{x}}u \times \hat{\mathbf{z}}B_0x_0) \cdot \hat{\mathbf{y}} dl = -uB_0x_0l. \end{aligned}$$

The length of the loop is related to u by $x_0 = ut$. Hence,

$$V_{\text{emf}}^m = -B_0u^2lt \quad (\text{V}). \quad (6.27)$$

(b) Total emf

Since $\mathbf{B}$ is static, $V_{\text{emf}}^{\text{tr}} = 0$ and $V_{\text{emf}} = V_{\text{emf}}^m$ only. To verify that the same result can be obtained by the general form of Faraday's law, we evaluate the flux Φ through the surface of the loop. Thus,

$$\Phi = \int_S \mathbf{B} \cdot d\mathbf{s} = \int_S (\hat{\mathbf{z}}B_0x) \cdot \hat{\mathbf{z}} dx dy = B_0l \int_0^{x_0} x dx = \frac{B_0lx_0^2}{2}. \quad (6.28)$$

导线中的（带负电的）电子会导致它们向 $-\mathbf{E}_m$ 方向漂移；即朝向图 6-7 中标记为 1 的线端。这又会在 1 端和 2 端之间产生电压差，其中 2 端处于较高电势。感应电压称为动电动势 V_{emf}^m ，定义为导线 2 端和 1 端之间 $\mathbf{E}_m$ 的线积分，

$$V_{\text{emf}}^m = V_{12} = \int_2^1 \mathbf{E}_m \cdot d\mathbf{l} = \int_2^1 (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l}. \quad (6.24)$$

对于导线， $\mathbf{u} \times \mathbf{B} = \hat{\mathbf{x}}u \times \hat{\mathbf{z}}B_0 = -\hat{\mathbf{y}}uB_0$ 和 $d\mathbf{l} = \hat{\mathbf{y}} dl$ 。因此，

$$V_{\text{emf}}^m = V_{12} = -uB_0l. \quad (6.25)$$

一般来说，如果具有轮廓 C 的闭合电路的任何一段以速度 u 穿过静磁场 $\mathbf{B}$ ，则感应运动电动势由下式给出

$$V_{\text{emf}}^m = \oint_C (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l}. \quad (\text{motional emf}) \quad (6.26)$$

► 只有与磁场线交叉的电路部分才会产生 V_{emf}^m 。 ◀

例 6-3：滑动条

图6-8所示的矩形环具有恒定的宽度 l ，但当导电棒在静磁场 $\mathbf{B} = \hat{\mathbf{z}}B_0x$ 中以匀速 u 滑动时，其长度 x_0 随着时间而增加。

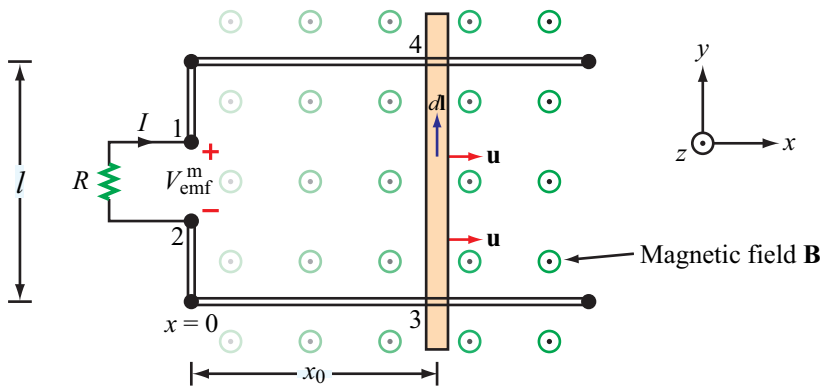


图 6-8 磁场中速度为 u 的滑动条，随 x 线性增加；即 $\mathbf{B} = \hat{\mathbf{z}}B_0x$ （例 6-3）。

请注意， B 随 x 线性增加。该条从 $t = 0$ 处的 $x = 0$ 开始。求端子 1 和 2 之间的运动电动势以及流过电阻器 R 的电流 I 。假设环路电阻 $R_l \ll R$ 。

解决方案：这个问题可以通过使用方程给出的运动电动势表达式来解决。（6.26）或应用法拉第定律的一般公式。我们现在证明这两种方法产生相同的结果。

(a) 运动电动势

滑动条是电路中与场 $\mathbf{B}$ 的线交叉的唯一部分，也是轮廓2341中对 V_{emf}^m 有贡献的唯一部分。因此，例如，在 $x = x_0$ 处，

$$\begin{aligned} V_{\text{emf}}^m = V_{12} = V_{43} &= \int_3^4 (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l} \\ &= \int_3^4 (\hat{\mathbf{x}}u \times \hat{\mathbf{z}}B_0x_0) \cdot \hat{\mathbf{y}} dl = -uB_0x_0l. \end{aligned}$$

循环的长度通过 $x_0 = ut$ 与 u 相关。因此，

$$V_{\text{emf}}^m = -B_0u^2lt \quad (\text{V}). \quad (6.27)$$

(b) 总电动势

由于 $\mathbf{B}$ 是静态的，因此仅 $V_{\text{emf}}^{\text{tr}} = 0$ 和 $V_{\text{emf}} = V_{\text{emf}}^m$ 。为了验证法拉第定律的一般形式是否可以得到相同的结果，我们评估了通过环路表面的通量 Φ 。因此，

$$\Phi = \int_S \mathbf{B} \cdot d\mathbf{s} = \int_S (\hat{\mathbf{z}}B_0x) \cdot \hat{\mathbf{z}} dx dy = B_0l \int_0^{x_0} x dx = \frac{B_0lx_0^2}{2}. \quad (6.28)$$

Substituting $x_0 = ut$ in Eq. (6.28) and then evaluating the negative of the derivative of the flux with respect to time gives

$$V_{\text{emf}} = -\frac{d\Phi}{dt} = -\frac{d}{dt} \left(\frac{B_0 l u^2 t^2}{2} \right) = -B_0 u^2 l t \quad (\text{V}), \quad (6.29)$$

which is identical with Eq. (6.27). Since V_{12} is negative, the current $I = B_0 u^2 l t / R$ flows in the direction shown in Fig. 6-8.

Example 6-4: Moving Loop

The rectangular loop shown in Fig. 6-9 is situated in the x - y plane and moves away from the origin with velocity $\mathbf{u} = \hat{y}5$ (m/s) in a magnetic field given by

$$\mathbf{B}(y) = \hat{z}0.2e^{-0.1y} \quad (\text{T}).$$

If $R = 5 \, \Omega$, find the current I at the instant that the loop sides are at $y_1 = 2$ m and $y_2 = 2.5$ m. The loop resistance may be ignored.

Solution: Since $\mathbf{u} \times \mathbf{B}$ is along $\hat{\mathbf{x}}$, voltages are induced across only the sides oriented along $\hat{\mathbf{x}}$, namely the sides linking point 1 to 2 and point 3 to 4. Had $\mathbf{B}$ been uniform, the induced voltages would have been the same, and the net voltage across the resistor would have been zero. In the present case, however, $\mathbf{B}$ decreases exponentially with y , thereby assuming a different value over side 1-2 than over side 3-4. Side 1-2 is at $y_1 = 2$ m, and the corresponding magnetic field is

$$\mathbf{B}(y_1) = \hat{z}0.2e^{-0.1y_1} = \hat{z}0.2e^{-0.2} \quad (\text{T}).$$

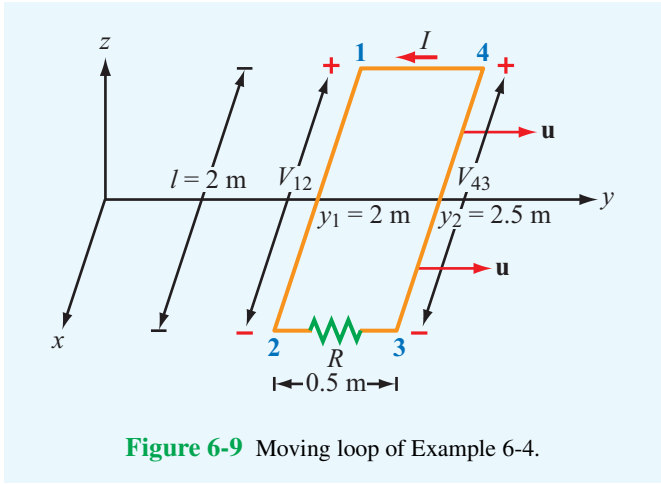


Figure 6-9 Moving loop of Example 6-4.

The induced voltage V_{12} is then given by

$$\begin{aligned} V_{12} &= \int_2^1 [\mathbf{u} \times \mathbf{B}(y_1)] \cdot d\mathbf{l} \\ &= \int_{l/2}^{-l/2} (\hat{y}5 \times \hat{z}0.2e^{-0.2}) \cdot \hat{\mathbf{x}} \, dx \\ &= -e^{-0.2}l \\ &= -2e^{-0.2} \\ &= -1.637 \quad (\text{V}). \end{aligned}$$

Similarly,

$$\begin{aligned} V_{43} &= -uB(y_2)l \\ &= -5 \times 0.2e^{-0.25} \times 2 \\ &= -1.558 \quad (\text{V}). \end{aligned}$$

Consequently, the current is in the direction shown in the figure and its magnitude is

$$I = \frac{V_{43} - V_{12}}{R} = \frac{0.079}{5} = 15.8 \, (\text{mA}).$$

Example 6-5: Moving Rod Next to a Wire

The wire shown in Fig. 6-10 carries a current $I = 10$ A. A 30-cm long metal rod moves with a constant velocity $\mathbf{u} = \hat{z}5$ m/s. Find V_{12} , where 1 and 2 are the two ends of the rod.

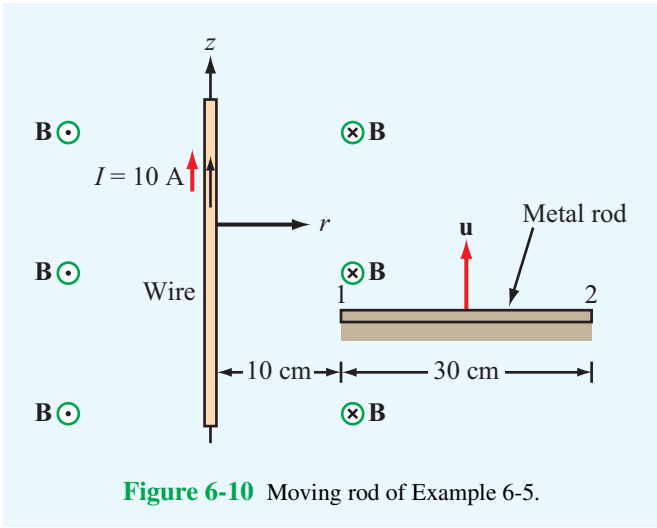


Figure 6-10 Moving rod of Example 6-5.

将 $x_0 = ut$ 代入等式：(6.28) 然后计算通量对时间的导数的负数给出

$$V_{\text{emf}} = -\frac{d\Phi}{dt} = -\frac{d}{dt} \left(\frac{B_0 l u^2 t^2}{2} \right) = -B_0 u^2 l t \quad (\text{V}), \quad (6.29)$$

这与等式相同。(6.27)。由于 V_{12} 为负，电流 $I=B_0 u^2 l t / R$ 沿图6-8所示的方向流动。

例 6-4：移动循环

图 6-9 所示的矩形环位于 x - y 平面上，并在磁场中以速度 $\mathbf{u} = \hat{y}5$ (m/s) 远离原点，该速度为：

$$\mathbf{B}(y) = \hat{z}0.2e^{-0.1y} \quad (\text{T}).$$

若 $R = 5 \, \Omega$ ，求环路边在 $y_1 = 2$ m和 $y_2 = 2.5$ m处瞬间的电流 I 。环路电阻可以忽略不计。

解：由于 $\mathbf{u} \times \mathbf{B}$ 沿 $\hat{\mathbf{x}}$ ，因此仅在沿 $\hat{\mathbf{x}}$ 定向的边（即连接点 1 到 2 和点 3 到 4 的边）上感应电压。如果 $\mathbf{B}$ 均匀，则感应电压将相同，电阻器两端的净电压将为零。然而，在本例中， $\mathbf{B}$ 随 y 呈指数下降，因此在 1-2 边上假设的值与在 3-4 边上的值不同。边 1-2 位于 $y_1 = 2$ m 处，对应的磁场为

$$\mathbf{B}(y_1) = \hat{z}0.2e^{-0.1y_1} = \hat{z}0.2e^{-0.2} \quad (\text{T}).$$

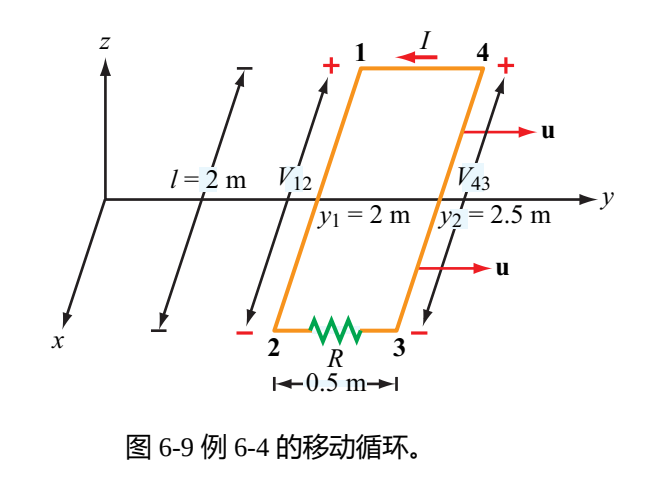


图 6-9 例 6-4 的移动循环。

感应电压 V_{12} 由下式给出

$$\begin{aligned} V_{12} &= \int_2^1 [\mathbf{u} \times \mathbf{B}(y_1)] \cdot d\mathbf{l} \\ &= \int_{l/2}^{-l/2} (\hat{y}5 \times \hat{z}0.2e^{-0.2}) \cdot \hat{\mathbf{x}} \, dx \\ &= -e^{-0.2}l \\ &= -2e^{-0.2} \\ &= -1.637 \quad (\text{V}). \end{aligned}$$

相似地，

$$\begin{aligned} V_{43} &= -uB(y_2)l \\ &= -5 \times 0.2e^{-0.25} \times 2 \\ &= -1.558 \quad (\text{V}). \end{aligned}$$

因此，电流的方向如图所示，其大小为

$$I = \frac{V_{43} - V_{12}}{R} = \frac{0.079}{5} = 15.8 \, (\text{mA}).$$

例 6-5：移动电线旁边的杆

图6-10所示的导线承载电流 $I = 10$ A。一根30厘米长的金属棒以恒定速度 $\mathbf{u} = \hat{z}5$ m/s移动。求 V_{12} ，其中 1 和 2 是杆的两端。

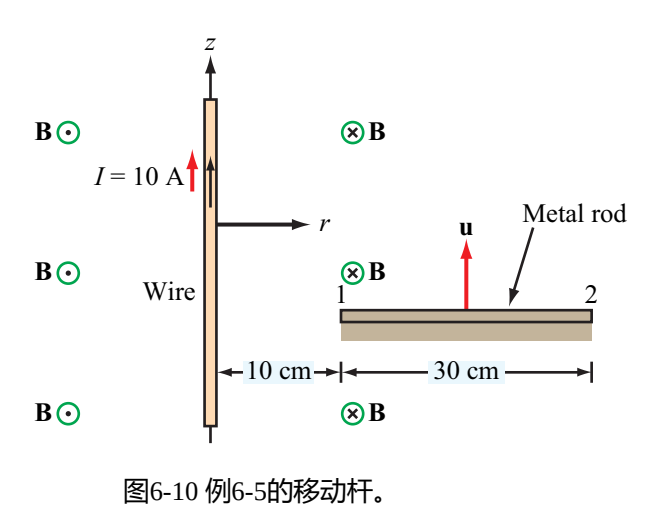
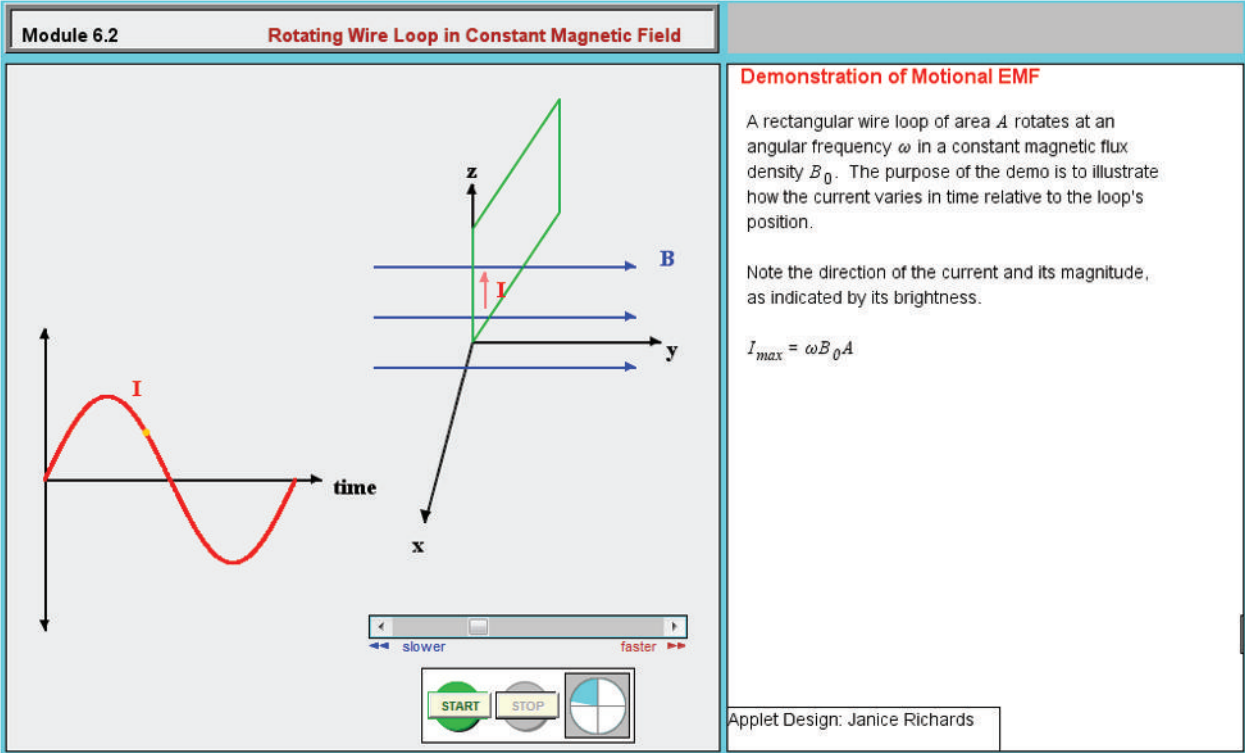


图6-10 例6-5的移动杆。

Module 6.2 Rotating Wire Loop in Constant Magnetic Field The principle of the electromagnetic generator is demonstrated by a rectangular loop rotating in the presence of a magnetic field.



Solution: The current I induces a magnetic field

$$\mathbf{B} = \hat{\phi} \frac{\mu_0 I}{2\pi r},$$

where r is the radial distance from the wire and the direction of $\hat{\phi}$ is into the page on the rod side of the wire. The movement of the rod in the presence of the field $\mathbf{B}$ induces a motional emf given by

$$\begin{aligned} V_{12} &= \int_{40 \text{ cm}}^{10 \text{ cm}} (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l} \\ &= \int_{40 \text{ cm}}^{10 \text{ cm}} \left(\hat{z} 5 \times \hat{\phi} \frac{\mu_0 I}{2\pi r} \right) \cdot \hat{r} dr \\ &= -\frac{5\mu_0 I}{2\pi} \int_{40 \text{ cm}}^{10 \text{ cm}} \frac{dr}{r} \\ &= -\frac{5 \times 4\pi \times 10^{-7} \times 10}{2\pi} \times \ln\left(\frac{10}{40}\right) \\ &= 13.9 \quad (\mu\text{V}). \end{aligned}$$

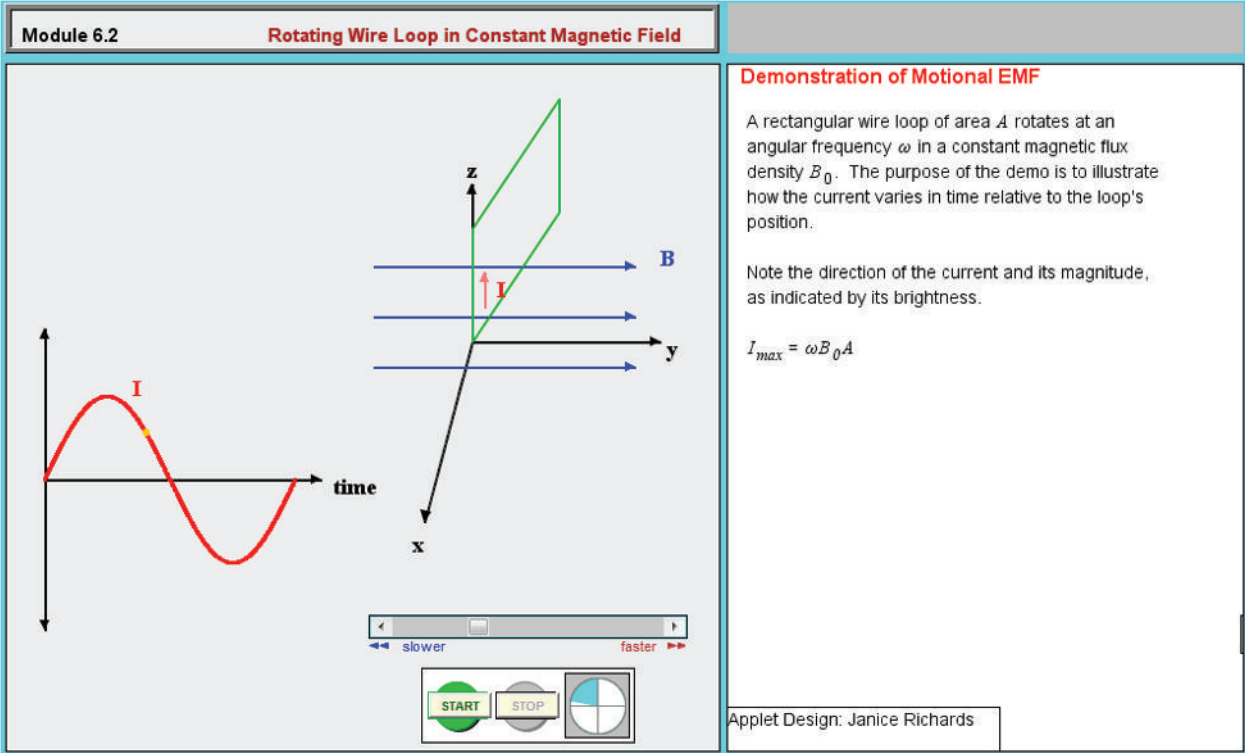
Concept Question 6-4: Suppose that no friction is involved in sliding the conducting bar of **Fig. 6-8** and that the horizontal arms of the circuit are very long. Hence, if the bar is given an initial push, it should continue moving at a constant velocity, and its movement generates electrical energy in the form of an induced emf, indefinitely. Is this a valid argument? If not, why not? Can we generate electrical energy without having to supply an equal amount of energy by other means?

Concept Question 6-5: Is the current flowing in the rod of **Fig. 6-10** a steady current? Examine the force on a charge q at ends 1 and 2 and compare.

Exercise 6-3: For the moving loop of **Fig. 6-9**, find I when the loop sides are at $y_1 = 4 \text{ m}$ and $y_2 = 4.5 \text{ m}$. Also, reverse the direction of motion such that $\mathbf{u} = -\hat{y}5 \text{ (m/s)}$.

Answer: $I = -13 \text{ (mA)}$. (See [EM](#).)

模块 6.2 恒定磁场中的旋转线圈 电磁发电机的原理通过在磁场中旋转的矩形线圈来演示。



解：电流 I 感应出磁场

$$\mathbf{B} = \hat{\phi} \frac{\mu_0 I}{2\pi r},$$

其中 r 是距线材的径向距离， $\hat{\phi}$ 的方向是线材杆侧进入页面的方向。在磁场 $\mathbf{B}$ 存在的情况下，杆的运动会产生由下式给出的运动电动势：

$$\begin{aligned} V_{12} &= \int_{40 \text{ cm}}^{10 \text{ cm}} (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l} \\ &= \int_{40 \text{ cm}}^{10 \text{ cm}} \left(\hat{z} 5 \times \hat{\phi} \frac{\mu_0 I}{2\pi r} \right) \cdot \hat{r} dr \\ &= -\frac{5\mu_0 I}{2\pi} \int_{40 \text{ cm}}^{10 \text{ cm}} \frac{dr}{r} \\ &= -\frac{5 \times 4\pi \times 10^{-7} \times 10}{2\pi} \times \ln\left(\frac{10}{40}\right) \\ &= 13.9 \quad (\mu\text{V}). \end{aligned}$$

概念问题 6-4：假设图 6-8 中的导电棒滑动时不存在摩擦力，并且电路的水平臂很长。因此，如果杆受到初始推力，它应该继续以恒定速度移动，并且其运动无限期地以感应电动势的形式产生电能。这是一个有效的论点吗？如果没有，为什么呢？我们能否在无需通过其他方式提供等量能源的情况下产生电能？

概念问题 6-5：图 6-10 棒中流动的电流是稳定电流吗？检查末端 1 和 2 处电荷 q 上的力并进行比较。

练习6-3：对于图6-9的移动环，求环边位于 $y_1 = 4 \text{ m}$ 和 $y_2 = 4.5 \text{ m}$ 处的 I 。另外，反转运动方向，使得 $\mathbf{u} = -\hat{y}5 \text{ (m/s)}$ 。

答案：我 $= -13 \text{ (毫安)}$ 。[\(参见^{EM}。\)](#)

Exercise 6-4: Suppose that we turn the loop of Fig. 6-9 so that its surface is parallel to the x - z plane. What would I be in that case?

Answer: $I = 0$. (See EM .)

6-5 The Electromagnetic Generator

The electromagnetic generator is the converse of the electromagnetic motor. The principles of operation of both instruments may be explained with the help of Fig. 6-11. A permanent magnet is used to produce a static magnetic field $\mathbf{B}$ in the slot between its two poles. When a current is passed through the conducting loop, as depicted in Fig. 6-11(a), the current flows in opposite directions in segments 1-2 and 3-4 of the loop. The induced magnetic forces on the two segments are also opposite, resulting in a torque that causes the loop to rotate about its axis. Thus, in a motor, electrical energy supplied by a voltage source is converted into mechanical energy in the form of a rotating loop, which can be coupled to pulleys, gears, or other movable objects.

If instead of passing a current through the loop to make it turn the loop is made to rotate by an external force, the movement of the loop in the magnetic field produces a motional emf, $V_{\text{emf}}^{\text{m}}$, as shown in Fig. 6-11(b). Hence, the motor has become a generator, and mechanical energy is getting converted into electrical energy.

Let us examine the operation of the electromagnetic generator in more detail using the coordinate system shown in Fig. 6-12. The magnetic field is

$$\mathbf{B} = \hat{\mathbf{z}}B_0, \quad (6.30)$$

and the axis of rotation of the conducting loop is along the x axis. Segments 1-2 and 3-4 of the loop are of length l each, and both cross the magnetic flux lines as the loop rotates. The other two segments are each of width w , and neither crosses the $\mathbf{B}$ lines when the loop rotates. Hence, only segments 1-2 and 3-4 contribute to the generation of the motional emf, $V_{\text{emf}}^{\text{m}}$.

As the loop rotates with an angular velocity ω about its own axis, segment 1-2 moves with velocity $\mathbf{u}$ given by

$$\mathbf{u} = \hat{\mathbf{n}}\omega \frac{w}{2}, \quad (6.31)$$

where $\hat{\mathbf{n}}$, the surface normal to the loop, makes an angle α with the z axis. Hence,

$$\hat{\mathbf{n}} \times \hat{\mathbf{z}} = \hat{\mathbf{x}} \sin \alpha. \quad (6.32)$$

Segment 3-4 moves with velocity $-\mathbf{u}$. Application of

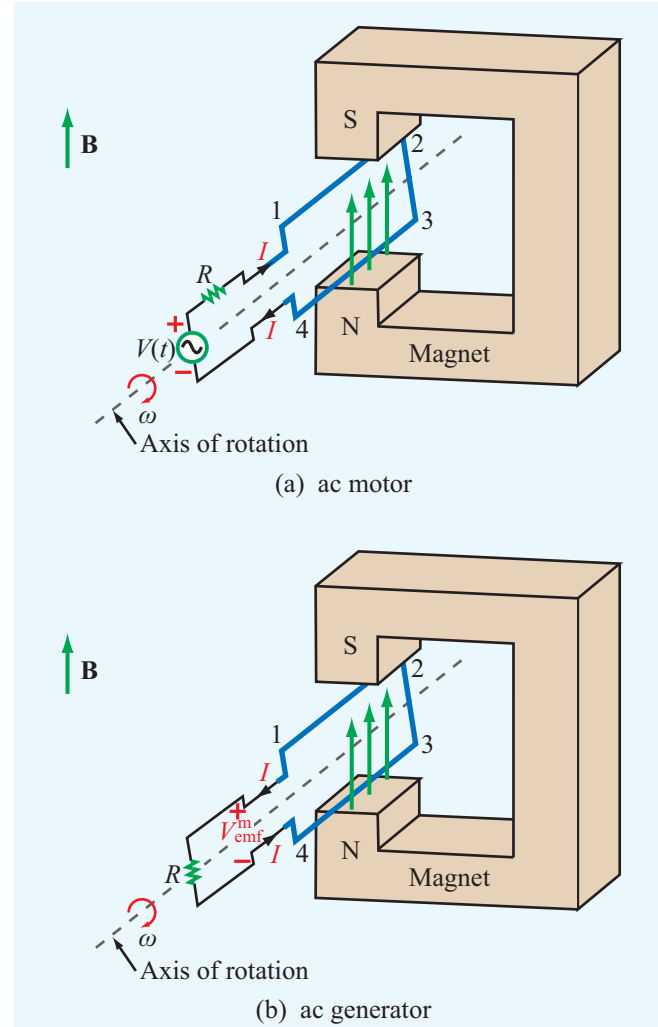


Figure 6-11 Principles of the ac motor and the ac generator. In (a) the magnetic torque on the wires causes the loop to rotate and in (b) the rotating loop generates an emf.

Eq. (6.26), consistent with our choice of $\hat{\mathbf{n}}$, gives

$$\begin{aligned} V_{\text{emf}}^{\text{m}} &= V_{14} = \int_2^1 (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l} + \int_4^3 (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l} \\ &= \int_{-l/2}^{l/2} \left[\left(\hat{\mathbf{n}}\omega \frac{w}{2} \right) \times \hat{\mathbf{z}}B_0 \right] \cdot \hat{\mathbf{x}} dx \\ &\quad + \int_{l/2}^{-l/2} \left[\left(-\hat{\mathbf{n}}\omega \frac{w}{2} \right) \times \hat{\mathbf{z}}B_0 \right] \cdot \hat{\mathbf{x}} dx. \end{aligned} \quad (6.33)$$

Using Eq. (6.32) in Eq. (6.33), we obtain the result

$$V_{\text{emf}}^{\text{m}} = w l \omega B_0 \sin \alpha = A \omega B_0 \sin \alpha, \quad (6.34)$$

练习 6-4: 假设我们转动图 6-9 中的环, 使其表面平行于 x - z 平面。那样的话我会怎样呢?

答案: 我 = 0。 (参见 EM 。)

6-5 电磁发生器

电磁发电机与电磁电机相反。两种仪器的工作原理可以借助图 6-11 进行解释。永磁体用于在其两极之间的槽中产生静磁场 $\mathbf{B}$ 。当电流通过导电回路时, 如图 6-11(a) 所示, 电流在回路的 1-2 和 3-4 段中以相反方向流动。两个部分上的感应磁力也是相反的, 从而产生使环绕其轴旋转的扭矩。因此, 在电机中, 由电压源提供的电能以旋转环的形式转换为机械能, 该旋转环可以耦合到滑轮、齿轮或其他可移动物体。

如果不是让电流通过环路使其转动, 而是通过外力使环路旋转, 则环路在磁场中的运动会产生运动电动势 $V_{\text{emf}}^{\text{m}}$, 如图 6-11(b) 所示。因此, 电动机变成了发电机, 机械能转化为电能。

让我们使用图 6-12 所示的坐标系更详细地检查电磁发电机的操作。磁场是

$$\mathbf{B} = \hat{\mathbf{z}}B_0, \quad (6.30)$$

导电环的旋转轴沿 x 轴。环路的 1-2 段和 3-4 段的长度各为 l , 并且在环路旋转时都穿过磁力线。另外两段的宽度均为 w , 当环旋转时, 都不与 $\mathbf{B}$ 线相交。因此, 只有段 1-2 和 3-4 有助于生成运动电动势 $V_{\text{emf}}^{\text{m}}$ 。

当环以角速度 ω 绕其自身轴旋转时, 段 1-2 以速度 $\mathbf{u}$ 移动, 速度 $\mathbf{u}$ 为:

$$\mathbf{u} = \hat{\mathbf{n}}\omega \frac{w}{2}, \quad (6.31)$$

其中, $\hat{\mathbf{n}}$ (与环垂直的表面) 与 z 轴形成角度 α 。因此,

$$\hat{\mathbf{n}} \times \hat{\mathbf{z}} = \hat{\mathbf{x}} \sin \alpha. \quad (6.32)$$

段 3-4 以速度 $-\mathbf{u}$ 移动。应用

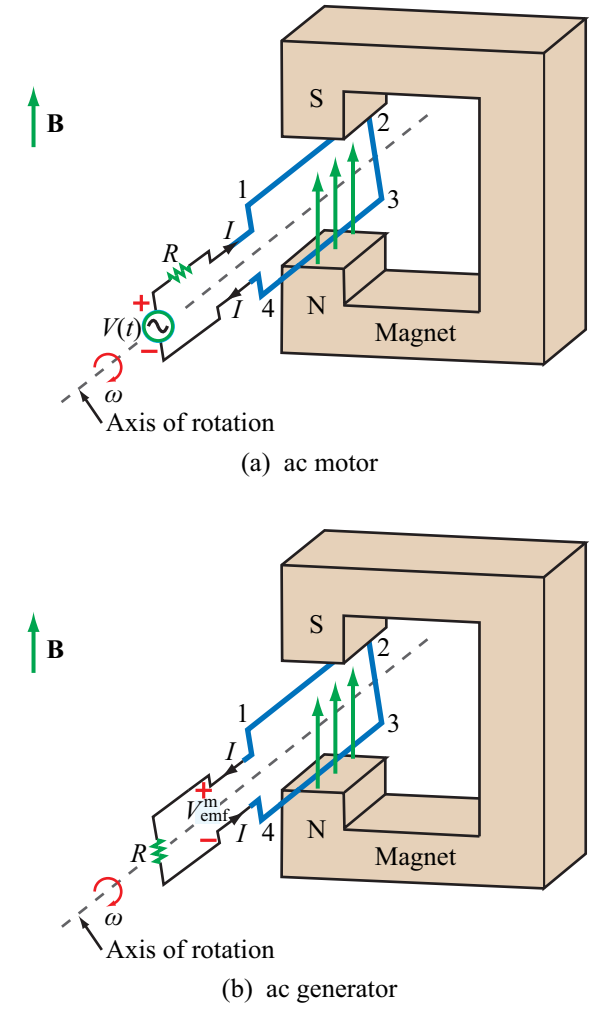


图 6-11 交流电动机和交流发电机的原理。在 (a) 中, 电线上的磁扭矩导致环路旋转, 在 (b) 中, 旋转环路产生电动势。

等式. (6.26) 与我们对 $\hat{\mathbf{n}}$ 的选择一致, 给出

$$\begin{aligned} V_{\text{emf}}^{\text{m}} &= V_{14} = \int_2^1 (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l} + \int_4^3 (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l} \\ &= \int_{-l/2}^{l/2} \left[\left(\hat{\mathbf{n}}\omega \frac{w}{2} \right) \times \hat{\mathbf{z}}B_0 \right] \cdot \hat{\mathbf{x}} dx \\ &\quad + \int_{l/2}^{-l/2} \left[\left(-\hat{\mathbf{n}}\omega \frac{w}{2} \right) \times \hat{\mathbf{z}}B_0 \right] \cdot \hat{\mathbf{x}} dx. \end{aligned} \quad (6.33)$$

使用方程式 (6.32) 式 (6.33), 我们得到结果

$$V_{\text{emf}}^{\text{m}} = w l \omega B_0 \sin \alpha = A \omega B_0 \sin \alpha, \quad (6.34)$$

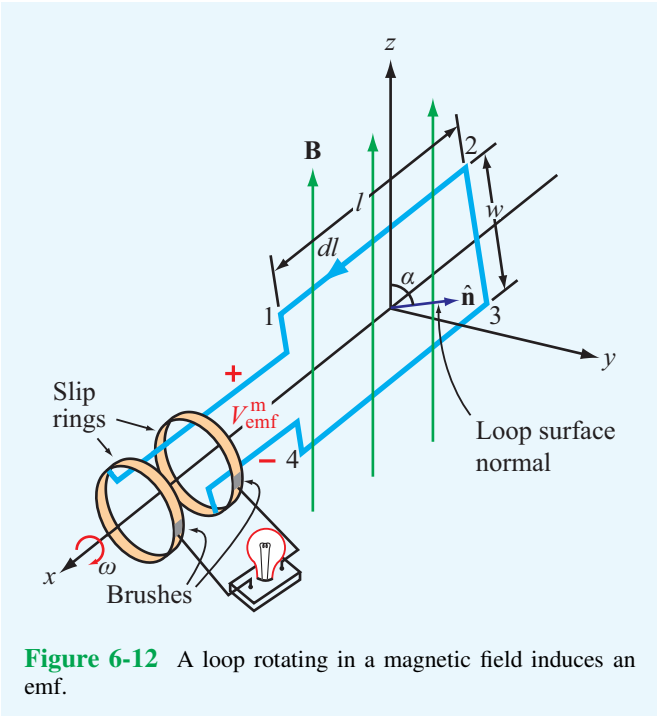


Figure 6-12 A loop rotating in a magnetic field induces an emf.

where $A = wl$ is the surface area of the loop. The angle α is related to ω by

$$\alpha = \omega t + C_0, \quad (6.35)$$

where C_0 is a constant determined by initial conditions. For example, if $\alpha = 0$ at $t = 0$, then $C_0 = 0$. In general,

$$V_{\text{emf}}^{\text{m}} = A\omega B_0 \sin(\omega t + C_0) \quad (\text{V}). \quad (6.36)$$

This same result can also be obtained by applying the general form of Faraday's law given by Eq. (6.6). The flux linking the surface of the loop is

$$\Phi = \int_S \mathbf{B} \cdot d\mathbf{s} = \int_S \hat{\mathbf{z}} B_0 \cdot \hat{\mathbf{n}} \, ds = B_0 A \cos \alpha = B_0 A \cos(\omega t + C_0), \quad (6.37)$$

and

$$V_{\text{emf}} = -\frac{d\Phi}{dt} = -\frac{d}{dt}[B_0 A \cos(\omega t + C_0)] = A\omega B_0 \sin(\omega t + C_0), \quad (6.38)$$

which is identical with the result given by Eq. (6.36).

► The voltage induced by the rotating loop is sinusoidal in time with an angular frequency ω equal to that of the rotating loop, and its amplitude is equal to the product of the surface area of the loop, the magnitude of the magnetic field generated by the magnet, and the angular frequency ω . ◀

Concept Question 6-6: Contrast the operation of an ac motor with that of an ac generator.

Concept Question 6-7: The rotating loop of Fig. 6-12 had a single turn. What would be the emf generated by a loop with 10 turns?

Concept Question 6-8: The magnetic flux linking the loop shown in Fig. 6-12 is maximum when $\alpha = 0$ (loop in x - y plane), yet according to Eq. (6.34), the induced emf is zero when $\alpha = 0$. Conversely, when $\alpha = 90^\circ$, the flux linking the loop is zero, but $V_{\text{emf}}^{\text{m}}$ is at a maximum. Is this consistent with your expectations? Why?

6-6 Moving Conductor in a Time-Varying Magnetic Field

For the general case of a single-turn conducting loop moving in a time-varying magnetic field, the induced emf is the sum of a transformer component and a motional component. Thus, the sum of Eqs. (6.8) and (6.26) gives

$$\begin{aligned} V_{\text{emf}} &= V_{\text{emf}}^{\text{tr}} + V_{\text{emf}}^{\text{m}} \\ &= -\int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s} + \oint_C (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l}. \end{aligned} \quad (6.39)$$

V_{emf} is also given by the general expression of Faraday's law:

$$V_{\text{emf}} = -\frac{d\Phi}{dt} = -\frac{d}{dt} \int_S \mathbf{B} \cdot d\mathbf{s} \quad (\text{total emf}). \quad (6.40)$$

In fact, it can be shown mathematically that the right-hand side of Eq. (6.39) is equivalent to the right-hand side of Eq. (6.40). For a particular problem, the choice between using Eq. (6.39) or Eq. (6.40) is usually made on the basis of which is the easier to apply. In either case, for an N -turn loop, the right-hand sides of Eqs. (6.39) and (6.40) should be multiplied by N .

Example 6-6: Electromagnetic Generator

Find the induced voltage when the rotating loop of the electromagnetic generator of Section 6-5 is in a magnetic field $\mathbf{B} = \hat{\mathbf{z}} B_0 \cos \omega t$. Assume that $\alpha = 0$ at $t = 0$.

Solution: The flux Φ is given by Eq. (6.37) with B_0 replaced with $B_0 \cos \omega t$. Thus,

$$\Phi = B_0 A \cos^2 \omega t,$$

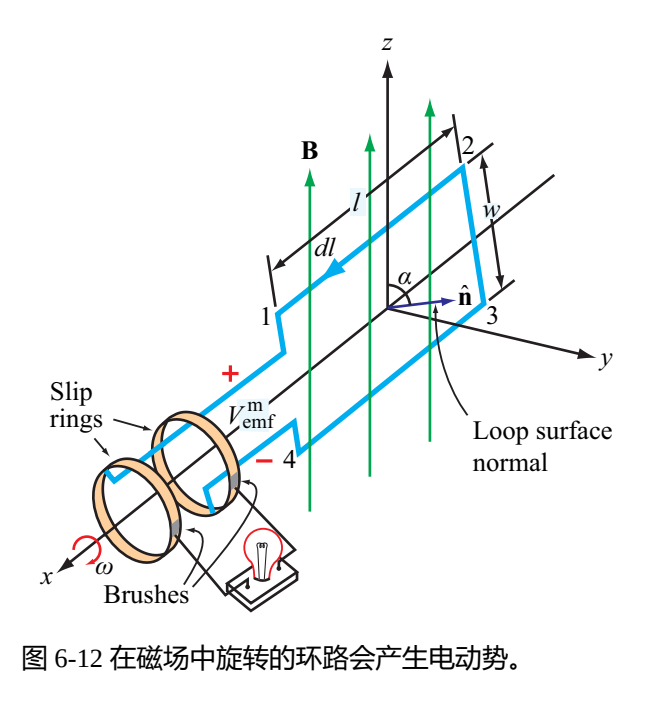


图 6-12 在磁场中旋转的环路会产生电动势。

其中 $A = wl$ 是环路的表面积。角度 α 与 ω 的关系为

$$\alpha = \omega t + C_0, \quad (6.35)$$

其中 C_0 是由初始条件确定的常数。例如，如果在 $t = 0$ 处 $\alpha = 0$ ，则 $C_0 = 0$ 。一般来说，

$$V_{\text{emf}}^{\text{m}} = A\omega B_0 \sin(\omega t + C_0) \quad (\text{V}). \quad (6.36)$$

通过应用等式给出的法拉第定律的一般形式也可以获得同样的结果。(6.6)。连接环路表面的通量为

$$\Phi = \int_S \mathbf{B} \cdot d\mathbf{s} = \int_S \hat{\mathbf{z}} B_0 \cdot \hat{\mathbf{n}} \, ds = B_0 A \cos \alpha = B_0 A \cos(\omega t + C_0), \quad (6.37)$$

和

$$V_{\text{emf}} = -\frac{d\Phi}{dt} = -\frac{d}{dt}[B_0 A \cos(\omega t + C_0)] = A\omega B_0 \sin(\omega t + C_0), \quad (6.38)$$

这与等式给出的结果相同。(6.36)。

► 旋转环感应的电压随时间呈正弦波，其角频率 ω 等于旋转环的电压，其振幅等于环的表面积、磁体产生的磁场强度和角频率 ω 的乘积。 ◀

概念问题 6-6：比较交流电动机和交流发电机的运行。

概念题 6-7：图 6-12 的旋转环只有一圈。10 匝环路产生的电动势是多少？

概念问题 6-8：当 $\alpha = 0$ (x - y 平面中的环路) 时，连接图 6-12 所示环路的磁通量最大，但根据方程：根据式(6.34)，当 α 为 0 时，感应电动势为零。相反，当 α 为 90° 时，连接环路的磁通为零，但 $V_{\text{emf}}^{\text{m}}$ 达到最大值。这符合您的预期吗？为什么？

6-6 时变磁场中移动的导体

对于在时变磁场中移动的单匝导电环路的一般情况，感应电动势是变压器分量和运动分量的总和。因此，等式的总和。(6.8) 和 (6.26) 给出

$$\begin{aligned} V_{\text{emf}} &= V_{\text{emf}}^{\text{tr}} + V_{\text{emf}}^{\text{m}} \\ &= -\int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s} + \oint_C (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l}. \end{aligned} \quad (6.39)$$

V_{emf} 也由法拉第定律的一般表达式给出：

$$V_{\text{emf}} = -\frac{d\Phi}{dt} = -\frac{d}{dt} \int_S \mathbf{B} \cdot d\mathbf{s} \quad (\text{total emf}). \quad (6.40)$$

事实上，可以从数学上证明方程的右侧：(6.39) 等价于式 (6.39) 的右侧。(6.40)。对于特定问题，可以选择使用方程。(6.39) 或等式。(6.40) 通常是在更容易应用的基础上制定的。在任何一种情况下，对于 N 匝循环，方程的右侧。(6.39) 和 (6.40) 应乘以 N 。

例 6-6：电磁发生器

求第 6-5 节的电磁发电机的旋转回路处于磁场 $\mathbf{B} = \hat{\mathbf{z}} B_0 \cos \omega t$ 中的感应电压。假设 $t = 0$ 时 $\alpha = 0$ 。

解：通量 Φ 由方程式给出。(6.37) 将 B_0 替换为 $B_0 \cos \omega t$ 。因此，

$$\Phi = B_0 A \cos^2 \omega t,$$

and

$$\begin{aligned} V_{\text{emf}} &= -\frac{d\Phi}{dt} \\ &= -\frac{d}{dt}(B_0 A \cos^2 \omega t) \\ &= 2B_0 A \omega \cos \omega t \sin \omega t = B_0 A \omega \sin 2\omega t. \end{aligned}$$

6-7 Displacement Current

Ampère's law in differential form is given by

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} \quad (\text{Ampère's law}). \quad (6.41)$$

Integrating both sides of Eq. (6.41) over an arbitrary open surface S with contour C , we have

$$\int_S (\nabla \times \mathbf{H}) \cdot d\mathbf{s} = \int_S \mathbf{J} \cdot d\mathbf{s} + \int_S \frac{\partial \mathbf{D}}{\partial t} \cdot d\mathbf{s}. \quad (6.42)$$

The surface integral of $\mathbf{J}$ equals the conduction current I_c flowing through S , and the surface integral of $\nabla \times \mathbf{H}$ can be converted into a line integral of $\mathbf{H}$ over the contour C bounding C by invoking Stokes's theorem. That is,

$$\begin{aligned} \int_S \mathbf{J} \cdot d\mathbf{s} &= I_c, \\ \int_S (\nabla \times \mathbf{H}) \cdot d\mathbf{s} &= \oint_C \mathbf{H} \cdot d\mathbf{l}, \end{aligned}$$

and

$$\oint_C \mathbf{H} \cdot d\mathbf{l} = I_c + \int_S \frac{\partial \mathbf{D}}{\partial t} \cdot d\mathbf{s}. \quad (6.43)$$

(Ampère's law)

The second term on the right-hand side of Eq. (6.43) of course has the same unit (amperes) as the current I_c , and because it is proportional to the time derivative of the electric flux density $\mathbf{D}$, which is also called the electric displacement, it is called the **displacement current** I_d . That is,

$$I_d = \int_S \mathbf{J}_d \cdot d\mathbf{s} = \int_S \frac{\partial \mathbf{D}}{\partial t} \cdot d\mathbf{s}, \quad (6.44)$$

where $\mathbf{J}_d = \partial \mathbf{D} / \partial t$ represents a **displacement current density**. In view of Eq. (6.44),

$$\oint_C \mathbf{H} \cdot d\mathbf{l} = I_c + I_d = I, \quad (6.45)$$

where I is the total current. In electrostatics, $\partial \mathbf{D} / \partial t = 0$; therefore, $I_d = 0$ and $I = I_c$. The concept of displacement current was first introduced in 1873 by James Clerk Maxwell when he formulated the unified theory of electricity and magnetism under time-varying conditions.

The parallel-plate capacitor is commonly used as an example with which to illustrate the physical meaning of the displacement current I_d . The simple circuit shown in **Fig. 6-13** consists of a capacitor and an ac source with voltage $V_s(t)$ given by

$$V_s(t) = V_0 \cos \omega t \quad (\text{V}). \quad (6.46)$$

According to Eq. (6.45), the total current flowing through any surface consists, in general, of a conduction current I_c and a displacement current I_d . Let us find I_c and I_d through each of the following two imaginary surfaces: (1) the cross section of the conducting wire, S_1 , and (2) the cross section of the capacitor S_2 (**Fig. 6-13**). We denote the conduction and displacement currents in the wire as I_{1c} and I_{1d} and those through the capacitor as I_{2c} and I_{2d} .

和

$$\begin{aligned} V_{\text{emf}} &= -\frac{d\Phi}{dt} \\ &= -\frac{d}{dt}(B_0 A \cos^2 \omega t) \\ &= 2B_0 A \omega \cos \omega t \sin \omega t = B_0 A \omega \sin 2\omega t. \end{aligned}$$

6-7 位移电流

微分形式的安培定律 由下式给出

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} \quad (\text{Ampère's law}). \quad (6.41)$$

对方程两边求积分。(6.41) 在具有轮廓 C 的任意开放曲面 S 上, 我们有

$$\int_S (\nabla \times \mathbf{H}) \cdot d\mathbf{s} = \int_S \mathbf{J} \cdot d\mathbf{s} + \int_S \frac{\partial \mathbf{D}}{\partial t} \cdot d\mathbf{s}. \quad (6.42)$$

$\mathbf{J}$ 的表面积分等于流过 S 的传导电流 I_c , 并且 $\nabla \times \mathbf{H}$ 的表面积分可以通过调用斯托克斯定理转换为 $\mathbf{H}$ 在以 C 为界的轮廓 C 上的线积分。那是,

$$\begin{aligned} \int_S \mathbf{J} \cdot d\mathbf{s} &= I_c, \\ \int_S (\nabla \times \mathbf{H}) \cdot d\mathbf{s} &= \oint_C \mathbf{H} \cdot d\mathbf{l}, \end{aligned}$$

和

$$\oint_C \mathbf{H} \cdot d\mathbf{l} = I_c + \int_S \frac{\partial \mathbf{D}}{\partial t} \cdot d\mathbf{s}. \quad (6.43)$$

(安培定律)

等式右侧的第二项。式 (6.43) 当然与电流 I_c 具有相同的单位 (安培), 并且由于它与电通量密度 $\mathbf{D}$ 的时间导数 (也称为电位移) 成正比, 因此称为位移电流 I_d 。那是,

$$I_d = \int_S \mathbf{J}_d \cdot d\mathbf{s} = \int_S \frac{\partial \mathbf{D}}{\partial t} \cdot d\mathbf{s}, \quad (6.44)$$

其中, $\mathbf{J}_d = \partial \mathbf{D} / \partial t$ 表示位移电流密度。鉴于方程。(6.44),

$$\oint_C \mathbf{H} \cdot d\mathbf{l} = I_c + I_d = I, \quad (6.45)$$

其中 I 是总电流。在静电学中, $\partial \mathbf{D} / \partial t = 0$; 因此, $I_d = 0$ 和 $I = I_c$ 。位移电流的概念由詹姆斯·克拉克·麦克斯韦 (James Clerk Maxwell) 于 1873 年首次提出, 当时他制定了时变条件下的电和磁的统一理论。

通常以平行板电容器为例来说明位移电流 I_d 的物理意义。图 6-13 所示的简单电路由一个电容器和一个交流电源组成, 电压 $V_s(t)$ 为

$$V_s(t) = V_0 \cos \omega t \quad (\text{V}). \quad (6.46)$$

根据方程。根据方程 (6.45), 流过任何表面的总电流通常由传导电流 I_c 和位移电流 I_d 组成。让我们通过以下两个假想表面分别求出 I_c 和 I_d : (1) 导线的横截面 S_1 , 以及 (2) 电容器的横截面 S_2 (。 6-13)。我们将导线中的传导电流和位移电流表示为 I_{1c} 和 I_{1d} , 将通过电容器的传导电流和位移电流表示为 I_{2c} 和 I_{2d} 。

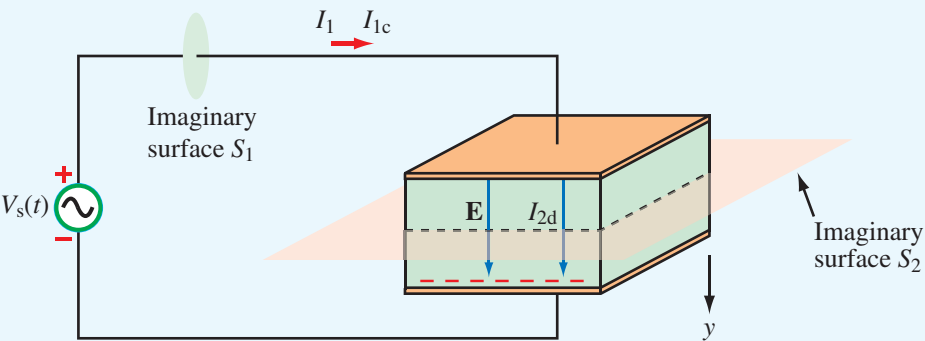
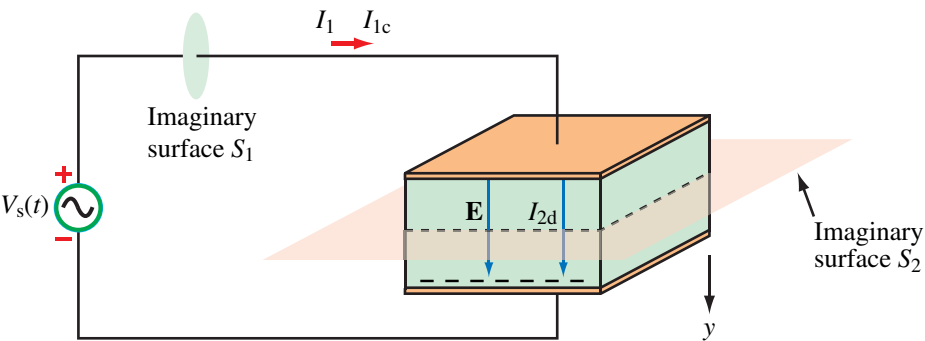


Figure 6-13 The displacement current I_{2d} in the insulating material of the capacitor is equal to the conducting current I_{1c} in the wire.



F图 6-13 电容器绝缘材料中的位移电流 I_{2d} 等于导线中的传导电流 I_{1c} 。

In the perfectly conducting wire, $\mathbf{D} = \mathbf{E} = 0$; hence, Eq. (6.44) gives $I_{1d} = 0$. As for I_{1c} , we know from circuit theory that it is related to the voltage across the capacitor V_C by

$$\begin{aligned} I_{1c} &= C \frac{dV_C}{dt} \\ &= C \frac{d}{dt} (V_0 \cos \omega t) \\ &= -CV_0 \omega \sin \omega t, \end{aligned} \tag{6.47}$$

where we used the fact that $V_C = V_s(t)$. With $I_{1d} = 0$, the total current in the wire is simply $I_1 = I_{1c} = -CV_0 \omega \sin \omega t$.

In the perfect dielectric with permittivity ϵ between the capacitor plates, $\sigma = 0$. Hence, $I_{2c} = 0$ because no conduction current exists there. To determine I_{2d} , we need to apply Eq. (6.44). From Example 4-14, the electric field $\mathbf{E}$ in the dielectric spacing is related to the voltage V_c across its plates by

$$\mathbf{E} = \hat{\mathbf{y}} \frac{V_c}{d} = \hat{\mathbf{y}} \frac{V_0}{d} \cos \omega t, \tag{6.48}$$

where d is the spacing between the plates and $\hat{\mathbf{y}}$ is the direction from the higher-potential plate toward the lower-potential plate at $t = 0$. The displacement current I_{2d} is obtained by applying Eq. (6.44) with $d\mathbf{s} = \hat{\mathbf{y}} \, ds$:

$$\begin{aligned} I_{2d} &= \int_S \frac{\partial \mathbf{D}}{\partial t} \cdot d\mathbf{s} \\ &= \int_A \left[\frac{\partial}{\partial t} \left(\hat{\mathbf{y}} \frac{\epsilon V_0}{d} \cos \omega t \right) \right] \cdot (\hat{\mathbf{y}} \, ds) \\ &= -\frac{\epsilon A}{d} V_0 \omega \sin \omega t \\ &= -CV_0 \omega \sin \omega t, \end{aligned} \tag{6.49}$$

where we used the relation $C = \epsilon A/d$ for the capacitance of the parallel-plate capacitor with plate area A . The expression for I_{2d} in the dielectric region between the conducting plates is identical with that given by Eq. (6.47) for the conduction current I_{1c} in the wire. The fact that these two currents are equal ensures the continuity of total current flow through the circuit.

► Even though the displacement current does not transport free charges, it nonetheless **behaves** like a real current. ◀

In the capacitor example, we treated the wire as a perfect conductor, and we assumed that the space between the capacitor plates was filled with a perfect dielectric. If the wire

has a finite conductivity σ_w , then $\mathbf{D}$ in the wire would not be zero; therefore, the current I_1 would consist of a conduction current I_{1c} as well as a displacement current I_{1d} ; that is, $I_1 = I_{1c} + I_{1d}$. By the same token, if the dielectric spacing material has a nonzero conductivity σ_d , then free charges would flow between the two plates, and I_{2c} would not be zero. In that case, the total current flowing through the capacitor would be $I_2 = I_{2c} + I_{2d}$. No matter the circumstances, the total capacitor current remains equal to the total current in the wire. That is, $I_1 = I_2$.

Example 6-7: Displacement Current Density

The conduction current flowing through a wire with conductivity $\sigma = 2 \times 10^7$ S/m and relative permittivity $\epsilon_r = 1$ is given by $I_c = 2 \sin \omega t$ (mA). If $\omega = 10^9$ rad/s, find the displacement current.

Solution: The conduction current $I_c = JA = \sigma EA$, where A is the cross section of the wire. Hence,

$$\begin{aligned} E &= \frac{I_c}{\sigma A} = \frac{2 \times 10^{-3} \sin \omega t}{2 \times 10^7 A} \\ &= \frac{1 \times 10^{-10}}{A} \sin \omega t \quad (\text{V/m}). \end{aligned}$$

Application of Eq. (6.44), with $D = \epsilon E$, leads to

$$\begin{aligned} I_d &= J_d A \\ &= \epsilon A \frac{\partial E}{\partial t} \\ &= \epsilon A \frac{\partial}{\partial t} \left(\frac{1 \times 10^{-10}}{A} \sin \omega t \right) \\ &= \epsilon \omega \times 10^{-10} \cos \omega t \\ &= 0.885 \times 10^{-12} \cos \omega t \quad (\text{A}), \end{aligned}$$

where we used $\omega = 10^9$ rad/s and $\epsilon = \epsilon_0 = 8.85 \times 10^{-12}$ F/m. Note that I_c and I_d are in phase quadrature (90° phase shift between them). Also, I_d is about nine orders of magnitude smaller than I_c , which is why the displacement current usually is ignored in good conductors.

Exercise 6-5: A poor conductor is characterized by a conductivity $\sigma = 100$ (S/m) and permittivity $\epsilon = 4\epsilon_0$. At what angular frequency ω is the amplitude of the conduction current density $\mathbf{J}$ equal to the amplitude of the displacement current density $\mathbf{J}_d$?

Answer: $\omega = 2.82 \times 10^{12}$ (rad/s). (See [EM](#).)

在理想导线中, $D = E = 0$; 因此, 等式(6.44) 给出 $I_{1d} = 0$ 。至于 I_{1c} , 我们从电路理论得知, 它与电容器 V_C 两端的电压有关:

$$\begin{aligned} I_{1c} &= C \frac{dV_C}{dt} \\ &= C \frac{d}{dt} (V_0 \cos \omega t) \\ &= -CV_0 \omega \sin \omega t, \end{aligned} \tag{6.47}$$

我们使用了 $V_C = V_s(t)$ 这一事实。当 $I_{1d} =$ 为 0 时, 导线中的总电流就是 $I_1 = I_{1c} = -CV_0 \omega \sin \omega t$ 。

在电容器板之间介电常数为 ϵ 的理想电介质中, $\sigma =$ 为 0。因此, $I_{2c} =$ 为 0, 因为那里不存在传导电流。为了确定 I_{2d} , 我们需要应用等式: (6.44)。从例 4-14 可以看出, 电介质间距中的电场 E 与其极板上的电压 V_c 的关系为:

$$\mathbf{E} = \hat{\mathbf{y}} \frac{V_c}{d} = \hat{\mathbf{y}} \frac{V_0}{d} \cos \omega t, \tag{6.48}$$

其中 d 是极板之间的间距, $\hat{\mathbf{y}}$ 是在 $t = 0$ 时从较高电势极板朝向较低电势极板的方向。位移电流 I_{2d} 通过应用等式获得。 (6.44) 与 $d\mathbf{s} = \hat{\mathbf{y}} \, ds$:

$$\begin{aligned} I_{2d} &= \int_S \frac{\partial \mathbf{D}}{\partial t} \cdot d\mathbf{s} \\ &= \int_A \left[\frac{\partial}{\partial t} \left(\hat{\mathbf{y}} \frac{\epsilon V_0}{d} \cos \omega t \right) \right] \cdot (\hat{\mathbf{y}} \, ds) \\ &= -\frac{\epsilon A}{d} V_0 \omega \sin \omega t \\ &= -CV_0 \omega \sin \omega t, \end{aligned} \tag{6.49}$$

其中, 我们使用关系式 $C = \epsilon A/d$ 来表示板面积为 A 的平行板电容器的电容。导电板之间介电区域中 I_{2d} 的表达式与等式 1 给出的相同。 (6.47) 为导线中的传导电流 I_{1c} 。这两个电流相等的事实确保了流过电路的总电流的连续性。

► 尽管位移电流不传输自由电荷, 但它的行为仍然像真实电流一样。 ◀

在电容器的例子中, 我们将电线视为理想的导体, 并假设电容器板之间的空间充满了理想的电介质。如果电线

具有有限电导率 σ_w , 则导线中的 D 不会为零; 因此, 电流 I_1 将由传导电流 I_{1c} 和位移电流 I_{1d} 组成; 也就是说, $I_1 = I_{1c} + I_{1d}$ 。出于同样的原因, 如果介电间隔材料具有非零电导率 σ_d , 则自由电荷将在两个板之间流动, 并且 I_{2c} 将不为零。在这种情况下, 流过电容器的总电流将为 $I_2 = I_{2c} + I_{2d}$ 。无论情况如何, 总电容器电流始终等于电线中的总电流。也就是说, $I_1 = I_2$ 。

例 6-7: 位移电流密度

流经电导率 $\sigma = 2 \times 10^7$ S/m 和相对介电常数 $\epsilon_r = 1$ 的导线的传导电流由 $I_c = 2 \sin \omega t$ (mA) 给出。如果 $\omega = 10^9$ rad/s, 求位移电流。

解: 传导电流 $I_c = JA = \sigma EA$, 其中 A 是导线的横截面。因此,

$$\begin{aligned} E &= \frac{I_c}{\sigma A} = \frac{2 \times 10^{-3} \sin \omega t}{2 \times 10^7 A} \\ &= \frac{1 \times 10^{-10}}{A} \sin \omega t \quad (\text{V/m}). \end{aligned}$$

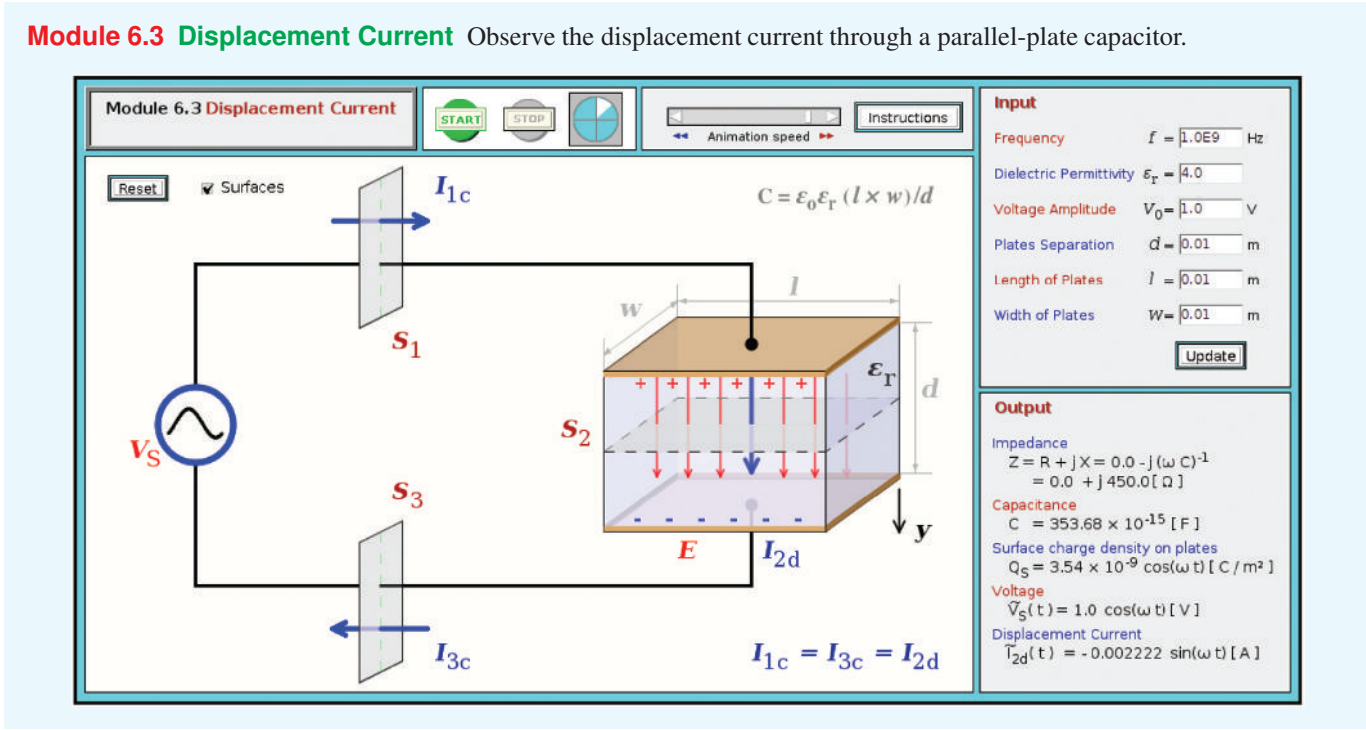
方程的应用 (6.44) , 与 $D = \epsilon E$, 导致

$$\begin{aligned} I_d &= J_d A \\ &= \epsilon A \frac{\partial E}{\partial t} \\ &= \epsilon A \frac{\partial}{\partial t} \left(\frac{1 \times 10^{-10}}{A} \sin \omega t \right) \\ &= \epsilon \omega \times 10^{-10} \cos \omega t \\ &= 0.885 \times 10^{-12} \cos \omega t \quad (\text{A}), \end{aligned}$$

其中我们使用 $\omega = 10^9$ rad/s 和 $\epsilon = \epsilon_0 = 8.85 \times 10^{-12}$ F/m。请注意, I_c 和 I_d 相位正交 (它们之间有 90° 相移)。此外, I_d 比 I_c 小约 9 个数量级, 这就是为什么在良导体中位移电流通常被忽略的原因。

练习 6-5: 不良导体的特征是电导率 $\sigma = 100$ (S/m) 和介电常数 $\epsilon = 4\epsilon_0$ 。在什么角频率 ω 处, 传导电流密度 $\mathbf{J}$ 的幅值等于位移电流密度 $\mathbf{J}_d$ 的幅值?

答案: $\omega = 2.82 \times 10^{12}$ (rad/s)。 (参见 [EM](#)。)



6-8 Boundary Conditions for Electromagnetics

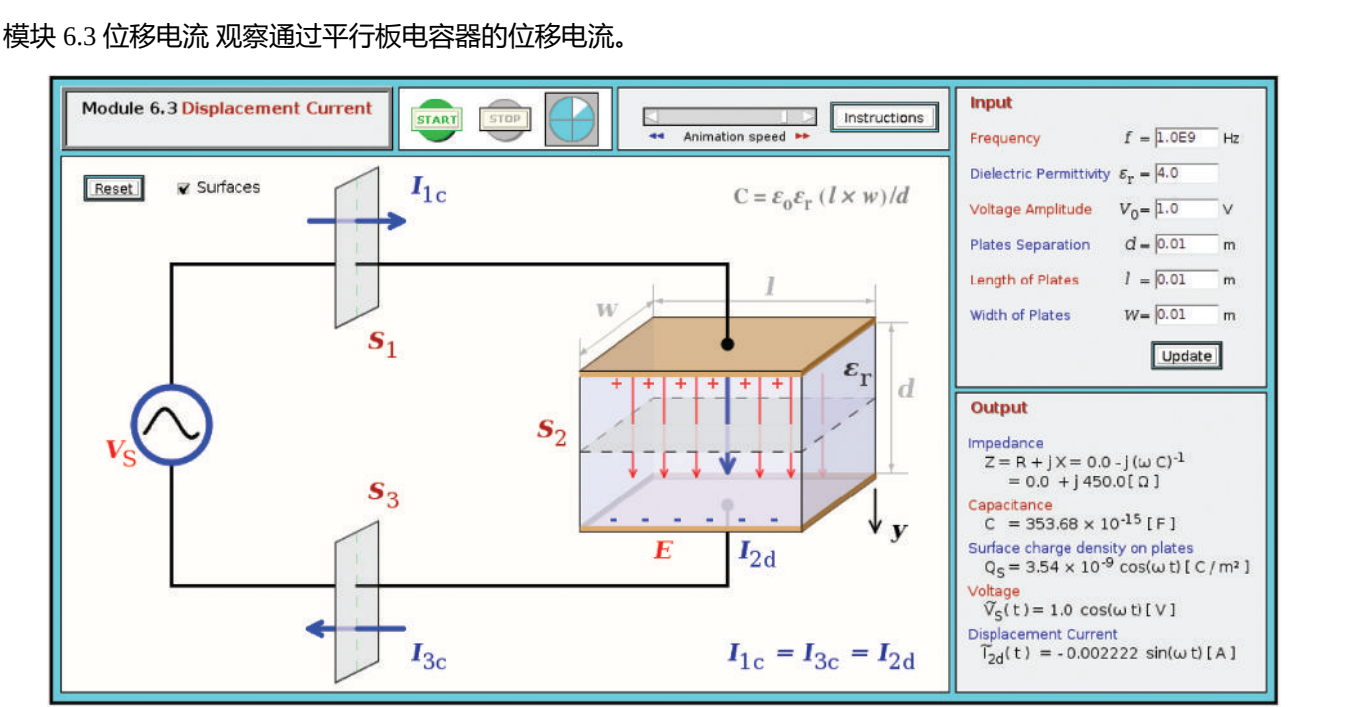
In Chapters 4 and 5, we applied the integral form of Maxwell's equations under static conditions to obtain boundary conditions applicable to the tangential and normal components of $\mathbf{E}$, $\mathbf{D}$, $\mathbf{B}$, and $\mathbf{H}$ on interfaces between contiguous media (Section 4-8 for $\mathbf{E}$ and $\mathbf{D}$ and in Section 5-6 for $\mathbf{B}$ and $\mathbf{H}$). In the dynamic case, Maxwell's equations (Table 6-1) include two new terms not accounted for in electrostatics and magnetostatics, namely, $\partial \mathbf{B} / \partial t$ in Faraday's law and $\partial \mathbf{D} / \partial t$ in Ampère's law.

► Nevertheless, the boundary conditions derived previously for electrostatic and magnetostatic fields remain valid for time-varying fields as well. ◀

This is because, if we were to apply the procedures outlined in the previously referenced sections for time-varying fields, we would find that the combination of the aforementioned terms vanish as the areas of the rectangular loops in Figs. 4-20 and 5-24 are made to approach zero.

The combined set of electromagnetic boundary conditions is summarized in Table 6-2.

- Concept Question 6-9:** When conduction current flows through a material, a certain number of charges enter the material on one end and an equal number leave on the other end. What's the situation like for the displacement current through a perfect dielectric?
- Concept Question 6-10:** Verify that the integral form of Ampère's law given by Eq. (6.43) leads to the boundary condition that the tangential component of $\mathbf{H}$ is continuous across the boundary between two dielectric media.



6-8 电磁学的边界条件

在第4章和第5章中，我们在静态条件下应用了麦克斯韦方程组的积分形式，以获得适用于连续介质之间的界面上的 $\mathbf{E}$ 、 $\mathbf{D}$ 、 $\mathbf{B}$ 和 $\mathbf{H}$ 的切向和法向分量的边界条件（ $\mathbf{E}$ 和 $\mathbf{D}$ 在4-8节中， $\mathbf{B}$ 和 $\mathbf{H}$ 在5-6节中）。在动态情况下，麦克斯韦方程组（表 6-1）包括静电学和静磁学中未考虑的两个新项，即法拉第定律中的 $\partial \mathbf{B} / \partial t$ 和安培定律中的 $\partial \mathbf{D} / \partial t$ 。

► 然而，先前针对静电场和静磁场导出的边界条件对于时变场仍然有效。 ◀

这是因为，如果我们将前面引用的部分中概述的过程应用于时变场，我们会发现上述术语的组合随着图 2 和 3 中矩形环的面积而消失。使4-20和5-24接近于零。

- 表 6-2 总结了电磁边界条件的组合集。
- 概念问题 6-9：当传导电流流过材料时，一定数量的电荷从一端进入材料，并在另一端离开相同数量的电荷。通过理想电介质的位移电流是什么样的情况？
- 概念问题 6-10：验证方程 1 给出的安培定律的积分形式。(6.43) 得出这样的边界条件： $\mathbf{H}$ 的切向分量在两个介电介质之间的边界上是连续的。

Table 6-2 Boundary conditions for the electric and magnetic fields.

Field Components	General Form	Medium 1 Dielectric	Medium 2 Dielectric	Medium 1 Dielectric	Medium 2 Conductor
Tangential E Normal D Tangential H Normal B	$\hat{\mathbf{n}}_2 \times (\mathbf{E}_1 - \mathbf{E}_2) = 0$ $\hat{\mathbf{n}}_2 \cdot (\mathbf{D}_1 - \mathbf{D}_2) = \rho_s$ $\hat{\mathbf{n}}_2 \times (\mathbf{H}_1 - \mathbf{H}_2) = \mathbf{J}_s$ $\hat{\mathbf{n}}_2 \cdot (\mathbf{B}_1 - \mathbf{B}_2) = 0$	$E_{1t} = E_{2t}$ $D_{1n} - D_{2n} = \rho_s$ $H_{1t} = H_{2t}$ $B_{1n} = B_{2n}$	$E_{1t} = E_{2t}$ $D_{1n} - D_{2n} = \rho_s$ $H_{1t} = H_{2t}$ $B_{1n} = B_{2n}$	$E_{1t} = E_{2t} = 0$ $D_{1n} = \rho_s$ $H_{1t} = J_s$ $B_{1n} = B_{2n} = 0$	$E_{1t} = E_{2t} = 0$ $D_{2n} = 0$ $H_{2t} = 0$ $B_{2n} = 0$

Notes: (1) ρ_s is the surface charge density at the boundary; (2) $\mathbf{J}_s$ is the surface current density at the boundary; (3) normal components of all fields are along $\hat{\mathbf{n}}_2$, the outward unit vector of medium 2; (4) $E_{1t} = E_{2t}$ implies that the tangential components are equal in magnitude and parallel in direction; (5) direction of $\mathbf{J}_s$ is orthogonal to $(\mathbf{H}_1 - \mathbf{H}_2)$.

6-9 Charge–Current Continuity Relation

Under static conditions, the charge density ρ_v and the current density $\mathbf{J}$ at a given point in a material are totally independent of one another. This is no longer true in the time-varying case. To show the connection between ρ_v and $\mathbf{J}$, we start by considering an arbitrary volume $\mathcal{V}$ bounded by a closed surface S (Fig. 6-14). The net positive charge contained in $\mathcal{V}$ is Q . Since, according to the law of conservation of electric charge (Section 1-3.2), charge can neither be created nor destroyed, the only way Q can increase is as a result of a net inward flow of positive charge into the volume $\mathcal{V}$. By the same token, for Q to decrease, there has to be a net outward flow of charge from $\mathcal{V}$. The inward and outward flow of charge constitute currents flowing across the surface S into and out of $\mathcal{V}$, respectively. We define I as the *net current flowing across S out of $\mathcal{V}$* . Accordingly, I is equal to the *negative* rate

of change of Q :

$$I = -\frac{dQ}{dt} = -\frac{d}{dt} \int_{\mathcal{V}} \rho_v \, d\mathcal{V}, \quad (6.50)$$

where ρ_v is the volume charge density in $\mathcal{V}$. According to Eq. (4.12), the current I is also defined as the outward flux of the current density $\mathbf{J}$ through the surface S . Hence,

$$\oint_S \mathbf{J} \cdot d\mathbf{s} = -\frac{d}{dt} \int_{\mathcal{V}} \rho_v \, d\mathcal{V}. \quad (6.51)$$

By applying the divergence theorem given by Eq. (3.98), we can convert the surface integral of $\mathbf{J}$ into a volume integral of its divergence $\nabla \cdot \mathbf{J}$, which then gives

$$\oint_S \mathbf{J} \cdot d\mathbf{s} = \int_{\mathcal{V}} \nabla \cdot \mathbf{J} \, d\mathcal{V} = -\frac{d}{dt} \int_{\mathcal{V}} \rho_v \, d\mathcal{V}. \quad (6.52)$$

For a stationary volume $\mathcal{V}$, the time derivative operates on ρ_v only. Hence, we can move it inside the integral and express it as a partial derivative of ρ_v :

$$\int_{\mathcal{V}} \nabla \cdot \mathbf{J} \, d\mathcal{V} = -\int_{\mathcal{V}} \frac{\partial \rho_v}{\partial t} \, d\mathcal{V}. \quad (6.53)$$

In order for the volume integrals on both sides of Eq. (6.53) to be equal for any volume $\mathcal{V}$, their integrands have to be equal at every point within $\mathcal{V}$. Hence,

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho_v}{\partial t}, \quad (6.54)$$

which is known as the *charge–current continuity relation*, or simply the *charge continuity equation*.

If the volume charge density within an elemental volume $\Delta\mathcal{V}$ (such as a small cylinder) is not a function of time

表 6-2 电场和磁场的边界条件。

Field Components	General Form	Medium 1 Dielectric	Medium 2 Dielectric	Medium 1 Dielectric	Medium 2 Conductor
Tangential E Normal D Tangential H Normal B	$\hat{\mathbf{n}}_2 \times (\mathbf{E}_1 - \mathbf{E}_2) = 0$ $\hat{\mathbf{n}}_2 \cdot (\mathbf{D}_1 - \mathbf{D}_2) = \rho_s$ $\hat{\mathbf{n}}_2 \times (\mathbf{H}_1 - \mathbf{H}_2) = \mathbf{J}_s$ $\hat{\mathbf{n}}_2 \cdot (\mathbf{B}_1 - \mathbf{B}_2) = 0$	$E_{1t} = E_{2t}$ $D_{1n} - D_{2n} = \rho_s$ $H_{1t} = H_{2t}$ $B_{1n} = B_{2n}$	$E_{1t} = E_{2t}$ $D_{1n} - D_{2n} = \rho_s$ $H_{1t} = J_s$ $B_{1n} = B_{2n}$	$E_{1t} = E_{2t} = 0$ $D_{1n} = \rho_s$ $H_{1t} = J_s$ $B_{1n} = B_{2n} = 0$	$E_{1t} = E_{2t} = 0$ $D_{2n} = 0$ $H_{2t} = 0$ $B_{2n} = 0$

Notes: (1) ρ_s is the surface charge density at the boundary; (2) $\mathbf{J}_s$ is the surface current density at the boundary; (3) normal components of all fields are along $\hat{\mathbf{n}}_2$, the outward unit vector of medium 2; (4) $E_{1t} = E_{2t}$ implies that the tangential components are equal in magnitude and parallel in direction; (5) direction of $\mathbf{J}_s$ is orthogonal to $(\mathbf{H}_1 - \mathbf{H}_2)$.

6-9 充电-电流连续性关系

在静态条件下，材料中给定点的电荷密度 ρ_v 和电流密度 $\mathbf{J}$ 完全彼此独立。在随时间变化的情况下，情况不再如此。为了显示 ρ_v 和 $\mathbf{J}$ 之间的联系，我们首先考虑由封闭曲面 S 界定的任意体积 $\mathcal{V}$ （图 6-14）。 $\mathcal{V}$ 中包含的净正电荷为 Q 。根据电荷守恒定律（第 1-3.2 节），电荷既不能产生也不能消失，因此 Q 增加的唯一方法是正电荷净向内流入体积 $\mathcal{V}$ 。出于同样的原因，为了使 Q 减小，必须有来自 $\mathcal{V}$ 的净向外电荷流。向内和向外的电荷流分别构成穿过表面 S 流入和流出 $\mathcal{V}$ 的电流。我们将 I 定义为 $\mathcal{V}$ 中流经 S 的净电流。因此， I 等于负利率

Q 的变化：

$$I = -\frac{dQ}{dt} = -\frac{d}{dt} \int_{\mathcal{V}} \rho_v \, d\mathcal{V}, \quad (6.50)$$

其中 ρ_v 是 $\mathcal{V}$ 中的体积电荷密度。根据方程。(4.12) 中，电流 I 也定义为通过表面 S 的电流密度 $\mathbf{J}$ 的向外通量。因此，

$$\oint_S \mathbf{J} \cdot d\mathbf{s} = -\frac{d}{dt} \int_{\mathcal{V}} \rho_v \, d\mathcal{V}. \quad (6.51)$$

通过应用方程给出的散度定理。(3.98)，我们可以将 $\mathbf{J}$ 的表面积分转换为其散度 $\nabla \cdot \mathbf{J}$ 的体积分，然后得到

$$\oint_S \mathbf{J} \cdot d\mathbf{s} = \int_{\mathcal{V}} \nabla \cdot \mathbf{J} \, d\mathcal{V} = -\frac{d}{dt} \int_{\mathcal{V}} \rho_v \, d\mathcal{V}. \quad (6.52)$$

对于固定体积 $\mathcal{V}$ ，时间导数仅对 ρ_v 起作用。因此，我们可以将其移入积分并将其表示为 ρ_v 的偏导数：

$$\int_{\mathcal{V}} \nabla \cdot \mathbf{J} \, d\mathcal{V} = -\int_{\mathcal{V}} \frac{\partial \rho_v}{\partial t} \, d\mathcal{V}. \quad (6.53)$$

为了求方程两边的体积分：(6.53) 对于任何体积 $\mathcal{V}$ 都相等，它们的被积数必须在 $\mathcal{V}$ 内的每个点都相等。因此，

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho_v}{\partial t}, \quad (6.54)$$

这称为电荷-电流连续性关系，或简称为电荷连续性方程。

如果元素体积 $\Delta\mathcal{V}$ （例如小圆柱体）内的体积电荷密度不是时间的函数

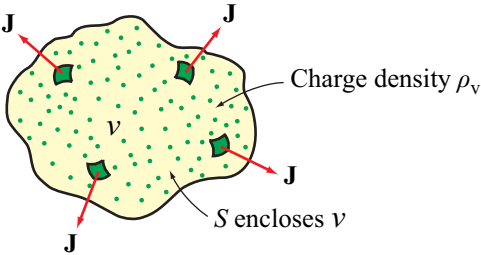


图 6-14 从体积 $\mathcal{V}$ 流出的总电流等于通过表面 S 的电流密度 $\mathbf{J}$ 的通量，而该通量又等于 $\mathcal{V}$ 中所包含的电荷的减少率。

Figure 6-14 The total current flowing out of a volume $\mathcal{V}$ is equal to the flux of the current density $\mathbf{J}$ through the surface S , which in turn is equal to the rate of decrease of the charge enclosed in $\mathcal{V}$.

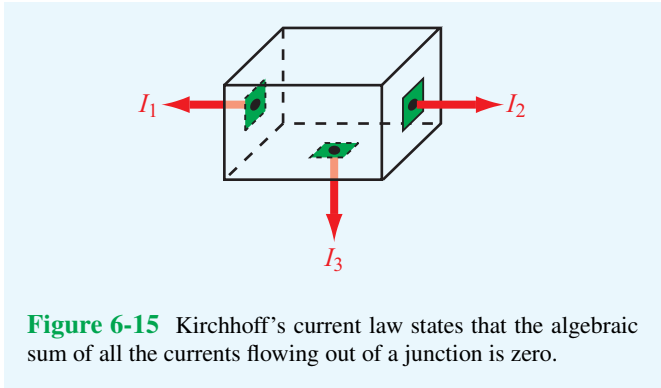


Figure 6-15 Kirchhoff’s current law states that the algebraic sum of all the currents flowing out of a junction is zero.

(i.e., $\partial \rho_v / \partial t = 0$), it means that the net current flowing out of $\Delta \mathcal{V}$ is zero or, equivalently, that the current flowing into $\Delta \mathcal{V}$ is equal to the current flowing out of it. In this case, Eq. (6.54) implies

$$\nabla \cdot \mathbf{J} = 0, \quad (6.55)$$

and its integral-form equivalent [from Eq. (6.51)] is

$$\oint_S \mathbf{J} \cdot d\mathbf{s} = 0. \quad \text{(Kirchhoff’s current law)} \quad (6.56)$$

Let us examine the meaning of Eq. (6.56) by considering a junction (or node) connecting two or more branches in an electric circuit. No matter how small, the junction has a volume $\mathcal{V}$ enclosed by a surface S . The junction shown in **Fig. 6-15** has been drawn as a cube, and its dimensions have been artificially enlarged to facilitate the present discussion. The junction has six faces (surfaces), which collectively constitute the surface S associated with the closed-surface integration given by Eq. (6.56). For each face, the integration represents the current flowing out through that face. Thus, Eq. (6.56) can be cast as

$$\sum_i I_i = 0, \quad \text{(Kirchhoff’s current law)} \quad (6.57)$$

where I_i is the current flowing outward through the i th face. For the junction of **Fig. 6-15**, Eq. (6.57) translates into $(I_1 + I_2 + I_3) = 0$. In its general form, Eq. (6.57) is an expression of **Kirchhoff’s current law**, which states that in an electric circuit *the sum of all the currents flowing out of a junction is zero*.

6-10 Free-Charge Dissipation in a Conductor

We stated earlier that current flow in a conductor is realized by the movement of loosely attached electrons under the in-

fluence of an externally applied electric field. These electrons, however, *are not excess charges*; their charge is balanced by an equal amount of positive charge in the atoms’ nuclei. In other words, the conductor material is electrically neutral, and the net charge density in the conductor is zero ($\rho_v = 0$). What happens then if an excess free charge q is introduced at some interior point in a conductor? The excess charge gives rise to an electric field, which forces the charges of the host material nearest to the excess charge to rearrange their locations, which in turn cause other charges to move, and so on. The process continues until neutrality is reestablished in the conductor material and a charge equal to q resides on the conductor’s surface.

How fast does the excess charge dissipate? To answer this question, let us introduce a volume charge density ρ_{v0} at the interior of a conductor and then find out the rate at which it decays down to zero. From Eq. (6.54), the continuity equation is given by

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho_v}{\partial t}. \quad (6.58)$$

In a conductor, the point form of Ohm’s law given by Eq. (4.73) states that $\mathbf{J} = \sigma \mathbf{E}$. Hence,

$$\sigma \nabla \cdot \mathbf{E} = -\frac{\partial \rho_v}{\partial t}. \quad (6.59)$$

Next, we use Eq. (6.1), $\nabla \cdot \mathbf{E} = \rho_v / \epsilon$, to obtain the partial differential equation

$$\frac{\partial \rho_v}{\partial t} + \frac{\sigma}{\epsilon} \rho_v = 0. \quad (6.60)$$

Given that $\rho_v = \rho_{v0}$ at $t = 0$, the solution of Eq. (6.60) is

$$\rho_v(t) = \rho_{v0} e^{-(\sigma/\epsilon)t} = \rho_{v0} e^{-t/\tau_r} \quad (\text{C/m}^3), \quad (6.61)$$

where $\tau_r = \epsilon / \sigma$ is called the **relaxation time constant**. We see from Eq. (6.61) that the initial excess charge ρ_{v0} decays exponentially at a rate τ_r . At $t = \tau_r$, the initial charge ρ_{v0} will have decayed to $1/e \approx 37\%$ of its initial value, and at $t = 3\tau_r$, it will have decayed to $e^{-3} \approx 5\%$ of its initial value at $t = 0$. For copper, with $\epsilon \approx \epsilon_0 = 8.854 \times 10^{-12}$ F/m and $\sigma = 5.8 \times 10^7$ S/m, $\tau_r = 1.53 \times 10^{-19}$ s. Thus, the charge dissipation process in a conductor is extremely fast. In contrast, the decay rate is very slow in a good insulator. For a material like mica with $\epsilon = 6\epsilon_0$ and $\sigma = 10^{-15}$ S/m, $\tau_r = 5.31 \times 10^4$ s, or approximately 14.8 hours.

Concept Question 6-11: Explain how the charge continuity equation leads to Kirchhoff’s current law.

Concept Question 6-12: How long is the relaxation time constant for charge dissipation in a perfect conductor? In a perfect dielectric?

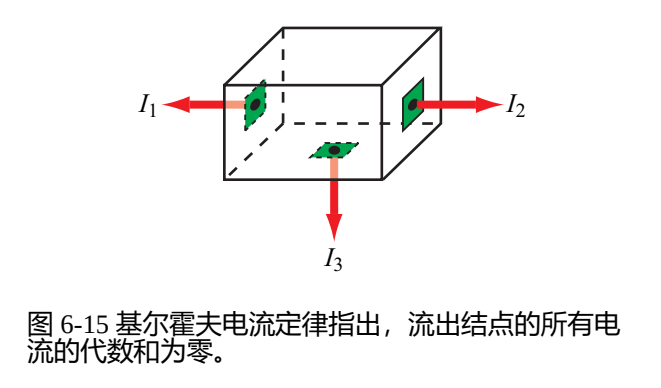


图 6-15 基尔霍夫电流定律指出，流出结点的所有电流的代数和为零。

(IE。 , $\partial \rho_v / \partial t = 0$) , 这意味着流出的净电流 $\Delta \mathcal{V}$ 为零, 或者等价地, 流入的电流 $\Delta \mathcal{V}$ 等于从中流出的电流。在这种情况下, 等式。 (6.54) 意味着

$$\nabla \cdot \mathbf{J} = 0, \quad (6.55)$$

及其积分形式等价物[来自方程。 (6.51)]是

$$\oint_S \mathbf{J} \cdot d\mathbf{s} = 0. \quad \text{(Kirchhoff’s current law)} \quad (6.56)$$

让我们检查一下等式的含义。 (6.56) 通过考虑连接电路中两个或多个分支的结点 (或节点) 。不管结有多少小, 其体积 $\mathcal{V}$ 都被表面 S 包围。图6-15中所示的结被画成立方体, 并且其尺寸被人为地放大以方便当前的讨论。连接点有六个面 (表面) , 它们共同构成了与方程 (1) 给出的闭表面积分相关的表面 S 。 (6.56) 。对于每个面, 积分代表通过该面流出的电流。因此, 等式。 (6.56) 可以表示为

$$\sum_i I_i = 0, \quad \text{(Kirchhoff’s current law)} \quad (6.57)$$

其中 I_i 是通过第 i 个面向外流动的电流。对于图 6-15 的连接点, 方程式(6.57) 转化为 $(I_1 + I_2 + I_3) = 0$ 。在其一般形式中, 方程: (6.57) 是基尔霍夫电流定律的表达式, 它指出在电路中流出结点的所有电流之和为零。

6-10 导体中的自由电荷耗散

我们之前说过, 导体中的电流是通过内部松散附着的电子的运动来实现的。

外部施加的电场的通量。然而, 这些电子并不是多余的电荷; 而是多余的电荷。它们的电荷由原子核中等量的正电荷平衡。换句话说, 导体材料是电中性的, 导体中的净电荷密度为零 ($\rho_v = 0$) 。如果在导体的某个内部点引入过量的自由电荷 q , 会发生什么情况? 多余的电荷会产生电场, 迫使最靠近多余电荷的主体材料的电荷重新排列其位置, 进而导致其他电荷移动, 依此类推。该过程持续进行, 直到导体材料重新建立中性并且导体表面上存在等于 q 的电荷。

多余电荷消散的速度有多快? 为了回答这个问题, 让我们在导体内部引入体积电荷密度 ρ_{v0} , 然后找出它衰减到零的速率。从方程。 (6.54), 连续性方程由下式给出

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho_v}{\partial t}. \quad (6.58)$$

在导体中, 欧姆定律的点形式由方程式给出。 (4.73) 指出 $\mathbf{J} = \sigma \mathbf{E}$ 。因此,

$$\sigma \nabla \cdot \mathbf{E} = -\frac{\partial \rho_v}{\partial t}. \quad (6.59)$$

接下来, 我们使用等式。 (6.1)、 $\nabla \cdot \mathbf{E} = \rho_v / \epsilon$, 得到偏微分方程

$$\frac{\partial \rho_v}{\partial t} + \frac{\sigma}{\epsilon} \rho_v = 0. \quad (6.60)$$

鉴于 $t = 0$ 时的 $\rho_v = \rho_{v0}$, 方程的解为: (6.60) 是

$$\rho_v(t) = \rho_{v0} e^{-(\sigma/\epsilon)t} = \rho_{v0} e^{-t/\tau_r} \quad (\text{C/m}^3), \quad (6.61)$$

其中 $\tau_r = \epsilon / \sigma$ 称为弛豫时间常数。我们从等式中看到。 (6.61) 初始过量电荷 ρ_{v0} 以速率 τ_r 呈指数衰减。在 $t = \tau_r$ 处, 初始电荷 ρ_{v0} 将衰减至其初始值的 $1/e \approx 37\%$, 在 $t = 3\tau_r$ 处, 它将衰减至其在 $t = 0$ 处初始值的 $e^{-3} \approx 5\%$ 。对于铜, $\epsilon \approx \epsilon_0 = 8.854 \times 10^{-12}$ F/m 和 $\sigma = 5.8 \times 10^7$ S/m、 $\tau_r = 1.53 \times 10^{-19}$ 秒。因此, 导体中的电荷耗散过程非常快。相反, 良好绝缘体的衰减率非常慢。对于像云母这样的材料, $\epsilon = 6\epsilon_0$ 和 $\sigma = 10^{-15}$ S/m、 $\tau_r = 5.31 \times 10^4$ s 或大约 14.8 小时。

概念问题 6-11: 解释电荷连续性方程如何推导出基尔霍夫电流定律。

概念问题 6-12: 理想导体中电荷耗散的弛豫时间常数是多长? 在理想的电介质中?

Technology Brief 12: EMF Sensors

An **electromotive force** (emf) sensor is a device that can generate an induced voltage in response to an external stimulus. Three types of emf sensors are profiled in this technical brief: the **piezoelectric transducer**, the **Faraday magnetic flux sensor**, and the **thermocouple**.

Piezoelectric Transducers

► **Piezoelectricity** is the property exhibited by certain crystals, such as quartz, that become electrically polarized when the crystal is subjected to mechanical pressure, thereby inducing a voltage across it. ◀

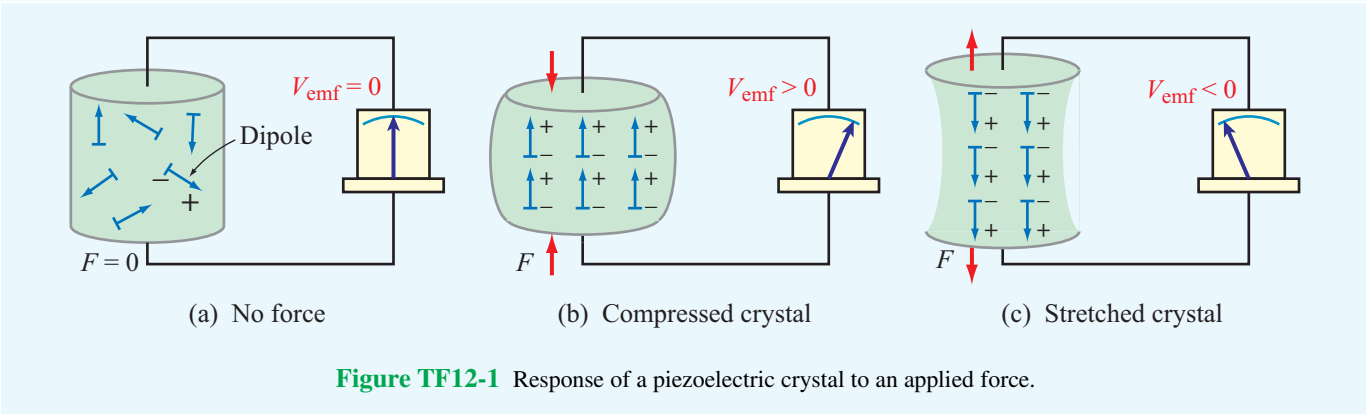
The crystal consists of polar domains represented by equivalent dipoles (Fig. TF12-1). Under the absence of an external force, the polar domains are randomly oriented throughout the material, but when **compressive** or **tensile** (stretching) stress is applied to the crystal, the polar domains align themselves along one of the principal axes of the crystal, leading to a net polarization (electric charge) at the crystal surfaces. Compression and stretching generate voltages of opposite polarity. The piezoelectric effect (**piezein** means to press or squeeze in Greek) was discovered by the **Curie brothers**, Pierre and Paul-Jacques, in 1880, and a year later, Lippmann predicted the converse property: If subjected to an electric field, the crystal would change in shape.

► The piezoelectric effect is a **reversible (bi-directional)** electromechanical process; application of force induces a voltage across the crystal, and conversely, application of a voltage changes the shape of the crystal. ◀

Piezoelectric crystals are used in **microphones** to convert mechanical vibrations (of the crystal surface) caused by acoustic waves into a corresponding electrical signal, and the converse process is used in **loudspeakers** to convert electrical signals into sound. In addition to having stiffness values comparable to that of steel, some piezoelectric materials exhibit very high sensitivity to the force applied upon them, with excellent linearity over a wide dynamic range. They can be used to measure surface deformations as small as **nanometers** (10^{-9} m), making them particularly attractive as positioning sensors in **scanning tunneling microscopes**. As **accelerometers**, they can measure acceleration levels as low as 10^{-4} g to as high as 100 g (where g is the acceleration due to gravity). Piezoelectric crystals and ceramics are used in cigarette lighters and gas grills as spark generators, in clocks and electronic circuitry as precision oscillators, in medical **ultrasound** diagnostic equipment as transducers (Fig. TF12-2), and in numerous other applications.

Faraday Magnetic Flux Sensor

According to Faraday's law [Eq. (6.6)], the emf voltage induced across the terminals of a conducting loop is directly proportional to the time rate of change of the



技术简介 12：EMF 传感器

电动势 (emf) 传感器是一种可以响应外部刺激而产生感应电压的装置。本技术简介介绍了三种类型的电动势传感器：压电传感器、法拉第磁通量传感器和热电偶。

压电换能器

► 压电是某些晶体（例如石英）所表现出的特性，当晶体受到机械压力时，晶体会发生电极化，从而在晶体上感应出电压。 ◀

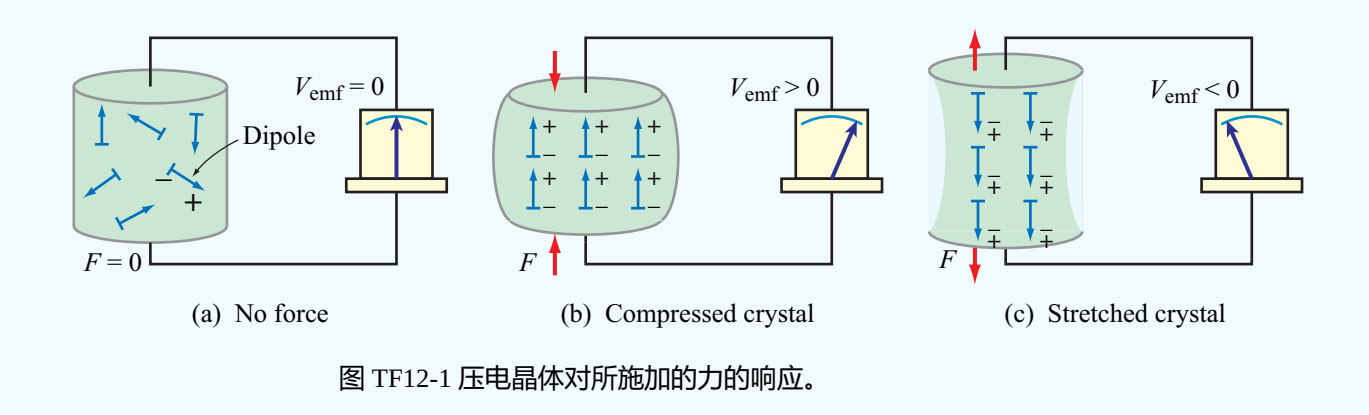
该晶体由由等效偶极子表示的极性域组成 (图TF12-1)。在没有外力的情况下，极域在整个材料中随机取向，但当向晶体施加压缩或拉伸应力时，极域沿着晶体的主轴之一排列，导致晶体表面出现净极化 (电荷)。压缩和拉伸会产生相反极性的电压。压电效应 (piezein 在希腊语中的意思是按压或挤压) 是由居里兄弟 Pierre 和 Paul-Jacques 于 1880 年发现的，一年后，Lippmann 预测了相反的特性：如果受到电场的影响，晶体的形状会发生变化。

► 压电效应是一种可逆 (双向) 机电过程；施加力会在晶体上感应出电压，相反，施加电压会改变晶体的形状。 ◀

压电晶体用于麦克风中，将声波引起的 (晶体表面的) 机械振动转换为相应的电信号，而相反的过程则用于扬声器中，将电信号转换为声音。除了具有与钢相当的刚度值之外，一些压电材料还对施加在其上的力表现出非常高的灵敏度，并且在宽动态范围内具有出色的线性度。它们可用于测量小至纳米 (10^{-9} m) 的表面变形，这使得它们作为扫描隧道显微镜中的定位传感器特别有吸引力。作为加速度计，它们可以测量低至 10^{-4} g 至高至 100 g 的加速度水平 (其中 g 是重力加速度)。压电晶体和陶瓷在打火机和燃气烤架中用作火花发生器，在时钟和电子电路中用作精密振荡器，在医疗超声诊断设备中用作传感器 (图TF12-2)，以及许多其他应用。

法拉第磁通传感器

根据法拉第定律 [Eq. (6.6)]，导电回路两端感应的电动势电压与电流变化率的时间成正比。



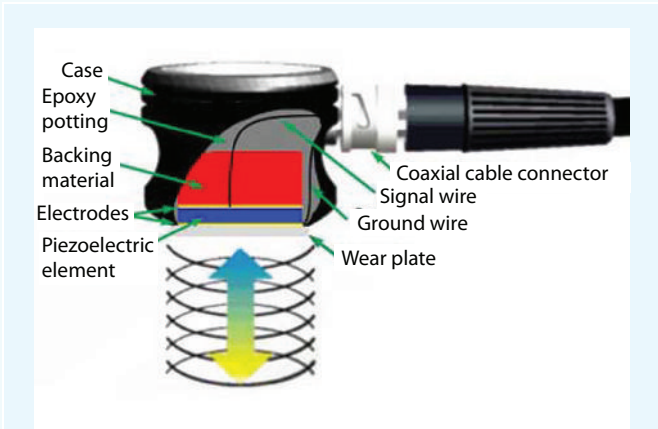


Figure TF12-2 The ultrasonic transducer uses piezoelectric crystals.

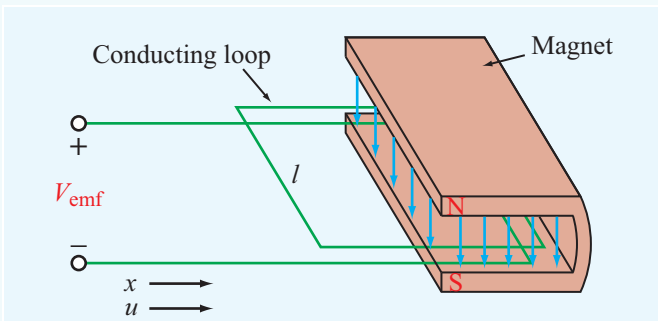


Figure TF12-3 In a Faraday accelerometer, the induced emf is directly proportional to the velocity of the loop (into and out of the magnet's cavity).

magnetic flux passing through the loop. For the configuration in Fig. TF12-3,

$$V_{\text{emf}} = -uB_0l,$$

where $u = dx/dt$ is the **velocity** of the loop (into or out of the magnet's cavity) with the direction of u defined as positive when the loop is moving inward into the cavity, B_0 is the magnetic field of the magnet, and l is the loop width. With B_0 and l being constant, the variation of $V_{\text{emf}}(t)$ with time t becomes a direct indicator of the time variation of $u(t)$. The time derivative of $u(t)$ provides the **acceleration** $a(t)$.

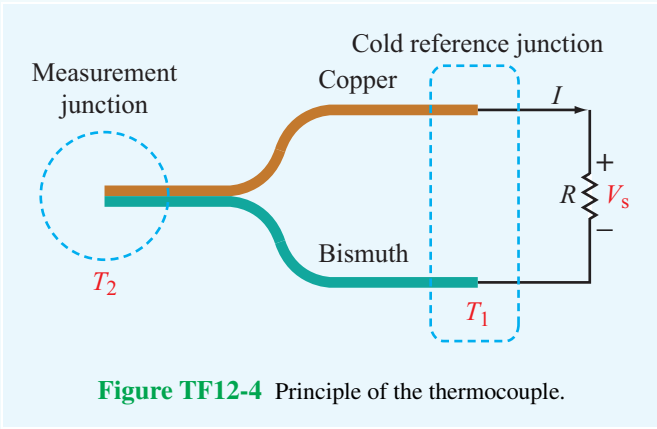
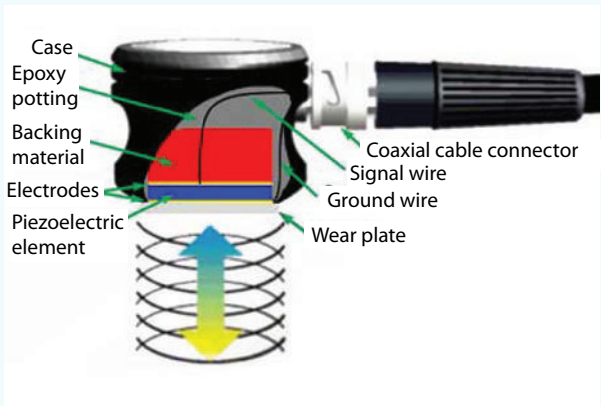


Figure TF12-4 Principle of the thermocouple.

Thermocouple

In 1821, **Thomas Seebeck** discovered that when a junction made of two different conducting materials, such as bismuth and copper, is heated it generates a thermally induced emf, which we now call the **Seebeck potential** V_S (Fig. TF12-4). When connected to a resistor, a current given by $I = V_S/R$ flows through the resistor.

This feature was advanced by **A. C. Becquerel** in 1826 as a means to measure the unknown temperature T_2 of a junction relative to a temperature T_1 of a (cold) reference junction. Today, such a generator of **thermoelectricity** is called a **thermocouple**. Initially, an ice bath was used to maintain T_1 at 0°C , but in today's temperature sensor designs, an artificial cold junction is used instead. The artificial junction is an electric circuit that generates a potential equal to that expected from a reference junction at temperature T_1 .



图TF12-2 超声波换能器采用压电晶体。

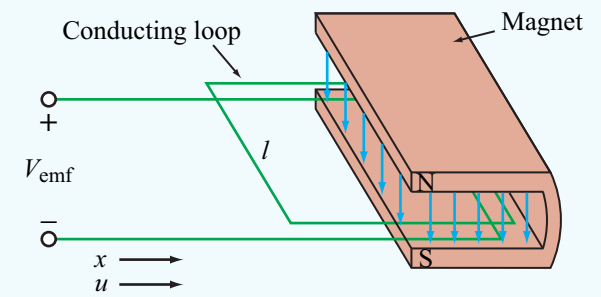
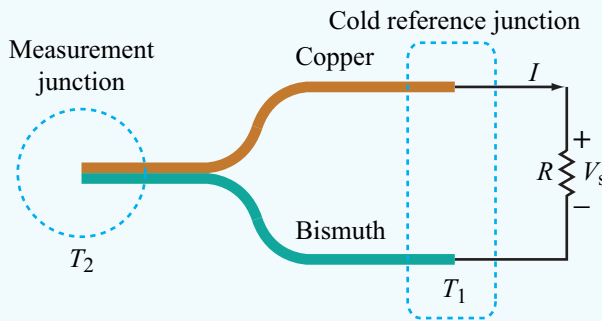


图 TF12-3 在法拉第加速度计中，感应电动势与回路速度（进出磁体腔）成正比。

通过环路的磁通量。对于图 TF12-3 中的配置，

$$V_{\text{emf}} = -uB_0l,$$

其中 $u = dx/dt$ 是环路的速度（进入或离开磁体空腔），当环路向内移动到空腔中时， u 的方向定义为正， B_0 是磁体的磁场， l 是环路宽度。当 B_0 和 l 恒定时， $V_{\text{emf}}(t)$ 随时间 t 的变化就成为 $u(t)$ 时间变化的直接指标。 $u(t)$ 的时间导数提供加速度 $a(t)$ 。



图TF12-4 热电偶原理。

热电偶

1821 年，托马斯·塞贝克 (Thomas Seebeck) 发现，当由两种不同的导电材料（例如铋和铜）制成的结被加热时，会产生热感应电动势，我们现在将其称为塞贝克电势 V_S (图 1. TF12-4)。当连接到电阻器时，由 $I = V_S/R$ 给出的电流流过电阻器。

这一功能由 A. C. Becquerel 于 1826 年提出，作为一种测量结点相对于（冷）参考结点温度 T_1 的未知温度 T_2 的方法。如今，这种热电发生器被称为热电偶。最初，使用冰浴将 T_1 维持在 0°C ，但在当今的温度传感器设计中，改用人工冷端。人工结是一种电路，它产生的电势等于温度 T_1 下参考结预期的电势。

Exercise 6-6: Determine (a) the relaxation time constant and (b) the time it takes for a charge density to decay to 1% of its initial value in quartz given that $\epsilon_r = 5$ and $\sigma = 10^{-17}$ S/m.

Answer: (a) $\tau_r = 51.2$ days, (b) 236 days. (See [EM](#).)

6-11 Electromagnetic Potentials

Our discussion of Faraday’s and Ampère’s laws revealed two aspects of the link between time-varying electric and magnetic fields. We now examine the implications of this interconnection on the electric scalar potential V and the vector magnetic potential $\mathbf{A}$.

In the static case, Faraday’s law reduces to

$$\nabla \times \mathbf{E} = 0, \quad \text{(static case)} \tag{6.62}$$

which states that the electrostatic field $\mathbf{E}$ is conservative. According to the rules of vector calculus, if a vector field $\mathbf{E}$ is conservative, it can be expressed as the gradient of a scalar. Hence, in Chapter 4, we defined $\mathbf{E}$ as

$$\mathbf{E} = -\nabla V. \quad \text{(electrostatics)} \tag{6.63}$$

In the dynamic case, Faraday’s law is

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}. \tag{6.64}$$

In view of the relation $\mathbf{B} = \nabla \times \mathbf{A}$, Eq. (6.64) can be expressed as

$$\nabla \times \mathbf{E} = -\frac{\partial}{\partial t} (\nabla \times \mathbf{A}), \tag{6.65}$$

which can be rewritten as

$$\nabla \times \left(\mathbf{E} + \frac{\partial \mathbf{A}}{\partial t} \right) = 0. \quad \text{(dynamic case)} \tag{6.66}$$

Let us for the moment define

$$\mathbf{E}' = \mathbf{E} + \frac{\partial \mathbf{A}}{\partial t}. \tag{6.67}$$

Using this definition, Eq. (6.66) becomes

$$\nabla \times \mathbf{E}' = 0. \tag{6.68}$$

Following the same logic that led to Eq. (6.63) from Eq. (6.62), we define

$$\mathbf{E}' = -\nabla V. \tag{6.69}$$

Upon substituting Eq. (6.67) for $\mathbf{E}'$ in Eq. (6.69) and then solving for $\mathbf{E}$, we have

$$\mathbf{E} = -\nabla V - \frac{\partial \mathbf{A}}{\partial t}. \quad \text{(dynamic case)} \tag{6.70}$$

Equation (6.70) reduces to Eq. (6.63) in the static case.

When the scalar potential V and the vector potential $\mathbf{A}$ are known, $\mathbf{E}$ can be obtained from Eq. (6.70), and $\mathbf{B}$ can be obtained from

$$\mathbf{B} = \nabla \times \mathbf{A}. \tag{6.71}$$

Next we examine the relations between the potentials, V and $\mathbf{A}$, and their sources: the charge and current distributions ρ_v and $\mathbf{J}$ in the time-varying case.

6-11.1 Retarded Potentials

Consider the situation depicted in [Fig. 6-16](#). A charge distribution ρ_v exists over a volume $\mathcal{V}'$ embedded in a perfect dielectric with permittivity ϵ . Were this a static charge distribution, then from Eq. (4.58a) the electric potential $V(\mathbf{R})$ at an observation point in space specified by the position vector $\mathbf{R}$ would be

$$V(\mathbf{R}) = \frac{1}{4\pi\epsilon} \int_{\mathcal{V}'} \frac{\rho_v(\mathbf{R}_i)}{R'} d\mathcal{V}', \tag{6.72}$$

where $\mathbf{R}_i$ denotes the position vector of an elemental volume $\Delta\mathcal{V}'$ containing charge density $\rho_v(\mathbf{R}_i)$, and $R' = |\mathbf{R} - \mathbf{R}_i|$ is the distance between $\Delta\mathcal{V}'$ and the observation point. If the charge

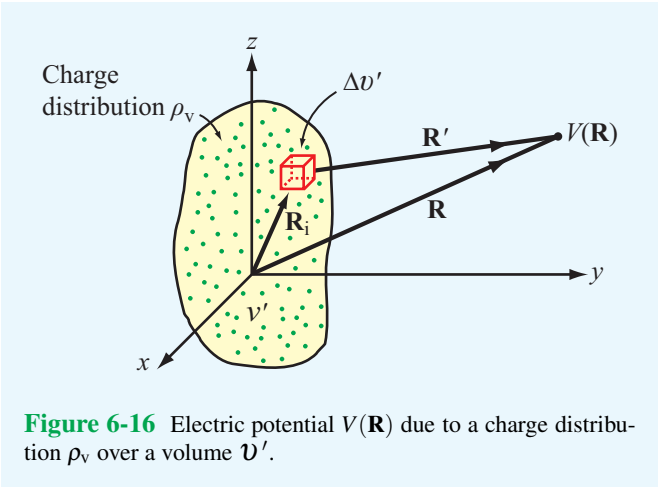


Figure 6-16 Electric potential $V(\mathbf{R})$ due to a charge distribution ρ_v over a volume $\mathcal{V}'$.

练习 6-6: 在 $\epsilon_r = 5$ 和 $\sigma = 10^{-17}$ S/m 的情况下, 确定 (a) 弛豫时间常数和 (b) 石英中电荷密度衰减到其初始值的 1% 所需的时间。

答案: (a) $\tau_r = 51.2$ 天, (b) 236 天。 (参见 [EM](#).)

6-11 电磁势

对法拉第定律和安培–麦克斯韦定律的讨论, 揭示了时变电场与磁场相互联系的两个方面。下面研究这种联系对电标量势 V 和磁矢势 $\mathbf{A}$ 的影响。

在静态情况下, 法拉第定律简化为

$$\nabla \times \mathbf{E} = 0, \quad \text{(static case)} \tag{6.62}$$

这说明静电场 $\mathbf{E}$ 是保守的。根据向量微积分的规则, 如果向量场 $\mathbf{E}$ 是保守的, 它可以表示为标量的梯度。因此, 在第 4 章中, 我们将 $\mathbf{E}$ 定义为

$$\mathbf{E} = -\nabla V. \quad \text{(electrostatics)} \tag{6.63}$$

在动态情况下, 法拉第定律为

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}. \tag{6.64}$$

鉴于关系 $\mathbf{B} = \nabla \times \mathbf{A}$, 等式 (6.64) 可以表示为

$$\nabla \times \mathbf{E} = -\frac{\partial}{\partial t} (\nabla \times \mathbf{A}), \tag{6.65}$$

可以重写为

$$\nabla \times \left(\mathbf{E} + \frac{\partial \mathbf{A}}{\partial t} \right) = 0. \quad \text{(dynamic case)} \tag{6.66}$$

让我们暂时定义

$$\mathbf{E}' = \mathbf{E} + \frac{\partial \mathbf{A}}{\partial t}. \tag{6.67}$$

使用这个定义, 方程 (6.66) 变为

$$\nabla \times \mathbf{E}' = 0. \tag{6.68}$$

遵循与方程相同的逻辑。 (6.63) 来自方程式 (6.62), 我们定义

$$\mathbf{E}' = -\nabla V. \tag{6.69}$$

代入方程式后。 (6.67) 对于方程中的 $\mathbf{E}'$ (6.69) 然后求解 $\mathbf{E}$, 我们有

$$\mathbf{E} = -\nabla V - \frac{\partial \mathbf{A}}{\partial t}. \quad \text{(dynamic case)} \tag{6.70}$$

方程 (6.70) 简化为方程 (6.63) 在静态情况下。当标量势 V 和矢量势 $\mathbf{A}$ 已知时, 可由式 (1) 求出 $\mathbf{E}$ 。 (6.70), $\mathbf{B}$ 可由下式获得

$$\mathbf{B} = \nabla \times \mathbf{A}. \tag{6.71}$$

接下来, 我们检查电势 V 和 $\mathbf{A}$ 及其来源之间的关系: 时变情况下的电荷和电流分布 ρ_v 和 $\mathbf{J}$ 。

6-11.1 迟滞电位

考虑图 6-16 中描述的情况。电荷分布 ρ_v 存在于嵌入介电常数为 ϵ 的理想电介质中的体积 $\mathcal{V}'$ 上。如果这是一个静电电荷分布, 则可以从方程式得出: (4.58a) 由位置向量 $\mathbf{R}$ 指定的空间观察点处的电势 $V(\mathbf{R})$ 为

$$V(\mathbf{R}) = \frac{1}{4\pi\epsilon} \int_{\mathcal{V}'} \frac{\rho_v(\mathbf{R}_i)}{R'} d\mathcal{V}', \tag{6.72}$$

其中, $\mathbf{R}_i$ 表示包含电荷密度 $\rho_v(\mathbf{R}_i)$ 的单元体积 $\Delta\mathcal{V}'$ 的位置向量, $R' = |\mathbf{R} - \mathbf{R}_i|$ 是 $\Delta\mathcal{V}'$ 与观测点之间的距离。如果电荷

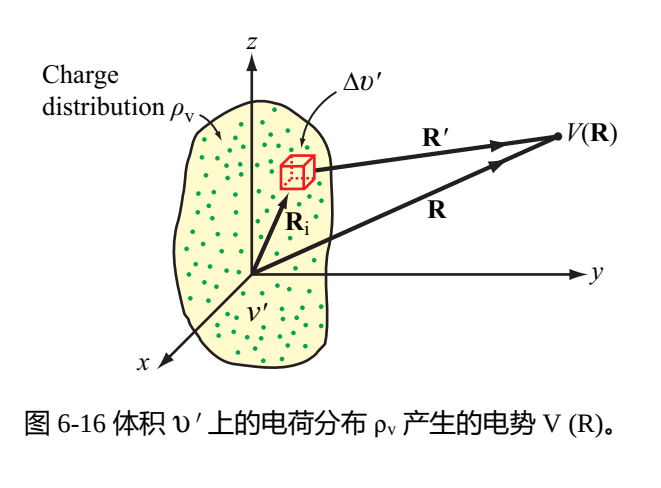


图 6-16 体积 $\mathcal{V}'$ 上的电荷分布 ρ_v 产生的电势 $V(\mathbf{R})$ 。

distribution is time-varying, we may be tempted to rewrite Eq. (6.72) for the dynamic case as

$$V(\mathbf{R}, t) = \frac{1}{4\pi\epsilon} \int_{\mathbf{v}'} \frac{\rho_v(\mathbf{R}_i, t)}{R'} d\mathbf{v}', \quad (6.73)$$

but such a form does not account for “reaction time.” If V_1 is the potential due to a certain distribution ρ_{v1} and ρ_{v1} were to suddenly change to ρ_{v2} , it will take a finite amount of time before V_1 a distance R' away changes to V_2 . *In other words, $V(\mathbf{R}, t)$ cannot change instantaneously.* The delay time is equal to $t' = R'/u_p$, where u_p is the velocity of propagation in the medium between the charge distribution and the observation point. Thus, $V(\mathbf{R}, t)$ at time t corresponds to ρ_v at an earlier time, that is, $(t - t')$. Hence, Eq. (6.73) should be rewritten as

$$V(\mathbf{R}, t) = \frac{1}{4\pi\epsilon} \int_{\mathbf{v}'} \frac{\rho_v(\mathbf{R}_i, t - R'/u_p)}{R'} d\mathbf{v}' \quad (\text{V}), \quad (6.74)$$

and $V(\mathbf{R}, t)$ is appropriately called the **retarded scalar potential**. If the propagation medium is vacuum, u_p is equal to the velocity of light c .

Similarly, the **retarded vector potential** $\mathbf{A}(\mathbf{R}, t)$ is related to the distribution of current density $\mathbf{J}$ by

$$\mathbf{A}(\mathbf{R}, t) = \frac{\mu}{4\pi} \int_{\mathbf{v}'} \frac{\mathbf{J}(\mathbf{R}_i, t - R'/u_p)}{R'} d\mathbf{v}' \quad (\text{Wb/m}). \quad (6.75)$$

This expression is obtained by extending the expression for the magnetostatic vector potential $\mathbf{A}(\mathbf{R})$ given by Eq. (5.65) to the time-varying case.

6-11.2 Time-Harmonic Potentials

The expressions given by Eqs. (6.74) and (6.75) for the retarded scalar and vector potentials are valid under both static and dynamic conditions and for any type of time dependence of the source functions ρ_v and $\mathbf{J}$. Because V and $\mathbf{A}$ depend linearly on ρ_v and $\mathbf{J}$, and as $\mathbf{E}$ and $\mathbf{B}$ depend linearly on V and $\mathbf{A}$, the relationships interconnecting all of these quantities obey the rules of linear systems. When analyzing linear systems, we can take advantage of sinusoidal-time functions to determine the system’s response to a source with arbitrary time dependence. As was noted in Section 1-7, if the time dependence is described by a (nonsinusoidal) periodic time function, it can always be expanded into a Fourier series of sinusoidal components, and if the time function is nonperiodic,

it can be represented by a Fourier integral. In either case, if the response of the linear system is known for all steady-state sinusoidal excitations, the principle of superposition can be used to determine its response to an excitation with arbitrary time dependence. Thus, the sinusoidal response of the system constitutes a fundamental building block that can be used to determine the response due to a source described by an arbitrary function of time. The term **time-harmonic** is often used in this context as a synonym for “steady-state sinusoidal time-dependent.”

In this subsection, we derive expressions for the scalar and vector potentials due to time-harmonic sources. Suppose that $\rho_v(\mathbf{R}_i, t)$ is a sinusoidal-time function with angular frequency ω given by

$$\rho_v(\mathbf{R}_i, t) = \rho_v(\mathbf{R}_i) \cos(\omega t + \phi). \quad (6.76)$$

Phasor analysis, which was first introduced in Section 1-7 and then used extensively in Chapter 2 to study wave propagation on transmission lines, is a useful tool for analyzing time-harmonic scenarios. A time harmonic charge distribution $\rho_v(\mathbf{R}_i, t)$ is related to its phasor $\tilde{\rho}_v(\mathbf{R}_i)$ as

$$\rho_v(\mathbf{R}_i, t) = \Re \left[\tilde{\rho}_v(\mathbf{R}_i) e^{j\omega t} \right], \quad (6.77)$$

Comparison of Eqs. (6.76) and (6.77) shows that in the present case $\tilde{\rho}_v(\mathbf{R}_i) = \rho_v(\mathbf{R}_i) e^{j\phi}$.

Next, we express the retarded charge density $\rho_v(\mathbf{R}_i, t - R'/u_p)$ in phasor form by replacing t with $(t - R'/u_p)$ in Eq. (6.77):

$$\begin{aligned} \rho_v(\mathbf{R}_i, t - R'/u_p) &= \Re \left[\tilde{\rho}_v(\mathbf{R}_i) e^{j\omega(t - R'/u_p)} \right] \\ &= \Re \left[\tilde{\rho}_v(\mathbf{R}_i) e^{-j\omega R'/u_p} e^{j\omega t} \right] \\ &= \Re \left[\tilde{\rho}_v(\mathbf{R}_i) e^{-jkR'} e^{j\omega t} \right], \end{aligned} \quad (6.78)$$

where

$$k = \frac{\omega}{u_p} \quad (6.79)$$

is called the **wavenumber** or phase constant of the propagation medium. (In general, the phase constant is denoted by the symbol “ β ”, but for lossless dielectric media, it is commonly denoted by the symbol “ k ” and called the wavenumber.) Similarly, we define the phasor $\tilde{V}(\mathbf{R})$ of the time function $V(\mathbf{R}, t)$ according to

$$V(\mathbf{R}, t) = \Re \left[\tilde{V}(\mathbf{R}) e^{j\omega t} \right]. \quad (6.80)$$

分布是随时间变化的，我们可能会想重写方程。（6.72）对于动态情况为

$$V(\mathbf{R}, t) = \frac{1}{4\pi\epsilon} \int_{\mathbf{v}'} \frac{\rho_v(\mathbf{R}_i, t)}{R'} d\mathbf{v}', \quad (6.73)$$

但这种形式并没有考虑“反应时间”。如果 V_1 是某个分布 ρ_{v1} 产生的势，并且 ρ_{v1} 突然变为 ρ_{v2} ，则距离 R' 远的 V_1 变为 V_2 需要有限的时间。换句话说， $V(\mathbf{R}, t)$ 不能瞬时改变。延迟时间等于 $t' = R'/u_p$ ，其中 u_p 是电荷分布和观察点之间的介质中的传播速度。因此，时间 t 处的 $V(\mathbf{R}, t)$ 对应于较早时间处的 ρ_v ，即 $(t - t')$ 。因此，等式（6.73）应重写为

$$V(\mathbf{R}, t) = \frac{1}{4\pi\epsilon} \int_{\mathbf{v}'} \frac{\rho_v(\mathbf{R}_i, t - R'/u_p)}{R'} d\mathbf{v}' \quad (\text{V}), \quad (6.74)$$

$V(\mathbf{R}, t)$ 被适当地称为延迟标量势。如果传播介质是真空， u_p 等于光速 c 。

类似地，延迟矢量势 $\mathbf{A}(\mathbf{R}, t)$ 与电流密度 $\mathbf{J}$ 的分布相关：

$$\mathbf{A}(\mathbf{R}, t) = \frac{\mu}{4\pi} \int_{\mathbf{v}'} \frac{\mathbf{J}(\mathbf{R}_i, t - R'/u_p)}{R'} d\mathbf{v}' \quad (\text{Wb/m}). \quad (6.75)$$

该表达式是通过扩展方程 1 给出的静磁矢量势 $\mathbf{A}(\mathbf{R})$ 的表达式而获得的。（5.65）到时变的情况。

6-11.2 时谐势

等式给出的表达式。对于延迟标量势和矢量势，方程 (6.74) 和 (6.75) 在静态和动态条件下以及对于源函数 ρ_v 和 $\mathbf{J}$ 的任何类型的时间依赖性都有效。因为 V 和 $\mathbf{A}$ 线性依赖于 ρ_v 和 $\mathbf{J}$ ，并且 $\mathbf{E}$ 和 $\mathbf{B}$ 线性依赖于 V 和 $\mathbf{A}$ ，所以所有这些量之间的相互关系遵循线性系统的规则。在分析线性系统时，我们可以利用正弦时间函数来确定系统对具有任意时间依赖性的源的响应。如第 1-7 节所述，如果时间依赖性由（非正弦）周期时间函数描述，则它始终可以展开为正弦分量的傅立叶级数，并且如果时间函数是非周期的，

它可以用傅里叶积分来表示。在任一情况下，如果线性系统的响应对于所有稳态正弦激励都是已知的，则可以使用叠加原理来确定其对具有任意时间依赖性的激励的响应。因此，系统的正弦响应构成了一个基本构建块，可用于确定由任意时间函数描述的源引起的响应。在这种情况下，术语“时谐”通常用作“稳态正弦时间相关”的同义词。

在本小节中，我们推导了时谐源引起的标量势和矢量势的表达式。假设 $\rho_v(\mathbf{R}_i, t)$ 是正弦时间函数，角频率 ω 为

$$\rho_v(\mathbf{R}_i, t) = \rho_v(\mathbf{R}_i) \cos(\omega t + \phi). \quad (6.76)$$

相量分析首先在第 1-7 节中介绍，然后在第 2 章中广泛用于研究传输线上的波传播，它是分析时谐波场景的有用工具。时间谐波电荷分布 $\rho_v(\mathbf{R}_i, t)$ 与其相量 $\tilde{\rho}_v(\mathbf{R}_i)$ 相关，如下所示

$$\rho_v(\mathbf{R}_i, t) = \Re \left[\tilde{\rho}_v(\mathbf{R}_i) e^{j\omega t} \right], \quad (6.77)$$

等式的比较(6.76) 和 (6.77) 表明在本例中 $\tilde{\rho}_v(\mathbf{R}_i) = \rho_v(\mathbf{R}_i) e^{j\phi}$ 。

接下来，我们将等式中的 t 替换为 $(t - R'/u_p)$ ，以相量形式表示延迟电荷密度 $\rho_v(\mathbf{R}_i, t - R'/u_p)$ 。（6.77）：

$$\begin{aligned} \rho_v(\mathbf{R}_i, t - R'/u_p) &= \Re \left[\tilde{\rho}_v(\mathbf{R}_i) e^{j\omega(t - R'/u_p)} \right] \\ &= \Re \left[\tilde{\rho}_v(\mathbf{R}_i) e^{-j\omega R'/u_p} e^{j\omega t} \right] \\ &= \Re \left[\tilde{\rho}_v(\mathbf{R}_i) e^{-jkR'} e^{j\omega t} \right], \end{aligned} \quad (6.78)$$

在哪里

$$k = \frac{\omega}{u_p} \quad (6.79)$$

称为传播介质的波数或相位常数。（一般情况下，相位常数用符号“ β ”表示，但对于无损电介质，通常用符号“ k ”表示，称为波数。）类似地，我们根据下式定义时间函数 $V(\mathbf{R}, t)$ 的相量 $\tilde{V}(\mathbf{R})$ ：

$$V(\mathbf{R}, t) = \Re \left[\tilde{V}(\mathbf{R}) e^{j\omega t} \right]. \quad (6.80)$$

Using Eqs. (6.78) and (6.80) in Eq. (6.74) gives

$$\Re \left[\tilde{V}(\mathbf{R}) e^{j\omega t} \right] = \Re \left[\frac{1}{4\pi\epsilon} \int_{\mathcal{V}'} \frac{\tilde{\rho}_v(\mathbf{R}_i) e^{-jkR'}}{R'} e^{j\omega t} d\mathcal{V}' \right]. \quad (6.81)$$

By equating the quantities inside the square brackets on both sides of Eq. (6.81) and canceling the common $e^{j\omega t}$ factor, we obtain the phasor-domain expression

$$\tilde{V}(\mathbf{R}) = \frac{1}{4\pi\epsilon} \int_{\mathcal{V}'} \frac{\tilde{\rho}_v(\mathbf{R}_i) e^{-jkR'}}{R'} d\mathcal{V}' \quad (\text{V}). \quad (6.82)$$

For any given charge distribution, Eq. (6.82) can be used to compute $\tilde{V}(\mathbf{R})$. Then the resultant expression can be used in Eq. (6.80) to find $V(\mathbf{R}, t)$. Similarly, the expression for $\mathbf{A}(\mathbf{R}, t)$ given by Eq. (6.75) can be transformed into

$$\mathbf{A}(\mathbf{R}, t) = \Re \left[\tilde{\mathbf{A}}(\mathbf{R}) e^{j\omega t} \right] \quad (6.83)$$

with

$$\tilde{\mathbf{A}}(\mathbf{R}) = \frac{\mu}{4\pi} \int_{\mathcal{V}'} \frac{\tilde{\mathbf{J}}(\mathbf{R}_i) e^{-jkR'}}{R'} d\mathcal{V}', \quad (6.84)$$

where $\tilde{\mathbf{J}}(\mathbf{R}_i)$ is the phasor function corresponding to $\mathbf{J}(\mathbf{R}_i, t)$. The magnetic field phasor $\tilde{\mathbf{H}}$ corresponding to $\tilde{\mathbf{A}}$ is given by

$$\tilde{\mathbf{H}} = \frac{1}{\mu} \nabla \times \tilde{\mathbf{A}}. \quad (6.85)$$

Recall that differentiation in the time domain is equivalent to multiplication by $j\omega$ in the phasor domain, and in a nonconducting medium ($\mathbf{J} = 0$), so Ampère’s law given by Eq. (6.41) becomes

$$\nabla \times \tilde{\mathbf{H}} = j\omega\epsilon\tilde{\mathbf{E}} \quad \text{or} \quad \tilde{\mathbf{E}} = \frac{1}{j\omega\epsilon} \nabla \times \tilde{\mathbf{H}}. \quad (6.86)$$

Hence, given a time-harmonic, current-density distribution with phasor $\tilde{\mathbf{J}}$, Eqs. (6.84) to (6.86) can be used successively

to determine both $\tilde{\mathbf{E}}$ and $\tilde{\mathbf{H}}$. The phasor vectors $\tilde{\mathbf{E}}$ and $\tilde{\mathbf{H}}$ also are related by the phasor form of Faraday’s law:

$$\begin{aligned} \nabla \times \tilde{\mathbf{E}} &= -j\omega\mu\tilde{\mathbf{H}} \\ \text{or} \quad \tilde{\mathbf{H}} &= -\frac{1}{j\omega\mu} \nabla \times \tilde{\mathbf{E}}. \end{aligned} \quad (6.87)$$

Example 6-8: Relating E to H

In a nonconducting medium with $\epsilon = 16\epsilon_0$ and $\mu = \mu_0$, the electric field intensity of an electromagnetic wave is

$$\mathbf{E}(z, t) = \hat{\mathbf{x}} 10 \sin(10^{10}t - kz) \quad (\text{V/m}). \quad (6.88)$$

Determine the associated magnetic field intensity $\mathbf{H}$ and find the value of k .

Solution: We begin by finding the phasor $\tilde{\mathbf{E}}(z)$ of $\mathbf{E}(z, t)$. Since $\mathbf{E}(z, t)$ is given as a sine function and phasors are defined in this book with reference to the cosine function, we rewrite Eq. (6.88) as

$$\begin{aligned} \mathbf{E}(z, t) &= \hat{\mathbf{x}} 10 \cos(10^{10}t - kz - \pi/2) \quad (\text{V/m}) \\ &= \Re \left[\tilde{\mathbf{E}}(z) e^{j\omega t} \right], \end{aligned} \quad (6.89)$$

with $\omega = 10^{10}$ (rad/s) and

$$\begin{aligned} \tilde{\mathbf{E}}(z) &= \hat{\mathbf{x}} 10 e^{-jkz} e^{-j\pi/2} \\ &= -\hat{\mathbf{x}} j 10 e^{-jkz}. \end{aligned} \quad (6.90)$$

To find both $\tilde{\mathbf{H}}(z)$ and k , we will perform a “circle”: We will use the given expression for $\tilde{\mathbf{E}}(z)$ in Faraday’s law to find $\tilde{\mathbf{H}}(z)$; then we will use $\tilde{\mathbf{H}}(z)$ in Ampère’s law to find $\tilde{\mathbf{E}}(z)$, which we will then compare with the original expression for $\tilde{\mathbf{E}}(z)$; and the comparison will yield the value of k . Application of Eq. (6.87) gives

$$\begin{aligned} \tilde{\mathbf{H}}(z) &= -\frac{1}{j\omega\mu} \nabla \times \tilde{\mathbf{E}} \\ &= -\frac{1}{j\omega\mu} \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ -j10e^{-jkz} & 0 & 0 \end{vmatrix} \\ &= -\frac{1}{j\omega\mu} \left[\hat{\mathbf{y}} \frac{\partial}{\partial z} (-j10e^{-jkz}) \right] \\ &= -\hat{\mathbf{y}} j \frac{10k}{\omega\mu} e^{-jkz}. \end{aligned} \quad (6.91)$$

使用方程式。式 (6.78) 和 (6.80) (6.74) 给出

$$\Re \left[\tilde{V}(\mathbf{R}) e^{j\omega t} \right] = \Re \left[\frac{1}{4\pi\epsilon} \int_{\mathcal{V}'} \frac{\tilde{\rho}_v(\mathbf{R}_i) e^{-jkR'}}{R'} e^{j\omega t} d\mathcal{V}' \right]. \quad (6.81)$$

通过使方程两边方括号内的数量相等： (6.81) 并取消公共 $e^{j\omega t}$ 因子，我们得到相量域表达式

$$\tilde{V}(\mathbf{R}) = \frac{1}{4\pi\epsilon} \int_{\mathcal{V}'} \frac{\tilde{\rho}_v(\mathbf{R}_i) e^{-jkR'}}{R'} d\mathcal{V}' \quad (\text{V}). \quad (6.82)$$

对于任何给定的电荷分布，方程。 (6.82) 可用于计算 $\tilde{V}(\mathbf{R})$ 。然后得到的表达式可以在等式中使用。 (6.80) 求 $V(\mathbf{R}, t)$ 。类似地， $\mathbf{A}(\mathbf{R}, t)$ 的表达式由方程式给出： (6.75) 可以转化为

$$\mathbf{A}(\mathbf{R}, t) = \Re \left[\tilde{\mathbf{A}}(\mathbf{R}) e^{j\omega t} \right] \quad (6.83)$$

和

$$\tilde{\mathbf{A}}(\mathbf{R}) = \frac{\mu}{4\pi} \int_{\mathcal{V}'} \frac{\tilde{\mathbf{J}}(\mathbf{R}_i) e^{-jkR'}}{R'} d\mathcal{V}', \quad (6.84)$$

其中 $\tilde{\mathbf{J}}(\mathbf{R}_i)$ 是对应于 $\mathbf{J}(\mathbf{R}_i, t)$ 的相量函数。对应于 $\tilde{\mathbf{A}}$ 的磁场相量 $\tilde{\mathbf{H}}$ 由下式给出

$$\tilde{\mathbf{H}} = \frac{1}{\mu} \nabla \times \tilde{\mathbf{A}}. \quad (6.85)$$

回想一下，时域中的微分相当于相量域中和非导电介质 ($\mathbf{J} = 0$) 中乘以 $j\omega$ ，因此安培定律 由方程 1 给出。 (6.41) 变为

$$\nabla \times \tilde{\mathbf{H}} = j\omega\epsilon\tilde{\mathbf{E}} \quad \text{or} \quad \tilde{\mathbf{E}} = \frac{1}{j\omega\epsilon} \nabla \times \tilde{\mathbf{H}}. \quad (6.86)$$

因此，给定具有相量 $\tilde{\mathbf{J}}$ 的时谐波电流密度分布，方程： (6.84)至(6.86)可依次使用

确定 $\tilde{\mathbf{E}}$ 和 $\tilde{\mathbf{H}}$ 。相量向量 $\tilde{\mathbf{E}}$ 和 $\tilde{\mathbf{H}}$ 也通过法拉第定律的相量形式相关：

$$\begin{aligned} \nabla \times \tilde{\mathbf{E}} &= -j\omega\mu\tilde{\mathbf{H}} \\ \text{or} \quad \tilde{\mathbf{H}} &= -\frac{1}{j\omega\mu} \nabla \times \tilde{\mathbf{E}}. \end{aligned} \quad (6.87)$$

例 6-8：将 E 与 H 联系起来

在具有 $\epsilon = 16\epsilon_0$ 和 $\mu = \mu_0$ 的非导电介质中，电磁波的电场强度为

$$\mathbf{E}(z, t) = \hat{\mathbf{x}} 10 \sin(10^{10}t - kz) \quad (\text{V/m}). \quad (6.88)$$

确定相关的磁场强度 $\mathbf{H}$ 并求出 k 值。

解：我们首先找到 $\mathbf{E}(z, t)$ 的相量 $\tilde{\mathbf{E}}(z)$ 。由于 $\mathbf{E}(z, t)$ 是作为正弦函数给出的，而本书中相量是参考余弦函数定义的，因此我们重写方程： (6.88) 作为

$$\begin{aligned} \mathbf{E}(z, t) &= \hat{\mathbf{x}} 10 \cos(10^{10}t - kz - \pi/2) \quad (\text{V/m}) \\ &= \Re \left[\tilde{\mathbf{E}}(z) e^{j\omega t} \right], \end{aligned} \quad (6.89)$$

$\omega = 10^{10}$ (rad/s) 和

$$\begin{aligned} \tilde{\mathbf{E}}(z) &= \hat{\mathbf{x}} 10 e^{-jkz} e^{-j\pi/2} \\ &= -\hat{\mathbf{x}} j 10 e^{-jkz}. \end{aligned} \quad (6.90)$$

为了找到 $\tilde{\mathbf{H}}(z)$ 和 k ，我们将执行一个 “循环”：我们将使用法拉第定律中给定的 $\tilde{\mathbf{E}}(z)$ 表达式来找到 $\tilde{\mathbf{H}}(z)$ ；然后我们将使用安培定律 中的 $\tilde{\mathbf{H}}(z)$ 来找到 $\tilde{\mathbf{E}}(z)$ ，然后将其与 $\tilde{\mathbf{E}}(z)$ 的原始表达式进行比较；比较将得出 k 的值。方程的应用(6.87) 给出

$$\begin{aligned} \tilde{\mathbf{H}}(z) &= -\frac{1}{j\omega\mu} \nabla \times \tilde{\mathbf{E}} \\ &= -\frac{1}{j\omega\mu} \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ -j10e^{-jkz} & 0 & 0 \end{vmatrix} \\ &= -\frac{1}{j\omega\mu} \left[\hat{\mathbf{y}} \frac{\partial}{\partial z} (-j10e^{-jkz}) \right] \\ &= -\hat{\mathbf{y}} j \frac{10k}{\omega\mu} e^{-jkz}. \end{aligned} \quad (6.91)$$

So far, we have used Eq. (6.90) for $\tilde{\mathbf{E}}(z)$ to find $\tilde{\mathbf{H}}(z)$, but k remains unknown. To find k , we use $\tilde{\mathbf{H}}(z)$ in Eq. (6.86) to find $\tilde{\mathbf{E}}(z)$:

$$\begin{aligned}\tilde{\mathbf{E}}(z) &= \frac{1}{j\omega\epsilon}\nabla\times\tilde{\mathbf{H}} \\ &= \frac{1}{j\omega\epsilon}\left[-\hat{\mathbf{x}}\frac{\partial}{\partial z}\left(-j\frac{10k}{\omega\mu}e^{-jkz}\right)\right] \\ &= -\hat{\mathbf{x}}j\frac{10k^2}{\omega^2\mu\epsilon}e^{-jkz}.\end{aligned}\tag{6.92}$$

Equating Eqs. (6.90) and (6.92) leads to

$$k^2 = \omega^2\mu\epsilon,$$

or

$$\begin{aligned}k &= \omega\sqrt{\mu\epsilon} \\ &= 4\omega\sqrt{\mu_0\epsilon_0} \\ &= \frac{4\omega}{c} = \frac{4\times 10^{10}}{3\times 10^8} = 133\text{ (rad/m)}.\end{aligned}\tag{6.93}$$

With k known, the instantaneous magnetic field intensity is then given by

$$\begin{aligned}\mathbf{H}(z,t) &= \Re\left[\tilde{\mathbf{H}}(z)e^{j\omega t}\right] \\ &= \Re\left[-\hat{\mathbf{y}}j\frac{10k}{\omega\mu}e^{-jkz}e^{j\omega t}\right] \\ &= \hat{\mathbf{y}}0.11\sin(10^{10}t - 133z)\text{ (A/m)}.\end{aligned}\tag{6.94}$$

We note that k has the same expression as the phase constant of a lossless transmission line [Eq. (2.49)].

Exercise 6-7: The magnetic field intensity of an electromagnetic wave propagating in a lossless medium with $\epsilon = 9\epsilon_0$ and $\mu = \mu_0$ is

$$\mathbf{H}(z,t) = \hat{\mathbf{x}}0.3\cos(10^8t - kz + \pi/4)\text{ (A/m)}.$$

Find $\mathbf{E}(z,t)$ and k .

Answer: $\mathbf{E}(z,t) = -\hat{\mathbf{y}}37.7\cos(10^8t - z + \pi/4)$ (V/m); $k = 1$ (rad/m). (See [EM](#).)

Chapter 6 Summary

Concepts

- Faraday’s law states that a voltage is induced across the terminals of a loop if the magnetic flux linking its surface changes with time.
- In an ideal transformer, the ratios of the primary to secondary voltages, currents, and impedances are governed by the turns ratio.
- Displacement current accounts for the “apparent” flow of charges through a dielectric. In reality, charges of opposite polarity accumulate along the two ends of a dielectric, giving the appearance of current flow through it.
- Boundary conditions for the electromagnetic fields at the interface between two different media are the same for both static and dynamic conditions.
- The charge continuity equation is a mathematical statement of the law of conservation of electric charge.
- Excess charges in the interior of a good conductor dissipate very quickly; through a rearrangement process, the excess charge is transferred to the surface of the conductor.
- In the dynamic case, the electric field $\mathbf{E}$ is related to both the scalar electric potential V and the magnetic vector potential $\mathbf{A}$.
- The retarded scalar and vector potentials at a given observation point take into account the finite time required for propagation between their sources, the charge and current distributions, and the location of the observation point.

到目前为止, 我们已经使用了等式。 (6.90) 让 $\tilde{\mathbf{E}}(z)$ 找到 $\tilde{\mathbf{H}}(z)$, 但 k 仍然未知。为了找到 k , 我们在等式中使用 $\tilde{\mathbf{H}}(z)$ 。 (6.86) 求 $\tilde{\mathbf{E}}(z)$:

$$\begin{aligned}\tilde{\mathbf{E}}(z) &= \frac{1}{j\omega\epsilon}\nabla\times\tilde{\mathbf{H}} \\ &= \frac{1}{j\omega\epsilon}\left[-\hat{\mathbf{x}}\frac{\partial}{\partial z}\left(-j\frac{10k}{\omega\mu}e^{-jkz}\right)\right] \\ &= -\hat{\mathbf{x}}j\frac{10k^2}{\omega^2\mu\epsilon}e^{-jkz}.\end{aligned}\tag{6.92}$$

等式(6.90) 和 (6.92) 导致

$$k^2 = \omega^2\mu\epsilon,$$

或者

$$\begin{aligned}k &= \omega\sqrt{\mu\epsilon} \\ &= 4\omega\sqrt{\mu_0\epsilon_0} \\ &= \frac{4\omega}{c} = \frac{4\times 10^{10}}{3\times 10^8} = 133\text{ (rad/m)}.\end{aligned}\tag{6.93}$$

当 k 已知时, 瞬时磁场强度由下式给出

$$\begin{aligned}\mathbf{H}(z,t) &= \Re\left[\tilde{\mathbf{H}}(z)e^{j\omega t}\right] \\ &= \Re\left[-\hat{\mathbf{y}}j\frac{10k}{\omega\mu}e^{-jkz}e^{j\omega t}\right] \\ &= \hat{\mathbf{y}}0.11\sin(10^{10}t - 133z)\text{ (A/m)}.\end{aligned}\tag{6.94}$$

我们注意到 k 与无损传输线的相位常数具有相同的表达式 [方程 1]。 (2.49)]。

练习 6-7: 在具有 $\epsilon = 9\epsilon_0$ 和 $\mu = \mu_0$ 的无损介质中传播的电磁波的磁场强度为

$$\mathbf{H}(z,t) = \hat{\mathbf{x}}0.3\cos(10^8t - kz + \pi/4)\text{ (A/m)}.$$

求 $\mathbf{E}(z,t)$ 和 k 。

答案: $\mathbf{E}(z,t) = -\hat{\mathbf{y}}37.7\cos(10^8t - z + \pi/4)$ (V/m); $k = 1$ (rad/m)。 (参见^{EM}。)

第六章总结

概念

- 法拉第定律指出, 如果连接环路表面的磁通量随时间变化, 则环路两端就会感应出电压。
- 在理想变压器中, 初级与次级电压、电流和阻抗之比由匝数比决定。
- 位移电流是指电荷通过电介质的“表观”流动。实际上, 相反极性的电荷沿着电介质的两端累积, 呈现出电流流过电介质的现象。
- 两种不同介质之间界面处的电磁场边界条件在静态和动态条件下都是相同的。
- 电荷连续性方程是电荷守恒定律的数学表述。
- 良导体内部的多余电荷很快就会消散。通过重新排列过程, 多余的电荷被转移到导体的表面。
- 在动态情况下, 电场 $\mathbf{E}$ 与标量电势 V 和磁矢量势 $\mathbf{A}$ 相关。
- 给定观察点处的延迟标量和矢量势考虑了它们的源之间传播所需的有限时间、电荷和电流分布以及观察点的位置。

Mathematical and Physical Models

Faraday's Law

$$V_{\text{emf}} = -\frac{d\Phi}{dt} = -\frac{d}{dt} \int_S \mathbf{B} \cdot d\mathbf{s} = V_{\text{emf}}^{\text{tr}} + V_{\text{emf}}^{\text{m}}$$

Transformer

$$V_{\text{emf}}^{\text{tr}} = -N \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s} \quad (N \text{ loops})$$

Motional

$$V_{\text{emf}}^{\text{m}} = \oint_C (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l}$$

Charge–Current Continuity

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho_v}{\partial t}$$

EM Potentials

$$\begin{aligned} \mathbf{E} &= -\nabla V - \frac{\partial \mathbf{A}}{\partial t} \\ \mathbf{B} &= \nabla \times \mathbf{A} \end{aligned}$$

Current Density

Conduction $\mathbf{J}_c = \sigma \mathbf{E}$

Displacement $\mathbf{J}_d = \frac{\partial \mathbf{D}}{\partial t}$

Conductor Charge Dissipation

$$\rho_v(t) = \rho_{v0} e^{-(\sigma/\epsilon)t} = \rho_{v0} e^{-t/\tau_r}$$

Important Terms

Provide definitions or explain the meaning of the following terms:

charge continuity equation	Faraday's law	retarded potential
charge dissipation	Kirchhoff's current law	transformer emf $V_{\text{emf}}^{\text{tr}}$
displacement current I_d	Lenz's law	wavenumber k
electromagnetic induction	motional emf $V_{\text{emf}}^{\text{m}}$	
electromotive force V_{emf}	relaxation time constant	

PROBLEMS

Sections 6-1 to 6-6: Faraday's Law and its Applications

*6.1 The switch in the bottom loop of Fig. P6.1 is closed at $t = 0$ and then opened at a later time t_1 . What is the direction of the current I in the top loop (clockwise or counterclockwise) at each of these two times?

6.2 The loop in Fig. P6.2 is in the x - y plane and $\mathbf{B} = \hat{\mathbf{z}}B_0 \sin \omega t$ with B_0 positive. What is the direction of I ($\hat{\phi}$ or $-\hat{\phi}$) at:

- (a) $t = 0$
 (b) $\omega t = \pi/4$
 (c) $\omega t = \pi/2$

6.3 A coil consists of 100 turns of wire wrapped around a square frame of sides 0.25 m. The coil is centered at the origin with each of its sides parallel to the x or y axis. Find the

* Answer(s) available in Appendix E.

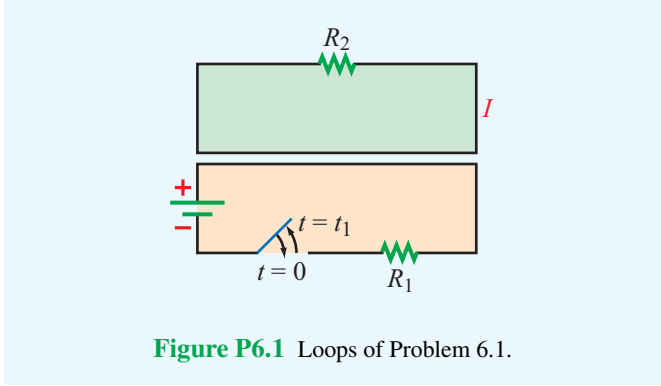


Figure P6.1 Loops of Problem 6.1.

induced emf across the open-circuited ends of the coil if the magnetic field is given by

- * (a) $\mathbf{B} = \hat{\mathbf{z}}20e^{-3t}$ (T)
 (b) $\mathbf{B} = \hat{\mathbf{z}}20 \cos x \cos 10^3 t$ (T)
 (c) $\mathbf{B} = \hat{\mathbf{z}}20 \cos x \sin 2y \cos 10^3 t$ (T)

数学与物理模型

法拉第定律

$$V_{\text{emf}} = -\frac{d\Phi}{dt} = -\frac{d}{dt} \int_S \mathbf{B} \cdot d\mathbf{s} = V_{\text{emf}}^{\text{tr}} + V_{\text{emf}}^{\text{m}}$$

感生电动势

$$V_{\text{emf}}^{\text{tr}} = -N \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s} \quad (\text{N 匝})$$

动生电动势

$$V_{\text{emf}}^{\text{m}} = \oint_C (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l}$$

电荷连续性方程

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho_v}{\partial t}$$

电磁势

$$\begin{aligned} \mathbf{E} &= -\nabla V - \frac{\partial \mathbf{A}}{\partial t} \\ \mathbf{B} &= \nabla \times \mathbf{A} \end{aligned}$$

电流密度

传导 $\mathbf{J}_c = \sigma \mathbf{E}$

位移 $\mathbf{J}_d = \frac{\partial \mathbf{D}}{\partial t}$

导体中的电荷弛豫

$$\rho_v(t) = \rho_{v0} e^{-(\sigma/\epsilon)t} = \rho_{v0} e^{-t/\tau_r}$$

重要术语

请给出下列术语的定义或解释其含义。

电荷连续性方程	法拉第定律	推迟势
电荷弛豫	基尔霍夫电流定律	感生电动势
位移电流 I_d	楞次定律	波数 k
电磁感应	动生电动势	
电动势 V_{emf}	弛豫时间常数	

问题

第 6-1 节至 6-6 节：法拉第定律及其应用

*6.1 图 P6.1 底部循环中的开关在 $t = 0$ 处闭合，然后在稍后时间 t_1 处打开。这两次时，顶部回路中的电流 I 的方向（顺时针或逆时针）是什么？

6.2 图 P6.2 中的环位于 x - y 平面中，且 $\mathbf{B} = \hat{\mathbf{z}}B_0 \sin \omega t$ 且 B_0 为正。 I ($\hat{\phi}$ 或 $-\hat{\phi}$) 的方向是什么：

- (a) $t = 0$
 (b) $\omega t = \pi/4$
 (c) $\omega t = \pi/2$

6.3 边长 0.25 m 的正方形线圈有 100 匝，以原点为中心，各边平行于 x 或 y 轴。求右栏所列各磁场下线圈开路两端的感应电动势。

* 标记的习题在附录 E 提供答案。

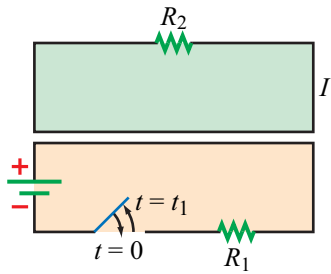


图 P6.1 习题 6.1 的两个回路。

习题 6.3 的磁通密度分别为：

- * (a) $\mathbf{B} = \hat{\mathbf{z}}20e^{-3t}$ (T)
 (b) $\mathbf{B} = \hat{\mathbf{z}}20 \cos x \cos 10^3 t$ (T)
 (c) $\mathbf{B} = \hat{\mathbf{z}}20 \cos x \sin 2y \cos 10^3 t$ (T)

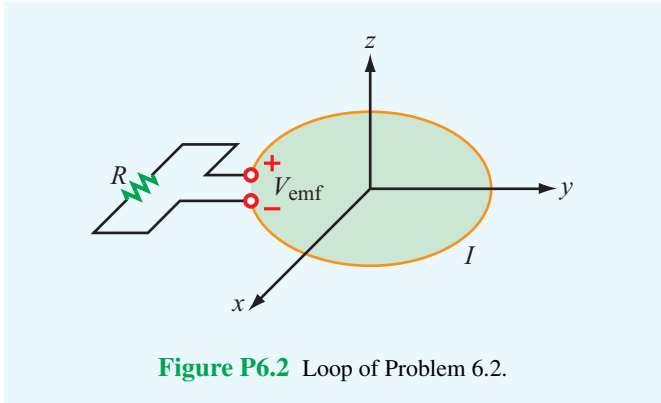


Figure P6.2 Loop of Problem 6.2.

6.4 A stationary conducting loop with an internal resistance of $0.5\ \Omega$ is placed in a time-varying magnetic field. When the loop is closed, a current of 5 A flows through it. What will the current be if the loop is opened to create a small gap and a $2\ \Omega$ resistor is connected across its open ends?

*6.5 A circular-loop TV antenna with 0.02 m^2 area is in the presence of a uniform-amplitude 300 MHz signal. When oriented for maximum response, the loop develops an emf with a peak value of 30 mV . What is the peak magnitude of $\mathbf{B}$ of the incident wave?

6.6 The square loop shown in Fig. P6.6 is coplanar with a long, straight wire carrying a current

$$I(t) = 5 \cos(2\pi \times 10^4 t) \quad (\text{A}).$$

- (a) Determine the emf induced across a small gap created in the loop.
- (b) Determine the direction and magnitude of the current that would flow through a $4\ \Omega$ resistor connected across the gap. The loop has an internal resistance of $1\ \Omega$.

*6.7 The rectangular conducting loop shown in Fig. P6.7 rotates at $6,000$ revolutions per minute in a uniform magnetic flux density given by

$$\mathbf{B} = \hat{\mathbf{y}} 50 \quad (\text{mT}).$$

Determine the current induced in the loop if its internal resistance is $0.5\ \Omega$.

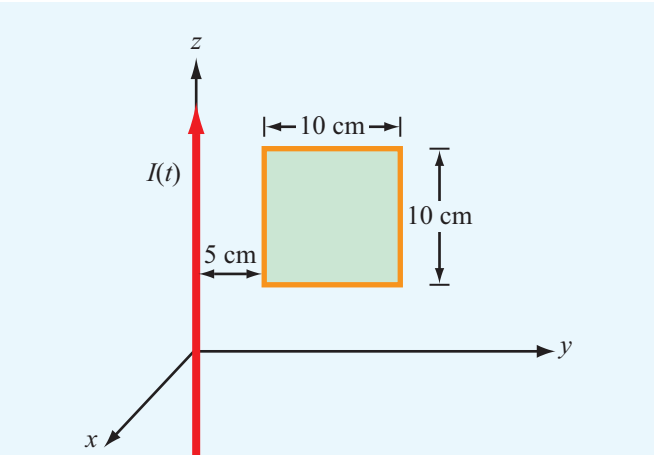


Figure P6.6 Loop coplanar with long wire (Problem 6.6).

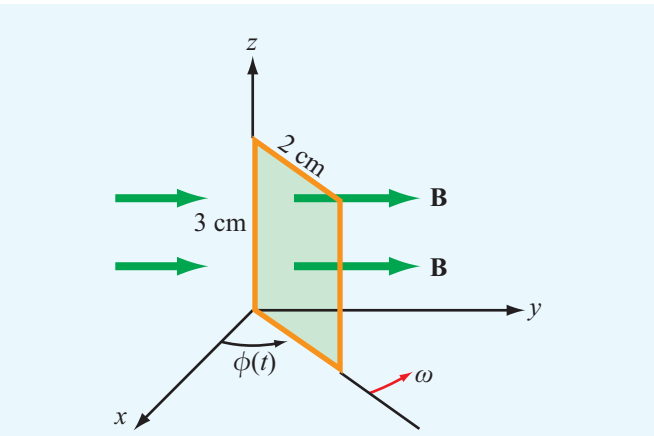


Figure P6.7 Rotating loop in a magnetic field (Problem 6.7).

6.8 The transformer shown in Fig. P6.8 consists of a long wire coincident with the z axis carrying a current $I = I_0 \cos \omega t$, coupling magnetic energy to a toroidal coil situated in the x - y plane and centered at the origin. The toroidal core uses iron material with relative permeability μ_r , around which 100 turns of a tightly wound coil serves to induce a voltage V_{emf} , as shown in the figure.

- (a) Develop an expression for V_{emf} .
- (b) Calculate V_{emf} for $f = 60\text{ Hz}$, $\mu_r = 4000$, $a = 5\text{ cm}$, $b = 6\text{ cm}$, $c = 2\text{ cm}$, and $I_0 = 50\text{ A}$.

6.9 A rectangular conducting loop $5\text{ cm} \times 10\text{ cm}$ with a small air gap in one of its sides is spinning at 7200 revolutions per minute. If the field $\mathbf{B}$ is normal to the loop axis and its

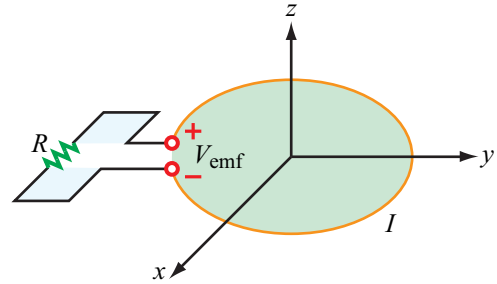


图 P6.2 问题 6.2 的循环。

6.4 将内阻为 $0.5\ \Omega$ 的静止导电环置于时变磁场中。当环路闭合时，有 5 A 的电流流过。如果环路打开形成一个小间隙，并且在其开路端连接一个 $2\ \Omega$ 电阻器，电流会是多少？

*6.5 面积为 0.02 m^2 的圆环电视天线存在均匀幅度 300 MHz 信号。当面向最大响应时，环路产生峰值为 30 mV 的电动势。入射波 $\mathbf{B}$ 的峰值幅度是多少？

6.6 图 P6.6 所示的方形环路与承载电流的长直导线共面

$$I(t) = 5 \cos(2\pi \times 10^4 t) \quad (\text{A}).$$

- (a) 确定环路中产生的小间隙上感应的电动势。(b) 确定流过连接在间隙上的 $4\ \Omega$ 电阻的电流方向和大小。环路的内阻为 $1\ \Omega$ 。

*6.7 图 P6.7 所示的矩形导电环以每分钟 $6,000$ 转的速度旋转，磁通密度由下式给出

$$\mathbf{B} = \hat{\mathbf{y}} 50 \quad (\text{mT}).$$

如果环路的内阻为 $0.5\ \Omega$ ，请确定环路中感应的电流。

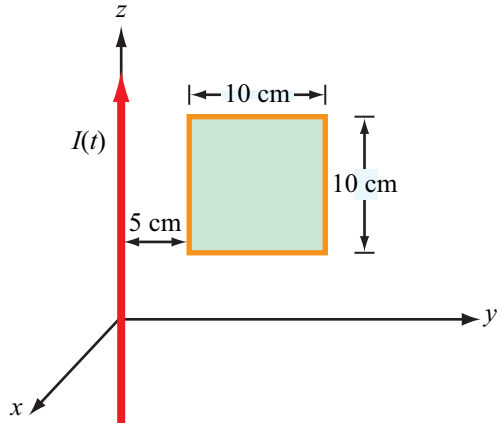


图 P6.6 环路与长导线共面 (问题 6.6)。

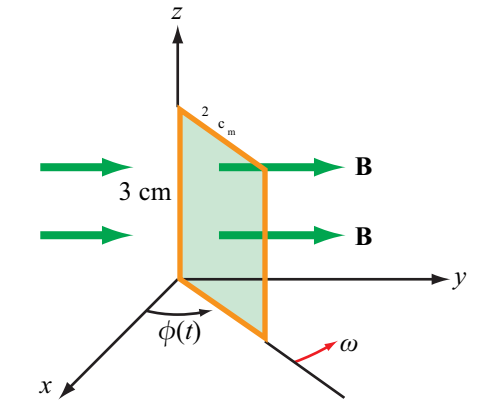


图 P6.7 磁场中的旋转环 (问题 6.7)。

6.8 图 P6.8 所示的变压器由一根与 z 轴重合的长导线组成，承载电流 $I = I_0 \cos \omega t$ ，将磁能耦合到位于 x - y 平面并以原点为中心的环形线圈。环形磁芯采用相对磁导率 μ_r 的铁材料，在其周围紧密缠绕 100 匝线圈，用于感应电压 V_{emf} ，如图所示。

- (a) 开发 V_{emf} 的表达式。(b) 计算 $f = 60\text{ Hz}$ 、 $\mu_r = 4000$ 、 $a = 5\text{ cm}$ 、 $b = 6\text{ cm}$ 、 $c = 2\text{ cm}$ 和 $I_0 = 50\text{ A}$ 的 V_{emf} 。

6.9 一个 $5\text{ cm} \times 10\text{ cm}$ 的矩形导电环，其一侧有一个小间隙，以每分钟 7200 转的速度旋转。如果场 $\mathbf{B}$ 垂直于循环轴且其

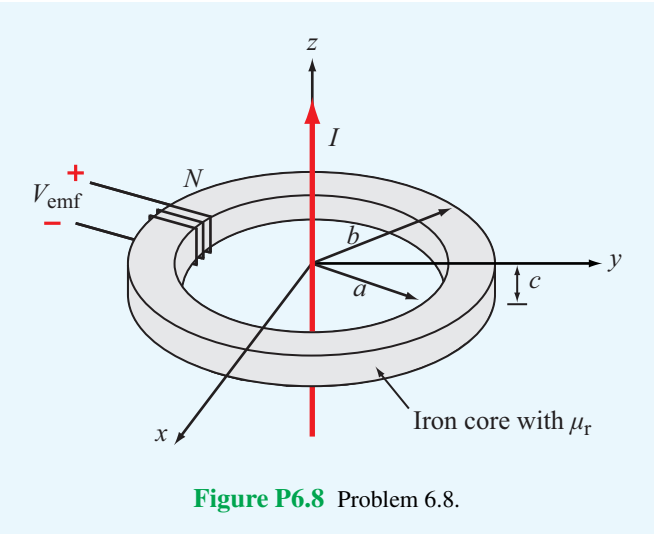


Figure P6.8 Problem 6.8.

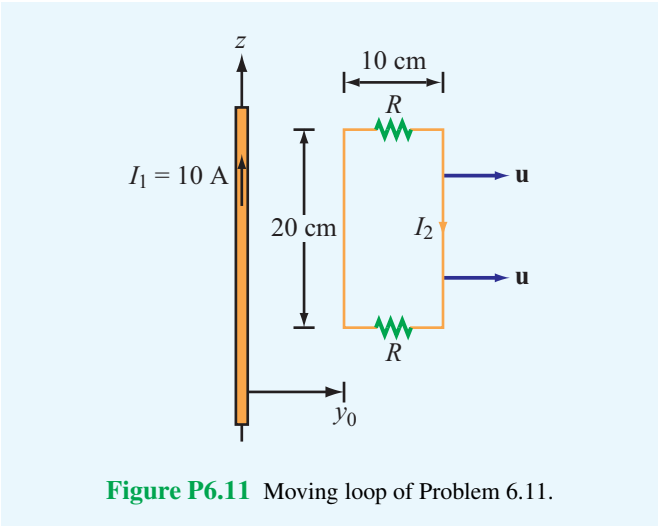


Figure P6.11 Moving loop of Problem 6.11.

magnitude is 6×10^{-6} T, what is the peak voltage induced across the air gap?

*6.10 A 50-cm long metal rod rotates about the z axis at 90 revolutions per minute with end 1 fixed at the origin as shown in Fig. P6.10. Determine the induced emf V_{12} if $\mathbf{B} = \hat{z}2 \times 10^{-4}$ T.

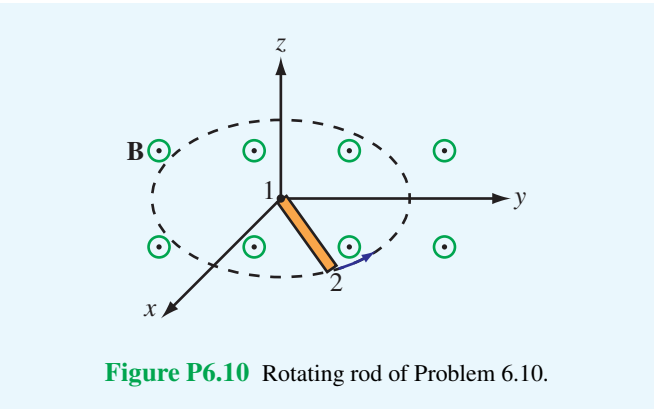


Figure P6.10 Rotating rod of Problem 6.10.

6.11 The loop shown in Fig. P6.11 moves away from a wire carrying a current $I_1 = 10$ A at a constant velocity $\mathbf{u} = \hat{y}7.5$ (m/s). If $R = 10 \Omega$ and the direction of I_2 is as defined in the figure, find I_2 as a function of y_0 , which is the distance between the wire and the loop. Ignore the internal resistance of the loop.

*6.12 The electromagnetic generator shown in Fig. 6-12 is connected to an electric bulb with a resistance of 150Ω . If the loop area is 0.1 m^2 and it rotates at 3,600 revolutions

per minute in a uniform magnetic flux density $B_0 = 0.4$ T, determine the amplitude of the current generated in the light bulb.

6.13 The circular, conducting disk shown in Fig. P6.13 lies in the x - y plane and rotates with uniform angular velocity ω about the z axis. The disk is of radius a and is present in a uniform magnetic flux density $\mathbf{B} = \hat{z}B_0$. Obtain an expression for the emf induced at the rim relative to the center of the disk.

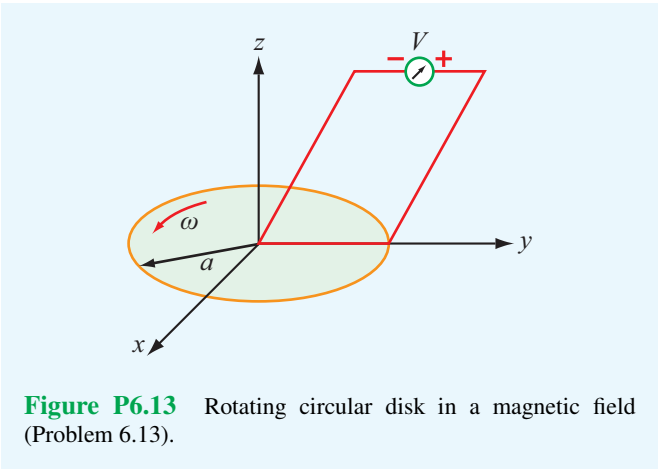


Figure P6.13 Rotating circular disk in a magnetic field (Problem 6.13).

Section 6-7: Displacement Current

6.14 The plates of a parallel-plate capacitor have areas of 10 cm^2 each and are separated by 2 cm. The capacitor is filled

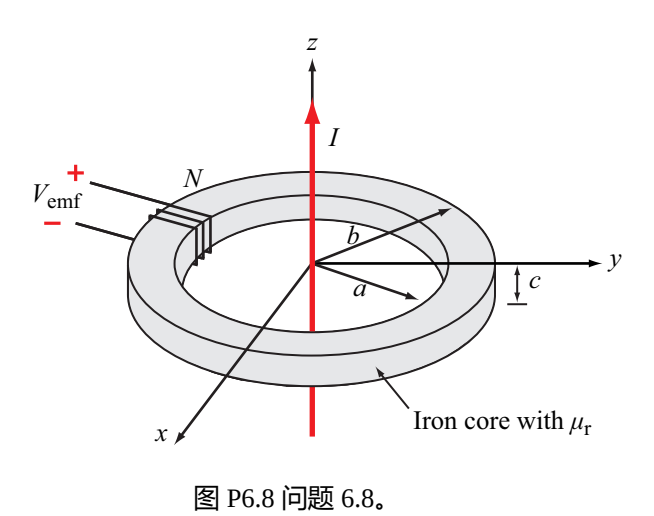


图 P6.8 问题 6.8。

幅度为 6×10^{-6} T, 气隙上感应的峰值电压是多少?

*6.10 一根 50 厘米长的金属棒绕 z 轴以每分钟 90 转的速度旋转, 一端固定在原点, 如图 P6.10 所示。如果 $\mathbf{B} = \hat{z}2 \times 10^{-4}$ T, 则确定感应电动势 V_{12} 。

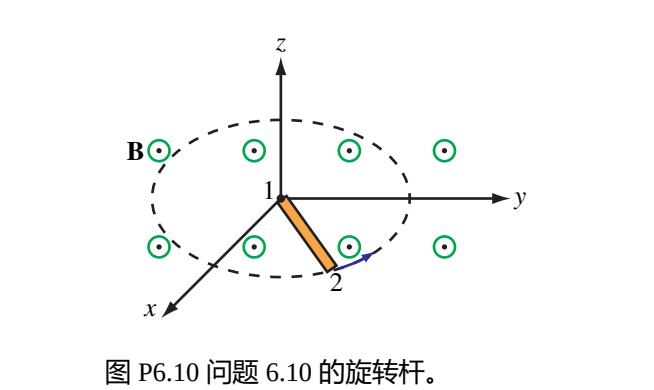


图 P6.10 问题 6.10 的旋转杆。

6.11 图 P6.11 所示的回路远离载有电流 I 的导线 $I_1 = 10$ A 恒速 $\mathbf{u} = \hat{y}7.5$ (米/秒)。如果 $R = 10 \Omega$ 和 I_2 的方向如图所示, 发现 I_2 作为 y 的函数, 这是电线和环路之间的距离。忽略环路的内阻。

*6.12 将图6-12所示的电磁发电机与电阻为 150Ω 的电灯泡相连。如果环路面积为 0.1 m^2 且转速为 3,600 转

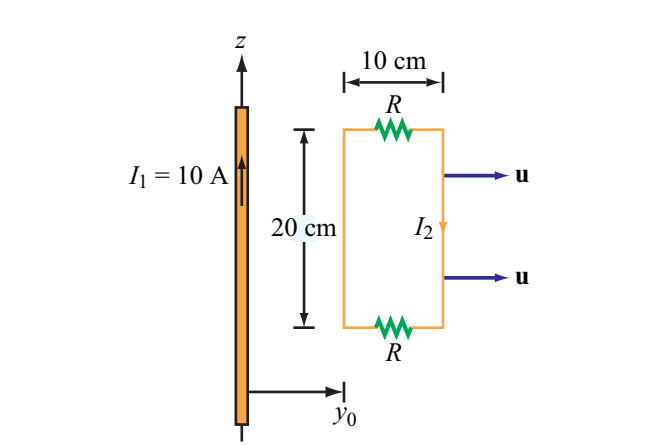


图 P6.11 问题 6.11 的移动循环。

每分钟在均匀磁通密度 $B_0 = 0.4$ T 下, 确定灯泡中产生的电流的幅度。

6.13 图 P6.13 所示的圆形导电盘位于 x - y 平面上, 并绕 z 轴以均匀角速度 ω 旋转。圆盘的半径为 a , 磁通密度均匀为 $\mathbf{B} = \hat{z}B_0$ 。获得相对于圆盘中心在边缘处感应的电动势的表达式。

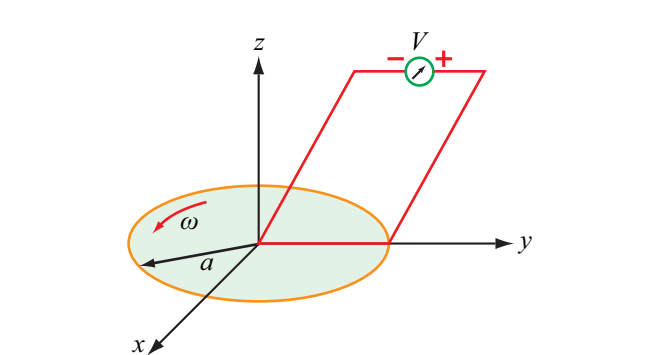


图 P6.13 在磁场中旋转圆盘 (问题 6.13)。

第 6-7 节: 位移电流

6.14 平行板电容器的极板面积为 10 cm^2 , 间距为 2 cm。电容已充满

with a dielectric material with $\epsilon = 4\epsilon_0$, and the voltage across it is given by $V(t) = 30 \cos 2\pi \times 10^6 t$ (V). Find the displacement current.

*6.15 A coaxial capacitor of length $l = 6$ cm uses an insulating dielectric material with $\epsilon_r = 9$. The radii of the cylindrical conductors are 0.5 cm and 1 cm. If the voltage applied across the capacitor is

$$V(t) = 50 \sin(120\pi t) \quad (\text{V}),$$

what is the displacement current?

6.16 The parallel-plate capacitor shown in Fig. P6.16 is filled with a lossy dielectric material of relative permittivity ϵ_r and conductivity σ . The separation between the plates is d and each plate is of area A . The capacitor is connected to a time-varying voltage source $V(t)$.

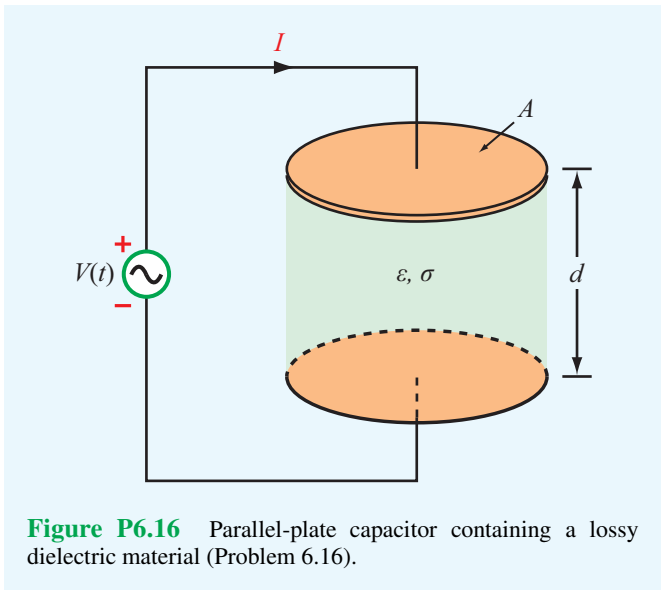


Figure P6.16 Parallel-plate capacitor containing a lossy dielectric material (Problem 6.16).

- (a) Obtain an expression for I_c , the conduction current flowing between the plates inside the capacitor, in terms of the given quantities.
- (b) Obtain an expression for I_d , the displacement current flowing inside the capacitor.
- (c) Based on your expressions for parts (a) and (b), give an equivalent-circuit representation for the capacitor.
- (d) Evaluate the values of the circuit elements for $A = 4 \text{ cm}^2$, $d = 0.5 \text{ cm}$, $\epsilon_r = 4$, $\sigma = 2.5 \text{ (S/m)}$, and

$$V(t) = 10 \cos(3\pi \times 10^3 t) \quad (\text{V}).$$

*6.17 In wet soil characterized by $\sigma = 10^{-2} \text{ (S/m)}$, $\mu_r = 1$, and $\epsilon_r = 36$, at what frequency is the conduction current density equal in magnitude to the displacement current density?

6.18 An electromagnetic wave propagating in seawater has an electric field with a time variation given by $\mathbf{E} = \hat{\mathbf{z}} E_0 \cos \omega t$. If the permittivity of water is $81\epsilon_0$ and its conductivity is 4 (S/m) , find the ratio of the magnitudes of the conduction current density to displacement current density at each of the following frequencies:

- (a) 1 kHz
(b) 1 MHz
(c) 1 GHz
(d) 100 GHz

Sections 6-9 and 6-10: Continuity Equation and Charge Dissipation

6.19 At $t = 0$, charge density ρ_{v0} was introduced into the interior of a material with a relative permittivity $\epsilon_r = 9$. If at $t = 1 \mu\text{s}$ the charge density has dissipated down to $10^{-3}\rho_{v0}$, what is the conductivity of the material?

*6.20 If the current density in a conducting medium is given by

$$\mathbf{J}(x, y, z; t) = (\hat{\mathbf{x}}z - \hat{\mathbf{y}}4y^2 + \hat{\mathbf{z}}2x) \cos \omega t,$$

determine the corresponding charge distribution $\rho_v(x, y, z; t)$.

6.21 In a certain medium, the direction of current density $\mathbf{J}$ points in the radial direction in cylindrical coordinates, and its magnitude is independent of both ϕ and z . Determine $\mathbf{J}$ given that the charge density in the medium is

$$\rho_v = \rho_0 r \cos \omega t \quad (\text{C/m}^3).$$

6.22 If we were to characterize how good a material is as an insulator by its resistance to dissipating charge, which of the following two materials is the better insulator?

Dry Soil:	$\epsilon_r = 2.5$,	$\sigma = 10^{-4} \text{ (S/m)}$
Fresh Water:	$\epsilon_r = 80$,	$\sigma = 10^{-3} \text{ (S/m)}$

Sections 6-11: Electromagnetic Potentials

6.23 The electric field of an electromagnetic wave propagating in air is given by

$$\begin{aligned} \mathbf{E}(z, t) = & \hat{\mathbf{x}} 4 \cos(6 \times 10^8 t - 2z) \\ & + \hat{\mathbf{y}} 3 \sin(6 \times 10^8 t - 2z) \quad (\text{V/m}). \end{aligned}$$

Find the associated magnetic field $\mathbf{H}(z, t)$.

采用具有 $\epsilon = 4\epsilon_0$ 的介电材料，其两端电压由 $V(t) = 30 \cos 2\pi \times 10^6 t$ (V) 给出。求位移电流。

*6.15 长度 $l = 6$ cm 的同轴电容器使用 $\epsilon_r = 9$ 的绝缘电介质材料。圆柱形导体的半径为 0.5 cm 和 1 cm。如果施加在电容器两端的电压为

$$V(t) = 50 \sin(120\pi t) \quad (\text{V}),$$

什么是位移电流？

6.16 图 P6.16 所示的平行板电容器填充有相对介电常数 ϵ_r 和电导率 σ 的有损介电材料。极板之间的间距为 d ，每个极板的面积为 A 。电容器连接到时变电压源 $V(t)$ 。

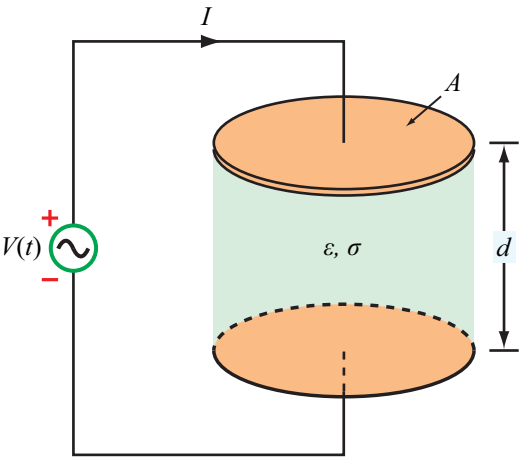


图 P6.16 包含有损耗介电材料的平行板电容器 (问题 6.16)。

(a) 根据给定的量，获得 I_c 的表达式，即电容器内部极板之间流动的传导电流。

(b) 获得电容器内部流动的位移电流 I_d 的表达式。

(c) 根据 (a) 和 (b) 部分的表达式，给出电容器的等效电路表示。

(d) 评估电路元件的值 $A = 4 \text{ cm}^2$ 、 $d = 0.5 \text{ cm}$ 、 $\epsilon_r = 4$ 、 $\sigma = 2.5 \text{ (S/m)}$ 和

$$V(t) = 10 \cos(3\pi \times 10^3 t) \quad (\text{V}).$$

*6.17 在以 $\sigma = 10^{-2} \text{ (S/m)}$ 、 $\mu_r = 1$ 和 $\epsilon_r = 36$ 为特征的湿土壤中，在什么频率下传导电流密度的大小等于位移电流密度？

6.18 在海水中传播的电磁波具有随时间变化的电场，其变化由 $\mathbf{E} = \hat{\mathbf{z}} E_0 \cos \omega t$ 给出。如果水的介电常数为 $81\epsilon_0$ ，电导率为 4 (S/m) ，求以下每个频率下传导电流密度与位移电流密度的大小之比：

- (a) 1 kHz
(b) 1 MHz
(c) 1 GHz
(d) 100 GHz

第 6-9 节和 6-10 节：连续性方程和电荷耗散

6.19 在 $t = 0$ 处，电荷密度 ρ_{v0} 被引入相对介电常数为 $\epsilon_r = 9$ 的材料内部。如果在 $t = 1 \mu\text{s}$ 处电荷密度已耗散至 $10^{-3}\rho_{v0}$ ，则该材料的电导率是多少？

*6.20 如果导电介质中的电流密度由下式给出

$$\mathbf{J}(x, y, z; t) = (\hat{\mathbf{x}}z - \hat{\mathbf{y}}4y^2 + \hat{\mathbf{z}}2x) \cos \omega t,$$

确定相应的电荷分布 $\rho_v(x, y, z; t)$ 。

6.21 在某种介质中，电流密度 $\mathbf{J}$ 的方向指向柱坐标中的径向，其大小与 ϕ 和 z 无关。鉴于介质中的电荷密度为 ρ_v ，确定 $\mathbf{J}$

$$\rho_v = \rho_0 r \cos \omega t \quad (\text{C/m}^3).$$

6.22 如果我们要通过材料对耗散电荷的抵抗力来表征材料作为绝缘体的好坏，那么以下两种材料中哪一种是最好的绝缘体？

Dry Soil:	$\epsilon_r = 2.5$,	$\sigma = 10^{-4} \text{ (S/m)}$
Fresh Water:	$\epsilon_r = 80$,	$\sigma = 10^{-3} \text{ (S/m)}$

第 6-11 节：电磁势

6.23 在空气中传播的电磁波的电场由下式给出

$$\begin{aligned} \mathbf{E}(z, t) = & \hat{\mathbf{x}} 4 \cos(6 \times 10^8 t - 2z) \\ & + \hat{\mathbf{y}} 3 \sin(6 \times 10^8 t - 2z) \quad (\text{V/m}). \end{aligned}$$

求出相关的磁场 $\mathbf{H}(z, t)$ 。

*6.24 The magnetic field in a dielectric material with $\epsilon = 4\epsilon_0$, $\mu = \mu_0$, and $\sigma = 0$ is given by

$$\mathbf{H}(y,t) = \hat{\mathbf{x}}5\cos(2\pi \times 10^7t + ky) \quad (\text{A/m}).$$

Find k and the associated electric field $\mathbf{E}$.

6.25 Given an electric field

$$\mathbf{E} = \hat{\mathbf{x}}E_0\sin ay\cos(\omega t - kz),$$

where E_0 , a , ω , and k are constants, find $\mathbf{H}$.

*6.26 The electric field radiated by a short dipole antenna is given in spherical coordinates by

$$\mathbf{E}(R,\theta;t) = \hat{\boldsymbol{\theta}}\frac{2 \times 10^{-2}}{R}\sin\theta\cos(6\pi \times 10^8t - 2\pi R) \quad (\text{V/m}).$$

Find $\mathbf{H}(R,\theta;t)$.

6.27 A Hertzian dipole is a short conducting wire carrying an approximately constant current over its length l . If such a dipole is placed along the z axis with its midpoint at the origin and if the current flowing through it is $i(t) = I_0\cos\omega t$, find the following:

(a) The retarded vector potential $\widetilde{\mathbf{A}}(R,\theta,\phi)$ at an observation point $Q(R,\theta,\phi)$ in a spherical coordinate system.

(b) The magnetic field phasor $\widetilde{\mathbf{H}}(R,\theta,\phi)$.

Assume l to be sufficiently small so that the observation point is approximately equidistant to all points on the dipole; that is, assume $R' \approx R$.

6.28 In free space, the magnetic field is given by

$$\mathbf{H} = \hat{\boldsymbol{\phi}}\frac{36}{r}\cos(6 \times 10^9t - kz) \quad (\text{mA/m}).$$

* (a) Determine k .

(b) Determine $\mathbf{E}$.

(c) Determine $\mathbf{J}_d$.

6.29 The magnetic field in a given dielectric medium is given by

$$\mathbf{H} = \hat{\mathbf{y}}6\cos 2z\sin(2 \times 10^7t - 0.1x) \quad (\text{A/m}),$$

where x and z are in meters.

(a) Determine $\mathbf{E}$.

(b) Determine the displacement current density $\mathbf{J}_d$.

(c) Determine the charge density ρ_v .

*6.24 $\epsilon = 4\epsilon_0$ 、 $\mu = \mu_0$ 和 $\sigma = 0$ 的介电材料中的磁场由下式给出

$$\mathbf{H}(y,t) = \hat{\mathbf{x}}5\cos(2\pi \times 10^7t + ky) \quad (\text{A/m}).$$

求 k 和相关电场 $\mathbf{E}$ 。

6.25 给定电场

$$\mathbf{E} = \hat{\mathbf{x}}E_0\sin ay\cos(\omega t - kz),$$

其中 E_0 、 a 、 ω 和 k 是常数，求 $\mathbf{H}$ 。

*6.26 短偶极子天线辐射的电场在球坐标中给出：

$$\mathbf{E}(R,\theta;t) = \hat{\boldsymbol{\theta}}\frac{2 \times 10^{-2}}{R}\sin\theta\cos(6\pi \times 10^8t - 2\pi R) \quad (\text{V/m}).$$

求 $\mathbf{H}(R,\theta;t)$ 。

6.27 赫兹偶极子是一根短导线，在其长度 l 上承载近似恒定的电流。如果这样的偶极子沿 z 轴放置，其中点位于原点，并且如果流过它的电流为 $i(t) = I_0\cos\omega t$ ，则找到以下内容：

(a) 球坐标系中观察点 $Q(R,\theta,\phi)$ 处的延迟矢量势 $\widetilde{\mathbf{A}}(R,\theta,\phi)$ 。

(b) 磁场相量 $\widetilde{\mathbf{H}}(R,\theta,\phi)$ 。

假设 l 足够小，使得观测点与偶极子上的所有点大致等距；也就是说，假设 $R' \approx R$ 。

6.28 在自由空间中，磁场由下式给出

$$\mathbf{H} = \hat{\boldsymbol{\phi}}\frac{36}{r}\cos(6 \times 10^9t - kz) \quad (\text{mA/m}).$$

* (一) 确定 k 。

(b) 确定 $\mathbf{E}$ 。

(c) 确定 $\mathbf{J}_d$ 。

6.29 给定电介质中的磁场由下式给出

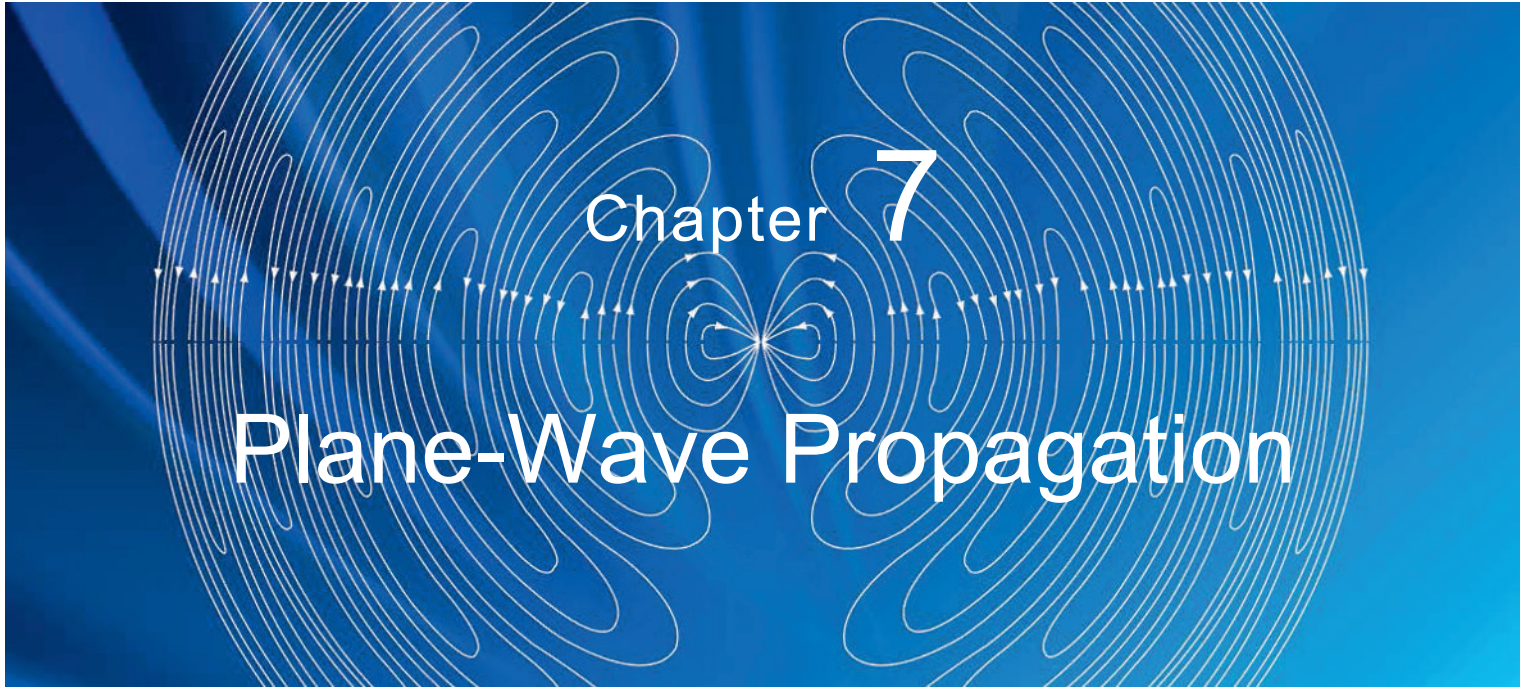
$$\mathbf{H} = \hat{\mathbf{y}}6\cos 2z\sin(2 \times 10^7t - 0.1x) \quad (\text{A/m}),$$

其中 x 和 z 的单位为米。

(a) 确定 $\mathbf{E}$ 。

(b) 确定位移电流密度 $\mathbf{J}_d$ 。

(c) 确定电荷密度 ρ_v 。



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Objectives

- Upon learning the material presented in this chapter, you should be able to:
- 1. Describe mathematically the electric and magnetic fields of TEM waves.
 - 2. Describe the polarization properties of an EM wave.
 - 3. Relate the propagation parameters of a wave to the constitutive parameters of the medium.
 - 4. Characterize the flow of current in conductors and use it to calculate the resistance of a coaxial cable.
 - 5. Calculate the rate of power carried by an EM wave in both lossless and lossy media.



章节内容

无界电磁波、302 7-1 时谐波场、303 7-2 无损介质中的平面波传播、304 7-3 波偏振、309 TB13 RFID 系统、313 7-4 有损介质中的平面波传播、317 TB14 液晶显示器 (LCD)、320 7-5 介质中的电流良导体、325 7-6 电磁功率密度、328 第 7 章总结、331 问题、333

目标

学习本章介绍的材料后，您应该能够：

- 1. 用数学方法描述 TEM 波的电场和磁场。
- 2. 描述电磁波的偏振特性。
- 3. 将波的传播参数与介质的本构参数联系起来。
- 4. 表征导体中的电流并用它来计算同轴电缆的电阻。
- 5. 计算电磁波在无损和有损介质中携带的功率率。

Unbounded EM Waves

It was established in Chapter 6 that a time-varying electric field produces a magnetic field and, conversely, a time-varying magnetic field produces an electric field. This cyclic pattern often results in electromagnetic (EM) waves propagating through free space and in material media. When a wave propagates through a homogeneous medium without interacting with obstacles or material interfaces, it is said to be **unbounded**. Light waves emitted by the sun and radio transmissions by antennas are good examples. Unbounded waves may propagate in both lossless and lossy media. Waves propagating in a **lossless medium** (e.g., air and perfect dielectrics) are similar to those on a lossless transmission line in that they do not attenuate. When propagating in a **lossy medium** (material with nonzero conductivity, such as water), part of the power carried by an EM wave gets converted into heat. A wave produced by a localized source, such as an antenna, expands outwardly in the form of a **spherical wave**, as depicted in Fig. 7-1(a). Even though an antenna may radiate more energy along some directions than along others, the spherical wave travels at the same speed in all directions. To an observer very far away from the source, however, the **wavefront** of the spherical wave appears approximately **planar**, as if it were part of a **uniform plane wave** with identical properties at all points in the plane tangent to the wavefront (Fig. 7-1(b)). Plane waves are easily described using a Cartesian coordinate system, which is mathematically easier to work with than the spherical coordinate system needed to describe spherical waves.

When a wave propagates along a material structure, it is said to be **guided**. The Earth's surface and ionosphere constitute parallel boundaries of a natural structure capable of guiding short-wave radio transmissions in the HF band* (3 to 30 MHz); indeed, the ionosphere is a good reflector at these frequencies, thereby allowing the waves to zigzag between the two boundaries (Fig. 7-2). When we discussed wave propagation on a transmission line in Chapter 2, we dealt with voltages and currents. For a transmission-line circuit such as that shown in Fig. 7-3, the ac voltage source excites an incident wave that travels down the coaxial line toward the load, and unless the load is matched to the line, part (or all) of the incident wave is reflected back toward the generator. At any point on the line, the instantaneous total voltage $v(z,t)$ is the sum of the incident and reflected waves—both of which vary sinusoidally with time. Associated with the voltage difference between the inner and outer conductors of the coaxial line is a radial electric field $E(z,t)$ that exists in the dielectric material between the conductors, and since $v(z,t)$ varies sinusoidally with time, so does $E(z,t)$. Furthermore, the current flowing through the inner

* See Fig. 1-17.

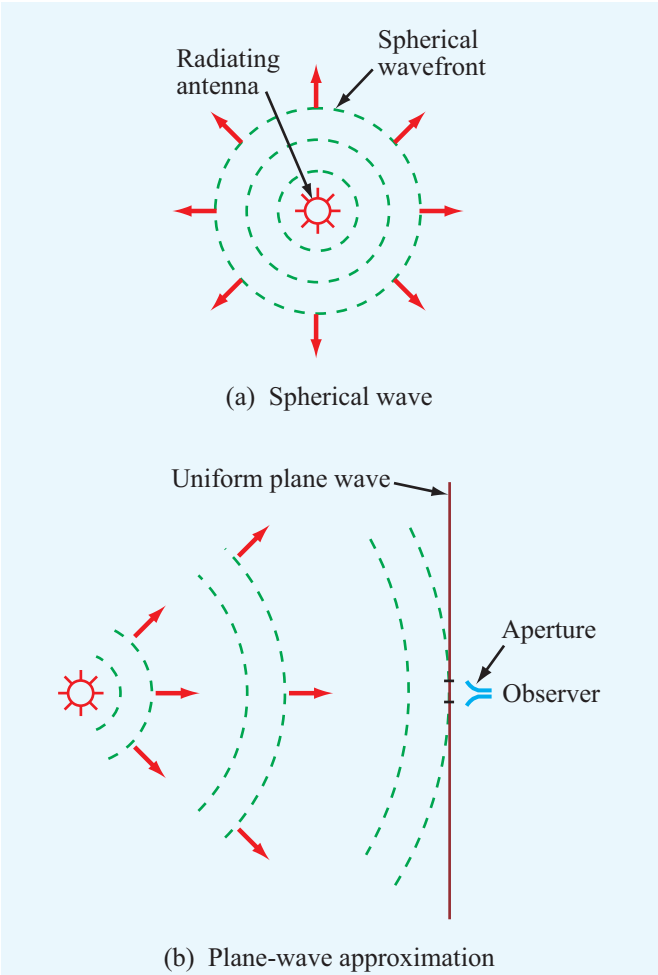


Figure 7-1 Waves radiated by an EM source, such as a light bulb or an antenna, have spherical wavefronts, as in (a); to a distant observer, however, the wavefront across the observer's aperture appears approximately planar, as in (b).

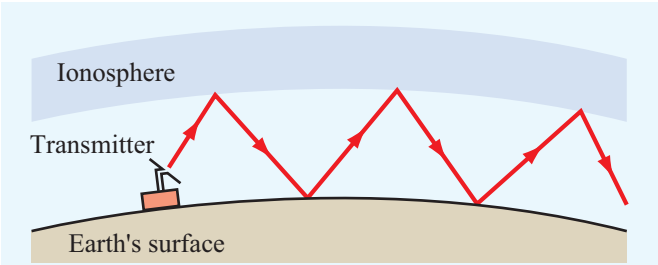


Figure 7-2 The atmospheric layer bounded by the ionosphere at the top and the Earth's surface at the bottom forms a guiding structure for the propagation of radio waves in the HF band.

无界电磁波

第六章确立了时变电场产生磁场，反之，时变磁场产生电场。这种循环模式通常会导致电磁 (EM) 波在自由空间和材料介质中传播。当波在均匀介质中传播而不与障碍物或材料界面相互作用时，称为无界波。太阳发射的光波和天线的无线电传输就是很好的例子。无界波可以在无损和有损介质中传播。在无损介质（例如空气和理想电介质）中传播的波与在无损传输线上传播的波相似，因为它们不会衰减。当在有损介质（具有非零电导率的材料，例如水）中传播时，电磁波携带的部分功率会转化为热量。由局部源（例如天线）产生的波以球面波的形式向外扩展，如图 7-1 (a) 所示。尽管天线沿某些方向辐射的能量可能比沿其他方向辐射的能量多，但球面波在所有方向上以相同的速度传播。然而，对于距离源很远的观察者来说，球面波的波前看起来近似平面，就好像它是均匀平面波的一部分，在与波前相切的平面上的所有点都具有相同的属性（图 7-1 (b) ）。使用笛卡尔坐标系可以轻松描述平面波，该坐标系在数学上比描述球面波所需的球面坐标系更容易使用。

当波沿着材料结构传播时，称为被引导。地球表面和电离层构成自然结构的平行边界，能够引导 HF 频段* (3 至 30 MHz) 的短波无线电传输；事实上，电离层在这些频率下是一个很好的反射器，从而允许波在两个边界之间曲折（图 7-2）。当我们在第二章中讨论传输线上的波传播时，我们讨论了电压和电流。对于如图 7-3 所示的传输线电路，交流电压源激发沿同轴轴向负载传播的入射波，除非负载与线路匹配，否则部分（或全部）入射波将反射回发生器。在线路上的任何一点，瞬时总电压 $v(z,t)$ 是入射波和反射波的总和，两者都随时间呈正弦变化。与同轴线内导体和外导体之间的电压差相关的是导体之间的介电材料中存在的径向电场 $E(z,t)$ ，并且由于 $v(z,t)$ 随时间呈正弦变化，因此 $E(z,t)$ 也随之变化。此外，流过内部的电流

* See Fig. 1-17.

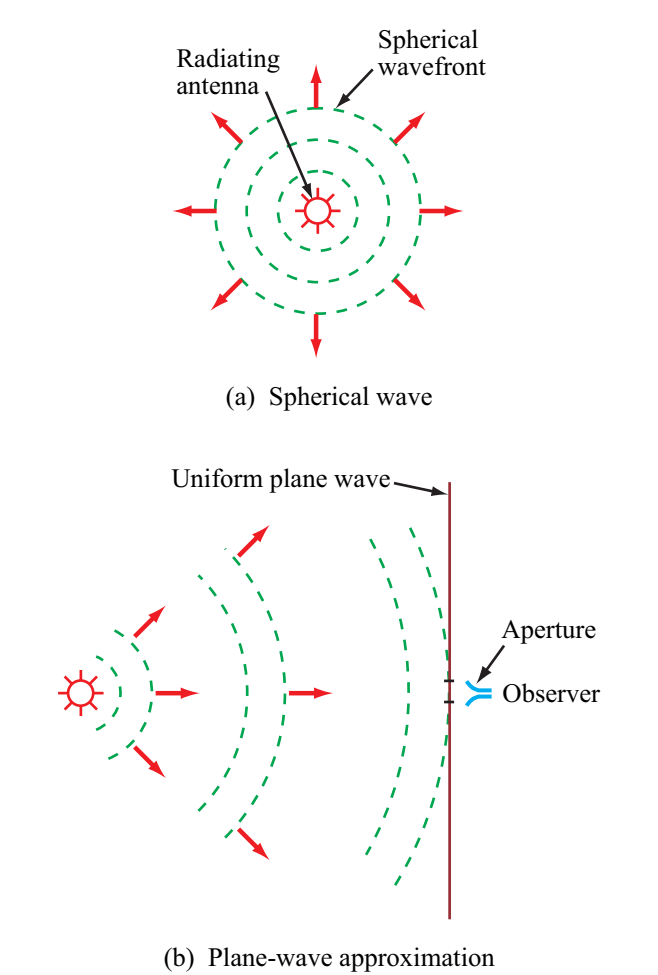


图 7-1 电磁源（例如灯泡或天线）辐射的波具有球形波前，如 (a) 所示；然而，对于远处的观察者来说，穿过观察者孔径的波前看起来近似平面，如 (b) 所示。

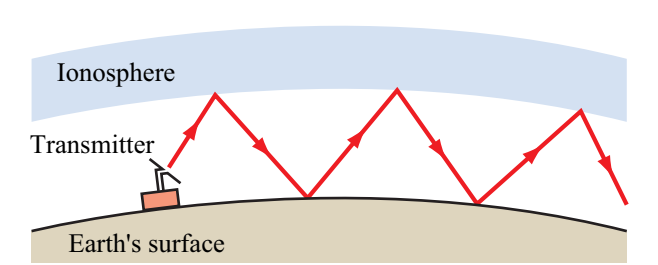


图 7-2 顶部为电离层、底部为地球表面的大气层形成了 HF 频段无线电波传播的引导结构。

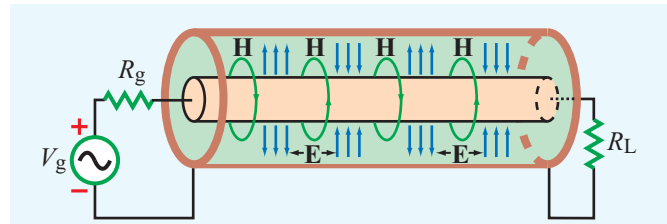


Figure 7-3 A guided electromagnetic wave traveling in a coaxial transmission line consists of time-varying electric and magnetic fields in the dielectric medium between the inner and outer conductors.

conductor induces an azimuthal magnetic field $\mathbf{H}(z, t)$ in the dielectric material surrounding it. These coupled fields, $\mathbf{E}(z, t)$ and $\mathbf{H}(z, t)$, constitute an electromagnetic wave. Thus, we can model wave propagation on a transmission line either in terms of the voltages across the line and the currents in its conductors or in terms of the electric and magnetic fields in the dielectric medium between the conductors.

In this chapter, we focus our attention on wave propagation in unbounded media. Unbounded waves have many practical applications in science and engineering. We consider both lossless and lossy media. Even though strictly speaking uniform plane waves cannot exist, we study them in this chapter to develop a physical understanding of wave propagation in lossless and lossy media. In Chapter 8, we examine how both planar and spherical waves are reflected by and transmitted through boundaries between dissimilar media. The processes of radiation and reception of waves by antennas are treated in Chapter 9.

7-1 Time-Harmonic Fields

Time-varying electric and magnetic fields ($\mathbf{E}$, $\mathbf{D}$, $\mathbf{B}$, and $\mathbf{H}$) and their sources (the charge density ρ_v and current density $\mathbf{J}$) generally depend on the spatial coordinates (x, y, z) and the time variable t . However, if their time variation is sinusoidal with angular frequency ω , then these quantities can be represented by a phasor that depends on (x, y, z) only. The vector phasor $\tilde{\mathbf{E}}(x, y, z)$ and the instantaneous field $\mathbf{E}(x, y, z; t)$ it describes are related as

$$\mathbf{E}(x, y, z; t) = \Re \left[\tilde{\mathbf{E}}(x, y, z) e^{j\omega t} \right]. \quad (7.1)$$

Similar definitions apply to $\mathbf{D}$, $\mathbf{B}$, and $\mathbf{H}$, as well as to ρ_v and $\mathbf{J}$. For a linear, isotropic, and homogeneous medium with electrical permittivity ϵ , magnetic permeability μ , and conductivity σ , Maxwell's equations (6.1) to (6.4) assume the

following form in the phasor domain.

$$\nabla \cdot \tilde{\mathbf{E}} = \tilde{\rho}_v / \epsilon, \quad (7.2a)$$

$$\nabla \times \tilde{\mathbf{E}} = -j\omega\mu\tilde{\mathbf{H}}, \quad (7.2b)$$

$$\nabla \cdot \tilde{\mathbf{H}} = 0, \quad (7.2c)$$

$$\nabla \times \tilde{\mathbf{H}} = \tilde{\mathbf{J}} + j\omega\epsilon\tilde{\mathbf{E}}. \quad (7.2d)$$

To derive these equations, we used $\mathbf{D} = \epsilon\mathbf{E}$ and $\mathbf{B} = \mu\mathbf{H}$ as well as the fact that, for time-harmonic quantities, differentiation in the time domain corresponds to multiplication by $j\omega$ in the phasor domain. These equations are the starting point for the subject matter treated in this chapter.

7-1.1 Complex Permittivity

In a medium with conductivity σ , the conduction current density $\tilde{\mathbf{J}}$ is related to $\tilde{\mathbf{E}}$ by $\tilde{\mathbf{J}} = \sigma\tilde{\mathbf{E}}$. Assuming no other current flows in the medium, Eq. (7.2d) may be written as

$$\nabla \times \tilde{\mathbf{H}} = \tilde{\mathbf{J}} + j\omega\epsilon\tilde{\mathbf{E}} = (\sigma + j\omega\epsilon)\tilde{\mathbf{E}} = j\omega \left(\epsilon - j\frac{\sigma}{\omega} \right) \tilde{\mathbf{E}}. \quad (7.3)$$

By defining the **complex permittivity** ϵ_c as

$$\epsilon_c = \epsilon - j\frac{\sigma}{\omega}, \quad (7.4)$$

Eq. (7.3) can be rewritten as

$$\nabla \times \tilde{\mathbf{H}} = j\omega\epsilon_c\tilde{\mathbf{E}}. \quad (7.5)$$

In a source-free medium, $\rho_v = 0$. Hence, Maxwell's equations become

$$\nabla \cdot \tilde{\mathbf{E}} = 0, \quad (7.6a)$$

$$\nabla \times \tilde{\mathbf{E}} = -j\omega\mu\tilde{\mathbf{H}}, \quad (7.6b)$$

$$\nabla \cdot \tilde{\mathbf{H}} = 0, \quad (7.6c)$$

$$\nabla \times \tilde{\mathbf{H}} = j\omega\epsilon_c\tilde{\mathbf{E}}. \quad (7.6d)$$

The complex permittivity ϵ_c given by Eq. (7.4) is often written in terms of a real part ϵ' and an imaginary part ϵ'' . Thus,

$$\epsilon_c = \epsilon - j\frac{\sigma}{\omega} = \epsilon' - j\epsilon'', \quad (7.7)$$

with

$$\epsilon' = \epsilon, \quad (7.8a)$$

$$\epsilon'' = \frac{\sigma}{\omega}. \quad (7.8b)$$

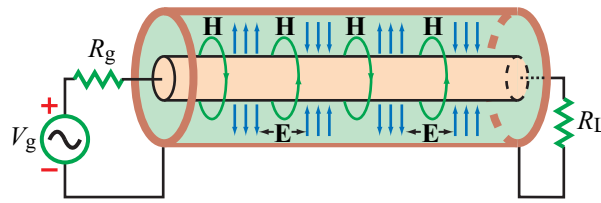


图 7-3 在同轴传输线中传播的引导电磁波由内导体和外导体之间的介电介质中的时变电场和磁场组成。

导体在其周围的介电材料中感应出方位角磁场 $H(z, t)$ 。这些耦合场 $E(z, t)$ 和 $H(z, t)$ 构成电磁波。因此，我们可以根据传输线上的电压和导体中的电流，或者根据导体之间介电介质中的电场和磁场来模拟传输线上的波传播。

在本章中，我们将重点关注无界介质中的波传播。无界波在科学和工程中有许多实际应用。我们考虑无损和有损媒体。尽管严格来说均匀平面波不存在，但我们在本章中研究它们是为了对波在无损和有损介质中传播的物理理解。在第八章中，我们研究平面波和球面波如何在不同介质之间的边界反射和传播。第 9 章讨论天线辐射和接收电波的过程。

7-1 时谐波场

时变电场和磁场 ($\mathbf{E}$ 、 $\mathbf{D}$ 、 $\mathbf{B}$ 和 $\mathbf{H}$) 及其源 (电荷密度 ρ_v 和电流密度 $\mathbf{J}$) 通常取决于空间坐标 (x, y, z) 和时间变量 t 。然而，如果它们的时间变化是具有角频率 ω 的正弦曲线，则这些量可以用仅依赖于 (x, y, z) 的相量来表示。矢量相量 $\tilde{\mathbf{E}}(x, y, z)$ 和它描述的瞬时场 $\mathbf{E}(x, y, z; t)$ 的关系为

$$\mathbf{E}(x, y, z; t) = \Re \left[\tilde{\mathbf{E}}(x, y, z) e^{j\omega t} \right]. \quad (7.1)$$

类似的定义适用于 $\mathbf{D}$ 、 $\mathbf{B}$ 和 $\mathbf{H}$ ，以及 ρ_v 和 $\mathbf{J}$ 。对于具有介电常数 ϵ 、磁导率 μ 和电导率 σ 的线性、各向同性和均匀介质，麦克斯韦方程 (6.1) 至 (6.4) 假设

相量域中的以下形式。

$$\nabla \cdot \tilde{\mathbf{E}} = \tilde{\rho}_v / \epsilon, \quad (7.2a)$$

$$\nabla \times \tilde{\mathbf{E}} = -j\omega\mu\tilde{\mathbf{H}}, \quad (7.2b)$$

$$\nabla \cdot \tilde{\mathbf{H}} = 0, \quad (7.2c)$$

$$\nabla \times \tilde{\mathbf{H}} = \tilde{\mathbf{J}} + j\omega\epsilon\tilde{\mathbf{E}}. \quad (7.2d)$$

为了推导这些方程，我们使用了 $\mathbf{D} = \epsilon\mathbf{E}$ 和 $\mathbf{B} = \mu\mathbf{H}$ 以及以下事实：对于时谐量，时域中的微分对应于相量域中 $j\omega$ 的乘法。这些方程是本章讨论的主题的起点。

7-1.1 复介电常数

在电导率为 σ 的介质中，传导电流密度 $\tilde{\mathbf{J}}$ 与 $\tilde{\mathbf{E}}$ 的关系为 $\tilde{\mathbf{J}} = \sigma\tilde{\mathbf{E}}$ 。假设介质中没有其他电流流动，方程：(7.2d) 可以写成

$$\nabla \times \tilde{\mathbf{H}} = \tilde{\mathbf{J}} + j\omega\epsilon\tilde{\mathbf{E}} = (\sigma + j\omega\epsilon)\tilde{\mathbf{E}} = j\omega \left(\epsilon - j\frac{\sigma}{\omega} \right) \tilde{\mathbf{E}}. \quad (7.3)$$

将复介电常数 ϵ_c 定义为

$$\epsilon_c = \epsilon - j\frac{\sigma}{\omega}, \quad (7.4)$$

等式。(7.3) 可以重写为

$$\nabla \times \tilde{\mathbf{H}} = j\omega\epsilon_c\tilde{\mathbf{E}}. \quad (7.5)$$

在无源介质中， $\rho_v = 0$ 。因此，麦克斯韦方程组变为

$$\nabla \cdot \tilde{\mathbf{E}} = 0, \quad (7.6a)$$

$$\nabla \times \tilde{\mathbf{E}} = -j\omega\mu\tilde{\mathbf{H}}, \quad (7.6b)$$

$$\nabla \cdot \tilde{\mathbf{H}} = 0, \quad (7.6c)$$

$$\nabla \times \tilde{\mathbf{H}} = j\omega\epsilon_c\tilde{\mathbf{E}}. \quad (7.6d)$$

复介电常数 ϵ_c 由方程式给出。(7.4) 通常用实部 ϵ' 和虚部 ϵ'' 来表示。因此，

$$\epsilon_c = \epsilon - j\frac{\sigma}{\omega} = \epsilon' - j\epsilon'', \quad (7.7)$$

和

$$\epsilon' = \epsilon, \quad (7.8a)$$

$$\epsilon'' = \frac{\sigma}{\omega}. \quad (7.8b)$$

For a lossless medium with $\sigma = 0$, it follows that $\epsilon'' = 0$ and $\epsilon_c = \epsilon' = \epsilon$.

7-1.2 Wave Equations

Next, we derive wave equations for $\tilde{\mathbf{E}}$ and $\tilde{\mathbf{H}}$ and then solve them to obtain explicit expressions for $\tilde{\mathbf{E}}$ and $\tilde{\mathbf{H}}$ as a function of the spatial variables (x, y, z) . To that end, we start by taking the curl of both sides of Eq. (7.6b) to get

$$\nabla \times (\nabla \times \tilde{\mathbf{E}}) = -j\omega\mu(\nabla \times \tilde{\mathbf{H}}).$$
(7.9)

Upon substituting Eq. (7.6d) into Eq. (7.9), we obtain

$$\nabla \times (\nabla \times \tilde{\mathbf{E}}) = -j\omega\mu(j\omega\epsilon_c\tilde{\mathbf{E}}) = \omega^2\mu\epsilon_c\tilde{\mathbf{E}}.$$
(7.10)

From Eq. (3.113), we know that the curl of the curl of $\tilde{\mathbf{E}}$ is

$$\nabla \times (\nabla \times \tilde{\mathbf{E}}) = \nabla(\nabla \cdot \tilde{\mathbf{E}}) - \nabla^2\tilde{\mathbf{E}},$$
(7.11)

where $\nabla^2\tilde{\mathbf{E}}$ is the Laplacian of $\tilde{\mathbf{E}}$, which in Cartesian coordinates is given by

$$\nabla^2\tilde{\mathbf{E}} = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) \tilde{\mathbf{E}}.$$
(7.12)

In view of Eq. (7.6a), the use of Eq. (7.11) in Eq. (7.10) gives

$$\nabla^2\tilde{\mathbf{E}} + \omega^2\mu\epsilon_c\tilde{\mathbf{E}} = 0,$$
(7.13)

which is known as the **homogeneous wave equation for $\tilde{\mathbf{E}}$** . By defining the **propagation constant** γ as

$$\gamma^2 = -\omega^2\mu\epsilon_c,$$
(7.14)

Eq. (7.13) can be written as

$$\nabla^2\tilde{\mathbf{E}} - \gamma^2\tilde{\mathbf{E}} = 0. \quad \textbf{(wave equation for } \tilde{\mathbf{E}} \textbf{)}$$
(7.15)

To derive Eq. (7.15), we took the curl of both sides of Eq. (7.6b) and then we used Eq. (7.6d) to eliminate $\tilde{\mathbf{H}}$ and obtain an equation in $\tilde{\mathbf{E}}$ only. If we reverse the process, that is, if we start by taking the curl of both sides of Eq. (7.6d) and then use Eq. (7.6b) to eliminate $\tilde{\mathbf{E}}$, we obtain a wave equation for $\tilde{\mathbf{H}}$:

$$\nabla^2\tilde{\mathbf{H}} - \gamma^2\tilde{\mathbf{H}} = 0. \quad \textbf{(wave equation for } \tilde{\mathbf{H}} \textbf{)}$$
(7.16)

Since the wave equations for $\tilde{\mathbf{E}}$ and $\tilde{\mathbf{H}}$ are of the same form, so are their solutions.

7-2 Plane-Wave Propagation in Lossless Media

The properties of an electromagnetic wave, such as its phase velocity u_p and wavelength λ , depend on the angular frequency ω and the medium's three constitutive parameters: ϵ , μ , and σ . If the medium is **nonconducting** ($\sigma = 0$), the wave does not suffer any attenuation as it travels; hence, the medium is said to be **lossless**. Because in a lossless medium $\epsilon_c = \epsilon$, Eq. (7.14) becomes

$$\gamma^2 = -\omega^2\mu\epsilon.$$
(7.17)

For lossless media, it is customary to define the **wavenumber** k as

$$k = \omega\sqrt{\mu\epsilon}.$$
(7.18)

In view of Eq. (7.17), $\gamma^2 = -k^2$ and Eq. (7.15) becomes

$$\nabla^2\tilde{\mathbf{E}} + k^2\tilde{\mathbf{E}} = 0.$$
(7.19)

7-2.1 Uniform Plane Waves

For an electric field phasor defined in Cartesian coordinates as

$$\tilde{\mathbf{E}} = \hat{\mathbf{x}}\tilde{E}_x + \hat{\mathbf{y}}\tilde{E}_y + \hat{\mathbf{z}}\tilde{E}_z,$$
(7.20)

substitution of Eq. (7.12) into Eq. (7.19) gives

$$\begin{aligned} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) (\hat{\mathbf{x}}\tilde{E}_x + \hat{\mathbf{y}}\tilde{E}_y + \hat{\mathbf{z}}\tilde{E}_z) \\ + k^2(\hat{\mathbf{x}}\tilde{E}_x + \hat{\mathbf{y}}\tilde{E}_y + \hat{\mathbf{z}}\tilde{E}_z) = 0. \end{aligned}$$
(7.21)

To satisfy Eq. (7.21), each vector component on the left-hand side of the equation must vanish. Hence,

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} + k^2 \right) \tilde{E}_x = 0,$$
(7.22)

and similar expressions apply to $\tilde{E}_y$ and $\tilde{E}_z$.

► A **uniform plane wave** is characterized by electric and magnetic fields that have uniform properties at all points across an infinite plane. ◀

If this happens to be the x - y plane, then $\mathbf{E}$ and $\mathbf{H}$ do not vary with x or y . Hence, $\partial\tilde{E}_x/\partial x = 0$ and $\partial\tilde{E}_x/\partial y = 0$, so Eq. (7.22) reduces to

$$\frac{d^2\tilde{E}_x}{dz^2} + k^2\tilde{E}_x = 0.$$
(7.23)

对于 $\sigma = 0$ 的无损介质，可以得出 $\epsilon'' = 0$ 和 $\epsilon_c = \epsilon' = \epsilon$ 。

7-1.2波动方程

接下来，我们推导 $\tilde{\mathbf{E}}$ 和 $\tilde{\mathbf{H}}$ 的波动方程，然后求解它们以获得 $\tilde{\mathbf{E}}$ 和 $\tilde{\mathbf{H}}$ 作为空间变量 (x, y, z) 的函数的显式表达式。为此，我们首先取方程两边的旋度。(7.6b) 得到

$$\nabla \times (\nabla \times \tilde{\mathbf{E}}) = -j\omega\mu(\nabla \times \tilde{\mathbf{H}}).$$
(7.9)

代入方程式后。(7.6d) 代入方程。(7.9)，我们得到

$$\nabla \times (\nabla \times \tilde{\mathbf{E}}) = -j\omega\mu(j\omega\epsilon_c\tilde{\mathbf{E}}) = \omega^2\mu\epsilon_c\tilde{\mathbf{E}}.$$
(7.10)

从方程。(3.113)，我们知道 $\tilde{\mathbf{E}}$ 的旋度的旋度是

$$\nabla \times (\nabla \times \tilde{\mathbf{E}}) = \nabla(\nabla \cdot \tilde{\mathbf{E}}) - \nabla^2\tilde{\mathbf{E}},$$
(7.11)

其中 $\nabla^2\tilde{\mathbf{E}}$ 是 $\tilde{\mathbf{E}}$ 的拉普拉斯算子，在笛卡尔坐标中由下式给出

$$\nabla^2\tilde{\mathbf{E}} = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) \tilde{\mathbf{E}}.$$
(7.12)

鉴于方程。(7.6a)，使用方程。(7.11) 式(7.10) 给出

$$\nabla^2\tilde{\mathbf{E}} + \omega^2\mu\epsilon_c\tilde{\mathbf{E}} = 0,$$
(7.13)

这被称为 $\tilde{\mathbf{E}}$ 的齐次波动方程。通过将传播常数 γ 定义为

$$\gamma^2 = -\omega^2\mu\epsilon_c,$$
(7.14)

等式。(7.13) 可以写成

$$\nabla^2\tilde{\mathbf{E}} - \gamma^2\tilde{\mathbf{E}} = 0. \quad \textbf{(wave equation for } \tilde{\mathbf{E}} \textbf{)}$$
(7.15)

推导出方程。(7.15)，我们取方程两边的旋度。(7.6b) 然后我们使用方程。(7.6d) 消去 $\tilde{\mathbf{H}}$ 并仅得到 $\tilde{\mathbf{E}}$ 中的方程。如果我们反转这个过程，也就是说，如果我们从方程两边的旋度开始。(7.6d)，然后使用方程。(7.6b) 消除 $\tilde{\mathbf{E}}$ ，我们得到 $\tilde{\mathbf{H}}$ 的波动方程：

$$\nabla^2\tilde{\mathbf{H}} - \gamma^2\tilde{\mathbf{H}} = 0. \quad \textbf{(wave equation for } \tilde{\mathbf{H}} \textbf{)}$$
(7.16)

由于 $\tilde{\mathbf{E}}$ 和 $\tilde{\mathbf{H}}$ 的波动方程具有相同的形式，因此它们的解也具有相同的形式。

7-2 无损介质中的平面波传播

电磁波的特性，例如相速度 u_p 和波长 λ ，取决于角频率 ω 和介质的三个本构参数： ϵ 、 μ 和 σ 。如果介质不导电 ($\sigma = 0$)，则波在传播时不会受到任何衰减；因此，该介质被称为无损介质。因为在无损介质 $\epsilon_c = \epsilon$ 中，等式：(7.14) 变为

$$\gamma^2 = -\omega^2\mu\epsilon.$$
(7.17)

对于无损介质，通常将波数 k 定义为

$$k = \omega\sqrt{\mu\epsilon}.$$
(7.18)

鉴于方程。(7.17), $\gamma^2 = -k^2$ 和方程。(7.15) 变为

$$\nabla^2\tilde{\mathbf{E}} + k^2\tilde{\mathbf{E}} = 0.$$
(7.19)

7-2.1 均匀平面波

对于笛卡尔坐标中定义的电场相量为

$$\tilde{\mathbf{E}} = \hat{\mathbf{x}}\tilde{E}_x + \hat{\mathbf{y}}\tilde{E}_y + \hat{\mathbf{z}}\tilde{E}_z,$$
(7.20)

代入方程式 (7.12) 代入等式。(7.19) 给出

$$\begin{aligned} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) (\hat{\mathbf{x}}\tilde{E}_x + \hat{\mathbf{y}}\tilde{E}_y + \hat{\mathbf{z}}\tilde{E}_z) \\ + k^2(\hat{\mathbf{x}}\tilde{E}_x + \hat{\mathbf{y}}\tilde{E}_y + \hat{\mathbf{z}}\tilde{E}_z) = 0. \end{aligned}$$
(7.21)

满足方程。根据 (7.21) 式，方程左侧的每个向量分量都必须消失。因此，

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} + k^2 \right) \tilde{E}_x = 0,$$
(7.22)

类似的表达式适用于 $\tilde{E}_y$ 和 $\tilde{E}_z$ 。

► 均匀平面波的特征是电场和磁场在无限平面上的所有点上具有均匀的特性。◀

如果这恰好是 x - y 平面，则 $\mathbf{E}$ 和 $\mathbf{H}$ 不随 x 或 y 变化。因此， $\partial\tilde{E}_x/\partial x = 0$ 和 $\partial\tilde{E}_x/\partial y = 0$ ，所以等式：(7.22) 减少为

$$\frac{d^2\tilde{E}_x}{dz^2} + k^2\tilde{E}_x = 0.$$
(7.23)

Similar expressions apply to $\tilde{E}_y$, $\tilde{H}_x$, and $\tilde{H}_y$. The remaining components of $\tilde{\mathbf{E}}$ and $\tilde{\mathbf{H}}$ are zero; that is, $\tilde{E}_z = \tilde{H}_z = 0$. To show that $\tilde{E}_z = 0$, let us consider the z component of Eq. (7.6d),

$$\hat{\mathbf{z}} \left(\frac{\partial \tilde{H}_y}{\partial x} - \frac{\partial \tilde{H}_x}{\partial y} \right) = \hat{\mathbf{z}} j\omega\epsilon \tilde{E}_z. \quad (7.24)$$

Since $\partial \tilde{H}_y / \partial x = \partial \tilde{H}_x / \partial y = 0$, it follows that $\tilde{E}_z = 0$. A similar examination involving Eq. (7.6b) reveals that $\tilde{H}_z = 0$.

► This means that a plane wave has no electric-field or magnetic-field components along its direction of propagation. ◀

For the phasor quantity $\tilde{E}_x$, the general solution of the ordinary differential equation given by Eq. (7.23) is

$$\tilde{E}_x(z) = \tilde{E}_x^+(z) + \tilde{E}_x^-(z) = E_{x0}^+ e^{-jkz} + E_{x0}^- e^{jkz}, \quad (7.25)$$

where E_{x0}^+ and E_{x0}^- are constants to be determined from boundary conditions. The solution given by Eq. (7.25) is similar in form to the solution for the phasor voltage $\tilde{V}(z)$ given by Eq. (2.54a) for the lossless transmission line. The first term in Eq. (7.25), containing the negative exponential e^{-jkz} , represents a wave with amplitude E_{x0}^+ traveling in the $+z$ direction. Likewise, the second term (with e^{jkz}) represents a wave with amplitude E_{x0}^- traveling in the $-z$ direction. Assume for the time being that $\tilde{\mathbf{E}}$ only has a component along x (i.e., $\tilde{E}_y = 0$) and that $\tilde{E}_x$ is associated with a wave traveling in the $+z$ direction only (i.e., $E_{x0}^- = 0$). Under these conditions,

$$\tilde{\mathbf{E}}(z) = \hat{\mathbf{x}} \tilde{E}_x^+(z) = \hat{\mathbf{x}} E_{x0}^+ e^{-jkz}. \quad (7.26)$$

To find the magnetic field $\tilde{\mathbf{H}}$ associated with this wave, we apply Eq. (7.6b) with $\tilde{E}_y = \tilde{E}_z = 0$:

$$\nabla \times \tilde{\mathbf{E}} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \tilde{E}_x^+(z) & 0 & 0 \end{vmatrix} = -j\omega\mu (\hat{\mathbf{x}} \tilde{H}_x + \hat{\mathbf{y}} \tilde{H}_y + \hat{\mathbf{z}} \tilde{H}_z). \quad (7.27)$$

For a uniform plane wave traveling in the $+z$ direction,

$$\partial E_x^+(z) / \partial x = \partial E_x^+(z) / \partial y = 0.$$

Hence, Eq. (7.27) gives

$$\tilde{H}_x = 0, \quad (7.28a)$$

$$\tilde{H}_y = \frac{1}{-j\omega\mu} \frac{\partial \tilde{E}_x^+(z)}{\partial z}, \quad (7.28b)$$

$$\tilde{H}_z = \frac{1}{-j\omega\mu} \frac{\partial E_x^+(z)}{\partial y} = 0. \quad (7.28c)$$

Use of Eq. (7.26) in Eq. (7.28b) gives

$$\tilde{H}_y(z) = \frac{k}{\omega\mu} E_{x0}^+ e^{-jkz} = H_{y0}^+ e^{-jkz}, \quad (7.29)$$

where H_{y0}^+ is the amplitude of $\tilde{H}_y(z)$ and is given by

$$H_{y0}^+ = \frac{k}{\omega\mu} E_{x0}^+. \quad (7.30)$$

For a wave traveling from the source toward the load on a transmission line, the amplitudes of its voltage and current phasors, V_0^+ and I_0^+ , are related by the characteristic impedance of the line, Z_0 . A similar connection exists between the electric and magnetic fields of an electromagnetic wave. The **intrinsic impedance** of a lossless medium is defined as

$$\eta = \frac{\omega\mu}{k} = \frac{\omega\mu}{\omega\sqrt{\mu\epsilon}} = \sqrt{\frac{\mu}{\epsilon}} \quad (\Omega), \quad (7.31)$$

where we used the expression for k given by Eq. (7.18).

In view of Eq. (7.31), the electric and magnetic fields of a $+z$ -propagating plane wave with $\mathbf{E}$ field along $\hat{\mathbf{x}}$ are

$$\tilde{\mathbf{E}}(z) = \hat{\mathbf{x}} \tilde{E}_x^+(z) = \hat{\mathbf{x}} E_{x0}^+ e^{-jkz}, \quad (7.32a)$$

$$\tilde{\mathbf{H}}(z) = \hat{\mathbf{y}} \frac{\tilde{E}_x^+(z)}{\eta} = \hat{\mathbf{y}} \frac{E_{x0}^+}{\eta} e^{-jkz}. \quad (7.32b)$$

► The electric and magnetic fields of a plane wave are perpendicular to each other, and both are perpendicular to the direction of wave travel (Fig. 7-4). These attributes qualify the wave as **transverse electromagnetic (TEM)**. ◀

Other examples of TEM waves include waves traveling on coaxial transmission lines ($\mathbf{E}$ is along $\hat{\mathbf{r}}$, $\mathbf{H}$ is along $\hat{\boldsymbol{\phi}}$, and the direction of travel is along $\hat{\mathbf{z}}$) and spherical waves radiated by antennas.

In the general case, E_{x0}^+ is a complex quantity with magnitude $|E_{x0}^+|$ and phase angle ϕ^+ . That is,

$$E_{x0}^+ = |E_{x0}^+| e^{j\phi^+}. \quad (7.33)$$

The instantaneous electric and magnetic fields therefore are

$$\mathbf{E}(z, t) = \Re \left[\tilde{\mathbf{E}}(z) e^{j\omega t} \right] = \hat{\mathbf{x}} |E_{x0}^+| \cos(\omega t - kz + \phi^+) \quad (\text{V/m}), \quad (7.34a)$$

类似的表达式适用于 $\tilde{E}_y$ 、 $\tilde{H}_x$ 和 $\tilde{H}_y$ 。 $\tilde{E}$ 和 $\tilde{H}$ 的其余分量为零；即， $\tilde{E}_z = \tilde{H}_z = 0$ 。为了证明 $\tilde{E}_z = 0$ ，让我们考虑方程的 z 分量。（7.6天），

$$\hat{\mathbf{z}} \left(\frac{\partial \tilde{H}_y}{\partial x} - \frac{\partial \tilde{H}_x}{\partial y} \right) = \hat{\mathbf{z}} j\omega\epsilon \tilde{E}_z. \quad (7.24)$$

由于 $\partial \tilde{H}_y / \partial x = \partial \tilde{H}_x / \partial y = 0$ ，因此 $\tilde{E}_z = 0$ 。涉及方程的类似检查。（7.6b）表明 $\tilde{H}_z = 0$ 。

► 这意味着平面波沿其传播方向没有电场或磁场分量。
◀

对于相量量 $\tilde{E}_x$ ，常微分方程的通解由式（1）给出。（7.23）是

$$\tilde{E}_x(z) = \tilde{E}_x^+(z) + \tilde{E}_x^-(z) = E_{x0}^+ e^{-jkz} + E_{x0}^- e^{jkz}, \quad (7.25)$$

其中 E_{x0}^+ 和 E_{x0}^- 是根据边界条件确定的常数。方程给出的解。（7.25）的形式与方程（7.25）给出的相量电压 $\tilde{V}(z)$ 的解类似。（2.54a）为无损传输线。等式中的第一项。式（7.25）包含负指数 e^{-jkz} ，表示沿 $+z$ 方向传播的振幅为 E_{x0}^+ 的波。同样，第二项（ e^{jkz} ）表示沿 $-z$ 方向传播的振幅为 E_{x0}^- 的波。暂时假设 $\tilde{E}$ 仅具有沿 x 的分量（即 $\tilde{E}_y = 0$ ），并且 $\tilde{E}_x$ 与仅沿 $+z$ 方向传播的波相关联（即 $E_{x0}^- = 0$ ）。在这些条件下，

$$\tilde{\mathbf{E}}(z) = \hat{\mathbf{x}} \tilde{E}_x^+(z) = \hat{\mathbf{x}} E_{x0}^+ e^{-jkz}. \quad (7.26)$$

为了找到与该波相关的磁场 $\tilde{H}$ ，我们应用方程：（7.6b）与 $\tilde{E}_y = \tilde{E}_z = 0$ ：

$$\nabla \times \tilde{\mathbf{E}} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \tilde{E}_x^+(z) & 0 & 0 \end{vmatrix} = -j\omega\mu (\hat{\mathbf{x}} \tilde{H}_x + \hat{\mathbf{y}} \tilde{H}_y + \hat{\mathbf{z}} \tilde{H}_z). \quad (7.27)$$

对于沿 $+z$ 方向传播的均匀平面波，

$$\partial E_x^+(z) / \partial x = \partial E_x^+(z) / \partial y = 0.$$

因此，等式。（7.27）给出

$$\tilde{H}_x = 0, \quad (7.28a)$$

$$\tilde{H}_y = \frac{1}{-j\omega\mu} \frac{\partial \tilde{E}_x^+(z)}{\partial z}, \quad (7.28b)$$

$$\tilde{H}_z = \frac{1}{-j\omega\mu} \frac{\partial E_x^+(z)}{\partial y} = 0. \quad (7.28c)$$

使用等式。（7.26）式（7.28b）给出

$$\tilde{H}_y(z) = \frac{k}{\omega\mu} E_{x0}^+ e^{-jkz} = H_{y0}^+ e^{-jkz}, \quad (7.29)$$

其中 H_{y0}^+ 是 $\tilde{H}_y(z)$ 的幅度，由下式给出

$$H_{y0}^+ = \frac{k}{\omega\mu} E_{x0}^+. \quad (7.30)$$

对于传输线上从源向负载传播的波，其电压和电流相量 V_0^+ 和 I_0^+ 的幅度与线路的特征阻抗 Z_0 相关。电磁波的电场和磁场之间也存在类似的联系。无损介质的固有阻抗定义为

$$\eta = \frac{\omega\mu}{k} = \frac{\omega\mu}{\omega\sqrt{\mu\epsilon}} = \sqrt{\frac{\mu}{\epsilon}} \quad (\Omega), \quad (7.31)$$

其中我们使用了等式给出的 k 表达式。（7.18）。鉴于方程。（7.31）中， $\mathbf{E}$ 场沿 $\hat{\mathbf{x}}$ 传播的 $+z$ 平面波的电场和磁场为

$$\tilde{\mathbf{E}}(z) = \hat{\mathbf{x}} \tilde{E}_x^+(z) = \hat{\mathbf{x}} E_{x0}^+ e^{-jkz}, \quad (7.32a)$$

$$\tilde{\mathbf{H}}(z) = \hat{\mathbf{y}} \frac{\tilde{E}_x^+(z)}{\eta} = \hat{\mathbf{y}} \frac{E_{x0}^+}{\eta} e^{-jkz}. \quad (7.32b)$$

► 平面波的电场和磁场相互垂直，并且都垂直于波的传播方向（图7-4）。这些属性将波定义为横向电磁波（TEM）。◀

TEM 波的其他示例包括在同轴传输线上传播的波（ $\mathbf{E}$ 沿 $\hat{\mathbf{r}}$ ， $\mathbf{H}$ 沿 $\hat{\boldsymbol{\phi}}$ ，传播方向沿 $\hat{\mathbf{z}}$ ）以及由天线辐射的球面波。

一般情况下， E_{x0}^+ 是一个复量，其大小为 $|E_{x0}^+|$ 和相位角 ϕ^+ 。那是，

$$E_{x0}^+ = |E_{x0}^+| e^{j\phi^+}. \quad (7.33)$$

因此瞬时电场和磁场为

$$\mathbf{E}(z, t) = \Re \left[\tilde{\mathbf{E}}(z) e^{j\omega t} \right] = \hat{\mathbf{x}} |E_{x0}^+| \cos(\omega t - kz + \phi^+) \quad (\text{V/m}), \quad (7.34a)$$